

Analytic Mechanics

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September 20, 2025

Contents

1	Introduction	1
2	Short introduction on Coordinate Systems	6
2.1	Constraints	17
3	Lagrangian Mechanics	27
3.1	The Euler-Lagrange Equations	32
3.2	Examples	40
3.3	Properties of the Euler-Lagrange Equations	44
3.4	Hamilton's Stationary Action Principle	50
4	Conservation Laws	55
4.1	Cyclic coordinates	56
4.2	Noether's Theorem	63
4.3	Jacobi Integral and Conservation of Energy	68
4.4	Central Force Problem – Hidden Symmetries	81
5	Perturbation Theory	92
A	Rotations in $\mathbb{R}^3$	101

1 Introduction

All those who are reading these notes, be it because they feel compelled to do it or because they do it out of their own volition for some inscrutable reason, should know that actually these are the first lines that I have written. As a matter of fact, I was trying to look for a compelling reason to convince you that one needs to go one step beyond Newtonian mechanics. The more I thought about it, the more I realized that the deep reasons do not lie in mathematical rigor, in the possibility of gaining deeper physical insights, but rather in sheer efficiency. Lagrangian

mechanics is not a better description of the classical world, it is *more efficient*. Let me try to convince you with a plain, simple, yet instructive example. Suppose I am asking you to build a toy model aimed at describing a satellite constrained to move in a geostationary orbit around Earth. Giving a full description of this system is beyond our grasp since among the many aspects that we would have to consider, we would have at least to take into account the following data:

- the satellite is a rigid body comprised of different mechanical and electric parts with internal forces accounting for their interactions,
- Earth is not round and therefore its gravitational field is neither homogeneous nor isotropic. On top of this, one would have to account for electromagnetic interactions since our planet possesses a huge magnetosphere shielding the surface from high energy cosmic radiation. These magnetic fields interact with the components of the satellite,
- The gravitational field close to Earth is modified by all the bodies in the solar system, in particular the Sun and the Moon. These influence the trajectory of the satellite hence hindering the possibility of keeping a geostationary orbit.

If you try to set up this problem using all you learned in your course of Mechanics, you would almost immediately feel the sheer difficulty of the task ahead of you since all these external forces would need to be accounted for. Therefore, it is the task of physics 1) to extrapolate which are the key and irreplaceable ingredients which influence the most the trajectory of the satellite, 2) to formulate a simpler yet effective model which can be studied both qualitatively and quantitatively and 3) to keep track of the approximations which have been introduced to build such a model. In particular we need to be able to quantify them to the extent that it is clear under which circumstances they stop being negligible features and they acquire significance. To this extent we can recognize that, to start with, we can make the following approximations:

1. the dimension of the satellite is small compared to all other scale lengths, *e.g.*, distance to the center of Earth. Hence we can assume that it is a pointwise object of a given mass m ,
2. the satellite is electrically neutral, hence all electromagnetic interactions can be discarded,
3. The distance from the surface of Earth is such that all deviations from the assumption that our planet is perfectly round play no significant role,
4. The gravitational force due to the interaction with the other bodies in the solar system, except Earth, is negligible.

It is rather intuitive that the last assumption is the strongest one, especially discarding the role of the Moon. For the time being we content ourselves with these hypotheses but we shall remark on the effectiveness of this assumption later in these notes. We might be prone to believe that the problem under scrutiny has been simplified to the extent that it has become very easy. Let us see if this is really true.

Example 1) Let us consider $\mathbb{R}^3$ as the physical space and let us endow it with standard Cartesian coordinates x, y, z . Let $t \in \mathbb{R}$ denote the time variable and consider the standard orthogonal reference system centered at the origin of $\mathbb{R}^3$. On top of this let us assume that Earth is modeled as a sphere of radius R and mass $M > 0$, centered at the origin, while the satellite is a point particle of mass $m > 0$ located at time $t = 0$ at a generic point (x_0, y_0, z_0) such that $r_0^2 \doteq x_0^2 + y_0^2 + z_0^2 \gg R^2$. Keeping in mind that we want the satellite to keep a geostationary orbit, according to Newton mechanics the position $(x(t), y(t), z(t))$ of the satellite is ruled by the following differential equations:

$$m\ddot{\mathbf{r}} = -\frac{GMm}{r_0^3}\mathbf{r} + \lambda\mathbf{r}, \quad (1)$$

where $\mathbf{r} \equiv \mathbf{r}(t)$ is the position vector of the particle at each instant of time. In other words $\mathbf{r}(t) = (x(t), y(t), z(t))$. Furthermore, on the right hand side of Equation (1), we account for two terms. The first one is the standard radial gravitational force, while the second one is a fictitious radial force which is added so to enforce the constraint that $\|\mathbf{r}(t)\| = \sqrt{x^2(t) + y^2(t) + z^2(t)} = r_0$. Hence $\lambda \equiv \lambda(t)$ is a Lagrange multiplier whose value can be derived by requiring that the radial component of the force vanishes, namely

$$m\ddot{\mathbf{r}} \cdot \mathbf{e}_{\mathbf{r}} = -\left(\frac{GMm}{r_0^3}\mathbf{r} - \lambda\mathbf{r}\right) \cdot \mathbf{e}_{\mathbf{r}}, \quad (2)$$

where $\cdot$ denotes the standard Euclidean scalar product on $\mathbb{R}^3$ while $\mathbf{e}_{\mathbf{r}} = \frac{\mathbf{r}}{\|\mathbf{r}\|}$. Observe that the constraint that the particle should not move radially in the radial direction entails that

$$\dot{\mathbf{r}} \cdot \mathbf{e}_{\mathbf{r}} = 0 \iff \dot{\mathbf{r}} \cdot \mathbf{r} = 0,$$

namely the velocity along the radial direction vanishes. This entails in turn that

$$\frac{d}{dt}(\dot{\mathbf{r}} \cdot \mathbf{r}) = \ddot{\mathbf{r}} \cdot \mathbf{r} + \|\dot{\mathbf{r}}\|^2 = 0 \iff \ddot{\mathbf{r}} \cdot \mathbf{e}_{\mathbf{r}} = -\frac{\|\dot{\mathbf{r}}\|^2}{\|\mathbf{r}\|}.$$

Replacing this result in Equation (2) yields

$$m\frac{\|\dot{\mathbf{r}}\|^2}{\|\mathbf{r}\|} = \frac{GMm}{r_0^3}\|\mathbf{r}\| - \lambda\|\mathbf{r}\| \implies \lambda = \frac{GMm}{r_0^3} - m\left(\frac{\|\dot{\mathbf{r}}\|}{\|\mathbf{r}\|}\right)^2.$$

Replacing this identity in Equation (1) and observing that $\|\mathbf{r}\| = r_0$, we end up with

$$\ddot{\mathbf{r}} = -\frac{\|\dot{\mathbf{r}}\|^2}{r_0}\mathbf{e}_{\mathbf{r}}. \quad (3)$$

The interpretation of this result is that all the gravitational forces are canceled out and the remaining term on the right hand side can be interpreted as the standard centripetal acceleration needed to keep the particle moving on a trajectory of constant radius. At first glance a reader

might think that, after all, this has been a rather simple and straightforward calculation starting from Equation (1) and ending with Equation (3). Yet, at a closer look, we should feel unsatisfied. On the one hand a very simple geometric constraint has been accounted for in a rather convoluted way inserting a fictitious force $\lambda \mathbf{r}$ whose physical interpretation and motivation is not that convincing. Furthermore Equation (3) has been obtained since we could find easily an explicit expression for λ , a feature not at all obvious in a more general scenario. Thirdly, Equation (3) is manifestly non linear, making it hard to solve especially when working in Cartesian coordinates. In this respect the geometry of the problem seems to suggest a priori that the motion of the system should take place on a sphere of a radius r_0 and, therefore, it would be highly natural to work with spherical coordinates. As an exercise let us see what would be the outcome. While Cartesian coordinates are $(x, y, z) \in \mathbb{R}^3$, spherical ones are (r, θ, φ) where $r \in (0, \infty)$, $\theta \in (0, \pi)$ while $\varphi \in [0, 2\pi)$. They are related by the transformation laws

$$\begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta \end{cases} . \quad (4)$$

Observe that the Jacobian of this transformation reads

$$J[r, \theta, \varphi] = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \varphi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \varphi} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \varphi} \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \varphi & r \cos \theta \cos \varphi & -r \sin \theta \sin \varphi \\ \sin \theta \sin \varphi & r \cos \theta \sin \varphi & r \sin \theta \cos \varphi \\ \cos \theta & -r \sin \theta & 0 \end{bmatrix} .$$

In order for Equation (4) to be a well defined coordinate transformation, it needs to be invertible, hence the determinant of the Jacobian must not vanish. A direct computation yields

$$\det(J) = r^2 \sin \theta ,$$

which vanishes only if $r = 0$. Hence *switching from Cartesian to polar coordinates* is not a globally defined transformation (think geometrically, $\mathbb{R}^3$ is not like working with $(0, \infty) \times \mathbb{S}^2$). Yet, we should not worry about this in the case in hand since we are imposing the constraint $r = r_0 > 0$, hence we are always far from the origin. Consequently, we can express the position of the point particle in terms of the spherical coordinates, *i.e.*

$$\mathbf{r}(t) = (x(t), y(t), z(t)) \longrightarrow (r_0, \theta(t), \varphi(t)).$$

The velocity reads

$$\begin{cases} \dot{x}(t) = r_0 \left(\cos \theta(t) \cos \varphi(t) \dot{\theta}(t) - \sin \theta(t) \sin \varphi(t) \dot{\varphi}(t) \right) \\ \dot{y}(t) = r_0 \left(\cos \theta(t) \sin \varphi(t) \dot{\theta}(t) + \sin \theta(t) \cos \varphi(t) \dot{\varphi}(t) \right) , \\ \dot{z}(t) = -r_0 \sin \theta(t) \dot{\theta}(t) \end{cases} , \quad (5)$$

where we have implemented that $\dot{r} = 0$ since $r(t) = r_0$ for all values of $t \in \mathbb{R}$. This entails in turn that

$$\|\dot{\mathbf{r}}(t)\|^2 = \dot{x}^2(t) + \dot{y}^2(t) + \dot{z}^2(t) = r_0^2 \left(\dot{\theta}^2(t) + \sin^2 \theta(t) \dot{\varphi}^2(t) \right) . \quad (6)$$

Taking the derivative with respect to t of Equation (5) we obtain the components of the acceleration vector in polar coordinates, namely

$$\begin{cases} \ddot{x} = r_0 \left[-\sin \theta \cos \phi \dot{\theta}^2 - 2 \cos \theta \sin \phi \dot{\theta} \dot{\phi} - \sin \theta \cos \phi \dot{\phi}^2 + \cos \theta \cos \phi \ddot{\theta} - \sin \theta \sin \phi \ddot{\phi} \right] \\ \ddot{y} = r_0 \left[-\sin \theta \sin \phi \dot{\theta}^2 + 2 \cos \theta \cos \phi \dot{\theta} \dot{\phi} - \sin \theta \sin \phi \dot{\phi}^2 + \cos \theta \sin \phi \ddot{\theta} + \sin \theta \cos \phi \ddot{\phi} \right], \\ \ddot{z} = r_0 \left(-\cos \theta \dot{\theta}^2 - \sin \theta \ddot{\theta} \right) \end{cases}, \quad (7)$$

where we have implemented that $\ddot{r} = 0$ since $r(t) = r_0$ for all values of $t \in \mathbb{R}$. Here, for the sake of conciseness we have left implicit the dependence on the time variable t . Inserting Equations (7) and Equation (6) in Equation (3) yields after a few algebraic manipulations the following set of coupled ordinary differential equations:

$$\begin{cases} \ddot{\theta}(t) - \sin \theta(t) \cos \theta(t) \dot{\phi}^2(t) = 0 \\ \sin^2 \theta(t) \ddot{\phi}(t) + 2 \sin \theta(t) \cos \theta(t) \dot{\theta}(t) \dot{\phi}(t) = \frac{d}{dt} (\sin^2 \theta(t) \dot{\phi}(t)) = 0 \end{cases} \quad (8)$$

As it will be discussed thoroughly in these notes, the second equation encodes the conservation of the component of the angular momentum along the z -axis, while the first one, known also as the *great circle equation* is a highly non linear ordinary differential equation.

Exercise 1. Derive explicitly Equation (8) from Equation (7).

The lesson that we learned and the questions which arise naturally are therefore the following

1. Given one or more geometric constraints, is it possible to write down the equations of motion induced by the laws of Newtonian mechanics in terms of coordinates which are subordinated by these constraints? In other words, if necessary can we avoid working from the very beginning with Cartesian coordinates?
2. Is it possible to avoid introducing fictitious forces and Lagrange multipliers, hence taking into account intrinsically the constraints that we might have to impose on the motion of the constituents of a physical system?
3. If the first two questions have a positive answer, is there a way to derive directly and efficiently the equations of motion in terms of the coordinates adapted to the underlying constraints? In other words can we get Equation (8) directly?

Yes, this is indeed **Lagrangian mechanics**

Remark 1. *I should be ending the section with the catch phrase above, but, wait a second! Let us go back to Equation (8). They describe the motion of a particle on a sphere and therefore, having written them, is by far not the end of the story. If we do not solve them, if we have no*

information whatsoever on the solutions for given initial data, what is the point? What have I learned about the physics of the system? Absolutely nothing! I am not talking about existence and uniqueness of the solution, which are important properties which hold true using for example the Picard-Lindelöf theorem. I am really asking you if you can infer how the solution looks like. This is all what the second part of this course is about! Stay tuned!

Important Notes for the Reader

Before starting the main part of these notes, the reader should be warned that the text contains several **colored boxes**. Colors are used with the following convention:

- **Red Boxes:** These are advanced mathematical comments which give you a deeper structural and technical understanding of the topic. Some if not all of these will be covered in other courses during the semester. Skipping them is possible without hindering a smooth reading, but it is not advised.
- **Blue Boxes:** These are basic topics which should have already been covered in other courses. Here they are recollected for the reader's convenience, presenting them with an updated language, tailored to this course.
- **Green Boxes:** Advanced versions of topics or of examples covered in the main text, to offer a more general viewpoint on some of the main concepts used in the course. A first reading of the course material does not require delving into them.
- **Black Boxes:** Herein we sketch paradigm shifts. These are covering concepts necessary to understand the limitation both of the formalism and of the techniques employed in these notes. They serve as a warning that, going beyond the topics covered here, might require changing completely both the physical and also the mathematical viewpoint.

2 Short introduction on Coordinate Systems

Is there a better description of a cube than that of its construction?

– The Brutalist

The example outlined in the previous section has highlighted the difficulty of studying the dynamics of one or more point particles using the laws of Newtonian mechanics once we impose additional constraints to their motion. Our task consists of looking for a novel formalism which allows us to take them into account efficiently while keeping untouched the building blocks of classical mechanics. It does not come as a surprise that the word *constraint* is too wide as it encompasses scenarios which are inherently too different to be dealt with within the same formalism. To convince yourselves, consider these two cases:

- a point particle of mass $m > 0$ is moving in the physical space $\mathbb{R}^3$ constrained to move on a plane, which, in standard Cartesian coordinates is parameterized as the locus $z = 0$.
- a point particle of mass $m > 0$ is moving in a room with impenetrable walls. This is modeled as $\Omega \subset \mathbb{R}^3$ where $\Omega \doteq [0, 1] \times [0, 1] \times [0, 1]$ and, as such, it entails that, considering the standard Cartesian coordinates (x_1, x_2, x_3) , these are constrained to abide by the inequalities $x_i \geq 0$ and $x_i \leq 1$, $i = 1, 2, 3$.

These are two totally reasonable idealizations of admissible physical scenarios, such as a ball moving on a table in the first case or a ball thrown at a wall. Yet the physics behind these two systems is drastically different since, in the second instance, we have to account for the bouncing against the wall, a rather extreme phenomenon from the viewpoint of classical mechanics. This can be seen considering the behaviour of the velocity vector of the particle which changes instantaneously its direction as soon as the particle hits the wall.

These two simple and idealized scenarios are thought as a way to convince you of the necessity of introducing a classification of the possible geometric constraints which can occur when discussing the dynamics of a physical system. The nomenclature that we shall use is by all means baroque, an heritage of a distant past or rather this is what you might think. It has been actually developed only 100 years ago by scientists such as Hertz, Appell and later Tullio Levi-Civita, Vito Volterra, and Jacques Hadamard. Enough words, time to be precise.

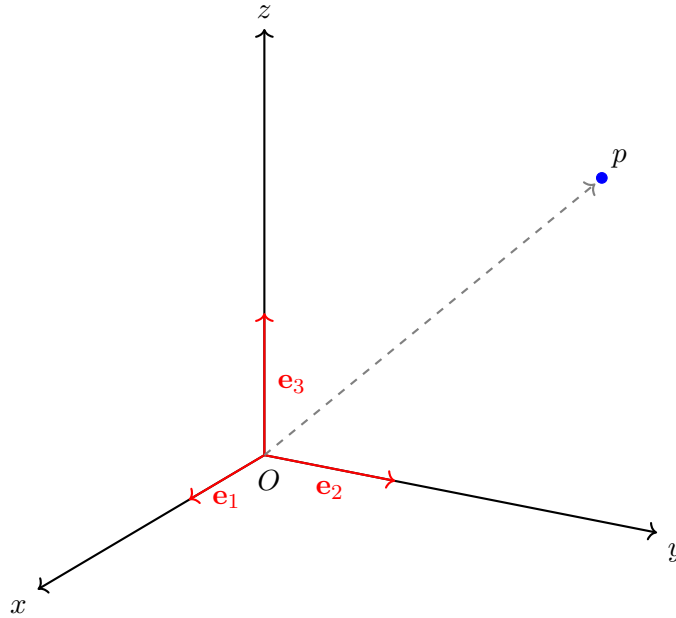


Figure 1: Standard picture of a reference system in $\mathbb{R}^3$ with O as the origin and with the three directions, pointing along the orthonormal basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$.

Throughout these notes we consider as ambient space, the *physical space* $\mathbb{R}^3$ realized as a three-dimensional real vector space where we have selected a **reference frame (system)** that is

1. a distinguished point O which plays the role of the origin,
2. an orthonormal basis of $\mathbb{R}^3$ which will be denoted by $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$.

Colloquially these three vectors are referred to as x -, y - and z -directions since it is common to depict a reference system in terms of three axis each intersecting at a point, namely O , and each parallel to one among these three unit vectors, see Figure 2.

Affine Space (Advanced Topic)

The attentive reader might be unhappy about our discussion of reference systems and coordinates therein since we have hidden under the carpet that we need to select a point O which is the origin, having the physical significance of representing the observer of the system itself. This is a wanted feature, but, if we interpret O as the origin of a Cartesian reference frame and hence as the 0-vector in $\mathbb{R}^n$, one might be wondering what does it mean to choose a different observer, being the 0 element of a vector space unique. Loosely speaking two observers O and O' would quarrel on who is who. This quandary is solved by the notion of affine spaces which is thoroughly discussed in [Mor20, Chap. 2]. Here we report the basic definition.

Definition 2. A collection of points $\mathbf{A}^n$ is called a real **affine** space of (finite) dimension $n \geq 1$ if there exists a real n -dimensional vector space V as well as a map

$$- : \mathbf{A}^n \times \mathbf{A}^n \rightarrow V, \quad (P, Q) \mapsto P - Q,$$

such that

1. for all $Q \in \mathbf{A}^n$ and for all $v \in V$, $\exists! P \in \mathbf{A}^n$ such that $P - Q = v$,
2. for all $P, Q, R \in \mathbf{A}^n$,

$$P - R = (P - Q) + (Q - R),$$

where the $+$ -operation is the standard addition in V .

In this respect, one can now infer that a reference system for classical mechanics is nothing but the assignment of an n -dimensional affine space $\mathbf{A}^n$ and of a distinguished point thereon $O \in \mathbf{A}^n$, called the *origin*. This identifies an n -dimensional vector space isomorphic to $\mathbb{R}^n$ as

$$V_O \doteq \{P - O \mid P \in \mathbf{A}^n\}.$$

This encompasses the freedom that we have in classical mechanics to choose an origin for a reference system. In the visual depiction of a vector as a segment with an arrow this corresponds to the freedom of choosing the base point.

Remark 2. In these notes and more generally throughout the course of analytic mechanics, it will be often convenient to consider simpler scenarios where $\mathbb{R}^3$ is replaced by $\mathbb{R}^2$ or even by $\mathbb{R}$. A reader can equivalently think of these cases as particular instances of the physical space where it has been imposed that all particles are constrained to move either on the plane $z = 0$ or on the line $y = z = 0$.

Consider now $p \in \mathbb{R}^3$. This simple statement has a twofold meaning:

- from the viewpoint of geometry, see Figure 2, this entails that we are selecting a point, which in the figure is depicted with the blue dot.
- from the viewpoint of linear algebra p identifies an element of the vector space $\mathbb{R}^3$.

It goes without saying that these two pictures are identical, since a point p identifies unambiguously the triplet of real numbers (a_p, b_p, c_p) with $a_p, b_p, c_p \in \mathbb{R}$. In turn we can associate to p the vector

$$p \longleftrightarrow v_p = a_p \mathbf{e}_1 + b_p \mathbf{e}_2 + c_p \mathbf{e}_3.$$

In other words we are exploiting that, given a vector space and a basis therein, we can decompose uniquely each element of such a space as a linear combination of the basis elements. From this viewpoint when we shall write *Let us consider a point $p \in \mathbb{R}^3$* , we refer equivalently to

1. we consider an element of $\mathbb{R}^3$ seen as $v_p = a_p \mathbf{e}_1 + b_p \mathbf{e}_2 + c_p \mathbf{e}_3$,
2. we consider the triple (a_p, b_p, c_p) with $a_p, b_p, c_p \in \mathbb{R}$. These numbers are also referred to as the **coordinates** of p . If the basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ is orthonormal, we say that a_p, b_p, c_p are the Cartesian coordinates of p and, often in the literature they are denoted by x_p, y_p, z_p .

Cartesian Coordinate Systems

Coordinates play a distinguished role in mechanics and more generally in physics, yet it is somehow hard to find in the literature a formal definition. Consequently this is often a blurred concept and this leads at a later stage to misconceptions. Therefore we try to avoid this deficiency, but the reader should be aware that we are not considering the most general scenario. More precisely we consider

- $\mathbb{R}^n$, $n \geq 1$, seen as an abstract vector space.

We call **coordinate system** on $\mathbb{R}^n$ a collection of n maps $x_i \in C^0(\mathbb{R}^n; \mathbb{R})$ such that

$$X : \mathbb{R}^n \rightarrow \underbrace{\mathbb{R} \times \cdots \times \mathbb{R}}_n, \quad \mathbb{R}^n \ni p \mapsto X(p) = (x_1(p), \dots, x_n(p)), \quad (9)$$

is injective. We say that a coordinate system is *Cartesian* with respect to an orthonormal basis $\mathbf{e}_i$, $i = 1, \dots, n$ if the maps in Equation (9) are such that for all $p \in \mathbb{R}^n$

$$p = \sum_{i=1}^n x_i(p) \mathbf{e}_i. \quad (10)$$

Observe that, since $\mathbf{e}_i \in \mathbb{R}^n$ we can compute its coordinates and it holds that

$$X(\mathbf{e}_i) = (0, 0, \dots, 1, \dots, 0),$$

where 1 appears at the i -th entry.

Our discussion is still unsatisfactory since Equation (9) is too abstract unless we work with a Cartesian coordinates system. A way to avoid this quandary is via the concept of *coordinate transformation*. Let us proceed step by step:

1. Consider once more $\mathbb{R}^n$, $n \geq 1$ and two Cartesian coordinates systems, X and X' , subordinated respectively to the orthonormal basis $\mathbf{e}_i$ and $\mathbf{e}'_j$, $i, j = 1, \dots, n$. Using linear algebra we can infer that there must exist an orthogonal matrix $O \in O(n)$, where

$$O(n) \doteq \{A \in M(n, \mathbb{R}) \mid A^\top = A^{-1}\},$$

denotes the collection of all orthogonal matrices, such that

$$\mathbf{e}'_j = \sum_{i=1}^n O_{ji} \mathbf{e}_i.$$

Hence, for all $p \in \mathbb{R}^n$,

$$p = \sum_{j=1}^n x'_j(p) \mathbf{e}'_j = \sum_{j=1}^n x'_j(p) \sum_{i=1}^n O_{ji} \mathbf{e}_i = \sum_{i=1}^n x_i(p) \mathbf{e}_i.$$

By linear independence this entails that

$$x_i(p) = \sum_{j=1}^n x'_j(p) O_{ji} \implies x_i = \sum_{j=1}^n x'_j O_{ji} \implies X = X'O, \quad (11)$$

where, in the last equality, we are interpreting both X and X' as maps yielding a row of n real numbers when applied to a point $p \in \mathbb{R}^n$. Hence $X'O$ is the product between a row vector and a matrix. One can therefore generalize this result by saying that all Cartesian coordinate systems are related by a linear transformation codified by an orthogonal matrix.

2. Under the same assumptions of the previous item, one can decide to give up on the request that $\mathbf{e}_i$ and $\mathbf{e}'_j$ are orthonormal, keeping only the requirement that they are bases so that for all $p \in \mathbb{R}^n$, we can still conclude that

$$p = \sum_{i=1}^n x_i(p) \mathbf{e}_i = \sum_{j=1}^n x'_j(p) \mathbf{e}'_j,$$

where $x_i(p), x'_j(p)$ are real numbers for all $i, j = 1, \dots, n$. By direct inspection we can conclude that we have individuated two coordinate systems X, X' as per Equation (9):

$$X^{(\iota)} : \mathbb{R}^n \rightarrow \underbrace{\mathbb{R} \times \dots \times \mathbb{R}}_n, \quad p \mapsto X^{(\iota)}(p) = (x_1^{(\iota)}(p), \dots, x_n^{(\iota)}(p)).$$

Still using standard results from linear algebra we can say that, introducing the collection of all invertible matrices

$$\text{GL}(n, \mathbb{R}) \doteq \{A \in M(n, \mathbb{R}) \mid \det A \neq 0\},$$

we can infer that there must exist $A \in \text{GL}(n, \mathbb{R})$ such that

$$\mathbf{e}'_j = \sum_{i=1}^n A_{ji} \mathbf{e}_i.$$

Following slavishly the previous scenario, we can infer that this induces the following transformation between coordinate systems:

$$x_i(p) = \sum_{j=1}^n x'_j(p) A_{ji} \implies x_i = \sum_{j=1}^n x'_j A_{ji} \implies X = X' A. \quad (12)$$

In particular if at least one among $\mathbf{e}_i$ and $\mathbf{e}'_j$, $i, j = 1, \dots, n$ is an orthonormal basis, we say that the matrix A individuates a *linear coordinate transformation*.

Example 3. Let us discuss a simple example of the previous analysis and, to this end, let us fix $n = 2$. Let us choose an orthonormal basis of $\mathbb{R}^2$, $\mathbf{e}_1, \mathbf{e}_2$ and let us introduce the associated coordinate system as

$$X : \mathbb{R}^2 \rightarrow \mathbb{R} \times \mathbb{R}, \quad p \mapsto X(p) = (x_1(p), x_2(p)) \equiv (x(p), y(p)),$$

where we have decided to use the familiar x,y symbols for the Cartesian coordinates. Suppose now that we choose the matrix $A \in \text{GL}(2, \mathbb{R})$ such that

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

This individuates a new basis of $\mathbb{R}^2$, namely

$$\mathbf{e}'_1 = 2\mathbf{e}_1 + \mathbf{e}_2, \quad \mathbf{e}'_2 = \mathbf{e}_1 + \mathbf{e}_2.$$

Denoting by (u, v) the coordinates associated to the new basis in place of x'_1, x'_2 , we can conclude that this coordinate system is related to (x, y) by the linear transformation as per Equation (12):

$$\begin{cases} x = 2u + v \\ y = u + v \end{cases} \implies \begin{cases} u = x - y \\ v = 2y - x \end{cases}$$

From a geometric viewpoint this operation is tantamount to choosing two different reference frames in $\mathbb{R}^2$ as depicted in Figure 3.

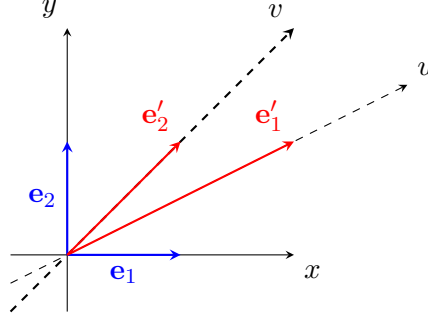


Figure 2: Graphical representation of the coordinate transformation as per Example 3.

Exercise 4. Let $n = 2$ and consider Cartesian coordinates (x, y) subordinated to an orthonormal basis $\mathbf{e}_1, \mathbf{e}_2$. Prove that the linear transformations

$$\begin{cases} x' = ax + by \\ y' = cx + dy \end{cases},$$

where $a, b, c, d \in \mathbb{R}$ individuate the most general linear coordinate transformation provided that $ad - bc \neq 0$.

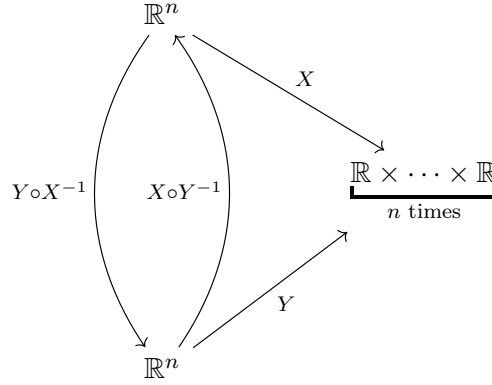
Our discussion on the role of coordinate systems is far from over. As a matter of fact, we have discussed only Cartesian coordinate systems and all those which are related to them by a linear transformation. Yet, already in the previous section, we have worked with polar coordinates (r, θ, φ) and it is not hard to see that they do not fit in our description. We need a more general concept which will be our last step.

3. Let us consider $\mathbb{R}^n$ and let us assume that X is a Cartesian coordinate system subordinated to an orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_n$. Loosely speaking the role of coordinates is to individuate unambiguously every point of $\mathbb{R}^n$ or of a subset thereof in terms of an n -tuple of numbers. In other words take any other injective map

$$Y \equiv (y_1, \dots, y_n) : \mathbb{R}^n \rightarrow \underbrace{\mathbb{R} \times \dots \times \mathbb{R}}_n,$$

where $y_i \in C^0(\mathbb{R}^n; \mathbb{R})$. We say that this is an *admissible coordinate system* provided that the *transition maps* $X \circ Y^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $Y \circ X^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ exist. In other words the following diagram is commutative (you can follow the arrows in any direction and the

result is the same):



Observe that, contrary to the previous points, there is no condition on $Y \circ X^{-1}$ or on $X \circ Y^{-1}$ to be implemented by matrices. In other words the coordinate transformation from X to Y and viceversa can be nonlinear.

For all practical purposes, this last item tells us that, in order to assign a new coordinate system, it is most convenient to define it in terms of how it transforms with respect to a Cartesian coordinate system. To convince you let us have a look at an example

Example 5. Let us discuss a simple example of the previous analysis and, to this end, let us fix $n = 2$. Let us choose an orthonormal basis of $\mathbb{R}^2$, $\mathbf{e}_1, \mathbf{e}_2$ and let us introduce the associated coordinate system as

$$X : \mathbb{R}^2 \rightarrow \mathbb{R} \times \mathbb{R}, \quad p \mapsto X(p) = (x_1(p), x_2(p)) \equiv (x(p), y(p)),$$

where we have decided to use the familiar x, y symbols for the Cartesian coordinates. We define now a new coordinate system

$$Y : \mathbb{R}^2 \rightarrow \mathbb{R} \times \mathbb{R} \mapsto Y(p) = (u(p), v(p)),$$

such that the map $X \circ Y^{-1}$ reads

$$\begin{cases} x = u \\ y = v - u^3 \end{cases}.$$

We can now construct by direct inspection the map $Y \circ X^{-1}$ since

$$\begin{cases} u = x \\ v = y + x^3 \end{cases}.$$

Disclaimer: In the previous discussion we have never required any special analytic property to the maps that we are considering. Yet, if we focus our attention to Cartesian coordinates, such as in Equation (11), we can infer that the map X is smooth, that is $X \in C^\infty(\mathbb{R}^n; \mathbb{R} \times \cdots \times \mathbb{R})$. Henceforth we shall always limit our attention to

coordinate systems Y such that the transformation maps $X \circ Y^{-1}$ and $Y \circ X^{-1}$ are smooth themselves, that is

$$X \circ Y^{-1}, Y \circ X^{-1} \in C^\infty(\mathbb{R}^n; \mathbb{R} \times \cdots \times \mathbb{R}).$$

Exercise 6. Prove that, if a coordinate transformation, *e.g.* $X \circ Y^{-1}$, is smooth and its Jacobian is invertible, then its inverse is smooth.

In many instances in classical mechanics, the geometry of the underlying physical system suggests to consider special classes of coordinates which are not everywhere defined on $\mathbb{R}^n$, in the sense that either the map $X \circ Y^{-1}$ or $Y \circ X^{-1}$ do not exist globally but only on suitable open subsets of $\mathbb{R}^n$. As mentioned above, this is not an issue provided one is aware of the limitations on the regions where these coordinates can be used. We discuss now three notable examples:

Example 7 (Spherical Coordinates). Consider on $\mathbb{R}^3$ the standard Cartesian coordinates which we denote as $X \equiv x, y, z$. We define a new set of coordinates, called *polar*, $r, \theta, \varphi : \mathbb{R}^3 \rightarrow \mathbb{R}$ and we define them by means of their transition maps. In terms of the nomenclature introduced above, if we denote by Y the new coordinate system, we have that $X \circ Y^{-1}$ is codified by Equation (4) which we recall here for the reader's convenience:

$$\begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta \end{cases} . \quad (13)$$

These are smooth maps since they are built out of polynomials and of trigonometric functions. Yet, as computed in the previous section the determinant of the Jacobian associated to this transformation is

$$\det(J) = r^2 \sin \theta,$$

which vanishes whenever $r = 0$ or $\theta = k\pi$, $k \in \mathbb{Z}$. In other words Equation (4) does not identify a good coordinate system on the whole $\mathbb{R}^3$ since it is not well-defined at the origin $(x, y, z) = (0, 0, 0)$ corresponding to $r = 0$ or on the z -axis $(0, 0, z)$, $z \in \mathbb{R}$, corresponding to $\theta = k\pi$. This is manifest inverting Equation (4) which is tantamount to constructing the map $Y \circ X^{-1}$ which reads

$$\begin{cases} r = \sqrt{x^2 + y^2 + z^2}, \\ \theta = \arccos \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right), \\ \phi = \arctan 2(y, x). \end{cases} , \quad (14)$$

where $\arctan 2$ is the 2-argument arctangent function defined as

$$\arctan 2(y, x) = \begin{cases} \arctan\left(\frac{y}{x}\right) & \text{if } x > 0 \\ \arctan\left(\frac{y}{x}\right) + \pi & \text{if } x < 0 \text{ and } y \geq 0 \\ \arctan\left(\frac{y}{x}\right) - \pi & \text{if } x < 0 \text{ and } y < 0 \\ +\frac{\pi}{2} & \text{if } x = 0 \text{ and } y > 0 \\ -\frac{\pi}{2} & \text{if } x = 0 \text{ and } y < 0 \\ \text{undefined} & \text{if } x = 0 \text{ and } y = 0 \end{cases}. \quad (15)$$

On the one hand observe that, from Equation (14), we can infer that the coordinate system $Y = (r, \theta, \varphi)$ is injective but not surjective since

$$r \geq 0, \quad \theta \in (0, \pi), \quad \varphi \in (0, 2\pi).$$

In addition observe that at the origin $(x, y, z) = (0, 0, 0)$ already the equation for $r = r(x, y, z)$ is not differentiable. At the same time, on the whole z -axis, namely the point $(0, 0, z)$, it is the function $\arctan 2(y, x)$ which is not smooth, confirming that spherical coordinates are well defined on $\mathbb{R}^3 \setminus \{(0, 0, z)\}, z \in \mathbb{R}$.

Spherical coordinates seem very natural to be used in the study of the dynamics of point particles subject to central forces as discussed in the previous section. Yet, in other circumstances, one might have to cope with directional forces, that is the underlying vector is pointing along a specific Cartesian direction. In this case, spherical coordinates do not seem to be the natural candidates and, in the following example, we outline a more natural choice.

Example 8 (Cylindrical Coordinates). Consider on $\mathbb{R}^3$ the standard Cartesian coordinates which we denote as $X \equiv x, y, z$. We define a new set of coordinates, called *cylindrical*, $r, \theta, z : \mathbb{R}^3 \rightarrow \mathbb{R}$ and we define them by means of their transition maps. In terms of the nomenclature introduce above, if we denote by Y the new coordinate system, we have that $X \circ Y^{-1}$ is codified by

$$\begin{cases} x = r \sin \theta \\ y = r \cos \theta \\ z = z \end{cases}. \quad (16)$$

These are smooth maps since they are built out of polynomials and of trigonometric functions. Yet the Jacobian associated to this transformation is

$$J = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial z} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial z} \end{bmatrix} = \begin{bmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

whose determinant is

$$\det(J) = r,$$

which vanishes at $r = 0$. In other words Equation (16) does not identify a good coordinate system on the whole $\mathbb{R}^3$ since it is not well-defined at the origin $(x, y, z) = (0, 0, z)$, $z \in \mathbb{R}$, corresponding to $r = 0$. This is manifest inverting Equation (4) which is tantamount to constructing the map $Y \circ X^{-1}$ which reads

$$\begin{cases} r = \sqrt{x^2 + y^2} \\ \phi = \arctan 2(y, x) \\ z = z \end{cases} \quad (17)$$

where the function $\arctan 2$ is defined as in Equation (15). On the one hand observe that, from Equation (17), we can infer that the coordinate system $Y = (r, \theta, z)$ is injective but not surjective since

$$r \geq 0, \quad \theta \in (0, 2\pi).$$

In addition observe that the equation for $r = r(x, y, z)$ is not differentiable on the whole z -axis. This confirms that cylindrical coordinates are well defined on $\mathbb{R}^3 \setminus \{(0, 0, z)\}$, $z \in \mathbb{R}$.

As a last example, we think of the motion of planets in the solar system. The reader should already be aware that, in agreement with the laws of Newtonian gravity, their orbits are elliptic and each orbit is planar. In this respect, it is clear that Cartesian coordinates do not appear to be the natural choice to account for this distinguished geometric feature. Here we will make implicit use of the information that the motion occurs on a plane.

Example 9 (Elliptic Coordinates). Consider on $\mathbb{R}^2$ the standard Cartesian coordinates which we denote as $X \equiv x, y$. We define a new set of coordinates, called *elliptic*, $\mu, \nu : \mathbb{R}^2 \rightarrow \mathbb{R}$ and we define them by means of their transition maps. In terms of the nomenclature introduced above, if we denote by Y the new coordinate system, we have that $X \circ Y^{-1}$ is codified by

$$\begin{cases} x = c \cosh \mu \cos \nu \\ y = c \sinh \mu \sin \nu \end{cases} \quad (18)$$

where $c > 0$ is an arbitrary but fixed, positive number. These are smooth maps since they are built out trigonometric and hyperbolic functions. Yet the Jacobian associated to this transformation is

$$J = \begin{bmatrix} \frac{\partial x}{\partial \mu} & \frac{\partial x}{\partial \nu} \\ \frac{\partial y}{\partial \mu} & \frac{\partial y}{\partial \nu} \end{bmatrix} = \begin{bmatrix} c \sinh \mu \cos \nu & -c \cosh \mu \sin \nu \\ c \cosh \mu \sin \nu & c \sinh \mu \cos \nu \end{bmatrix}$$

whose determinant is

$$\det(J) = c^2(\sinh^2 \mu + \sin^2 \nu),$$

which vanishes if and only if $\mu = 0$ while $\nu = k\pi$, $k \in \mathbb{Z}$. In other words Equation (18) does not identify a good coordinate system on the whole $\mathbb{R}^2$ since it is not well-defined at the origin $(x, y) = (\pm c, 0)$. Before trying to interpret this conclusion, let us invert Equation (18) which is tantamount to constructing the map $Y \circ X^{-1}$ which reads

$$\begin{cases} \mu = \operatorname{arccosh} \left(\frac{r_1 + r_2}{2c} \right) \\ \nu = \arccos \left(\frac{r_1 - r_2}{2c \sinh \mu} \right) \end{cases} \quad (19)$$

where

$$r_1 = \sqrt{(x-c)^2 + y^2}, \quad r_2 = \sqrt{(x+c)^2 + y^2}.$$

On the one hand observe that, from Equation (18), we can infer that the coordinate system $Y = (\mu, \nu)$ is injective but not surjective setting $\mu \in (0, \infty)$ but $\nu \in (0, \pi)$. In addition observe that at $(x, y) = (\pm c, 0)$, Equation (19) is not differentiable. This confirms that cylindrical coordinates are well defined on $\mathbb{R}^2 \setminus \{(-c, 0) \cup (c, 0)\}$. In order to grasp the role of c , observe that Equation (18) entails that

$$\frac{x^2}{c^2 \cosh^2 \mu} + \frac{y^2}{c^2 \sinh^2 \mu} = 1,$$

which is tantamount to saying that, for any but fixed value of $\mu > 0$, we obtain the equation of an ellipse whose focal distance (distance between the two foci) is c .

In addition to a reference system, since we are interested in discussing the dynamics of a physical system, we introduce

- a **time variable (coordinate)** $t \in \mathbb{R}$.

Also in this instance and in agreement with Newtonian mechanics, t is not uniquely defined, but we have the freedom of an affine transformation, much in the spirit of Definition 2. In other words we can equivalently consider any coordinate $t' = at + b$ with $a \neq 0$, while $b \in \mathbb{R}$. Yet we shall never exploit this possibility and we shall content ourselves with always denoting by t any but fixed choice of a time coordinate.

Having established the underlying analytic building blocks, in the physical space we consider for simplicity a set of n point particles, $n \geq 1$ and we are interested in studying their motion. To this end we fix the following

Nomenclature 1. *Given n point particles, $n \geq 1$ in the physical space $\mathbb{R}^3$, we call **configuration** the assignment of the position of each particle as a function of the time variable t , that is a collection of $3n$ functions $x_i, y_i, z_i \in C^\infty(\mathbb{R})$, $i = 1, \dots, n$ where $(x_i(t), y_i(t), z_i(t))$ are the coordinates of the i -th particle at time $t \in \mathbb{R}$. Despite the use of the symbols x, y, z we are not obliged to consider only Cartesian coordinates, but, for each i we can choose any admissible coordinate system.*

2.1 Constraints

The discussion of the previous section has highlighted two difficulties connected to Newtonian mechanics when dealing with specific physical systems which might be subject to constraints. We can summarize them heuristically as follows:

1. At a kinematic level, in absence of constraints, the motion of each point particle is best described in terms of global coordinates, the most natural being the Cartesian ones. Yet,

the presence of specific constraints might confine the motion to regions of $\mathbb{R}^3$, *e.g.*, a sphere or a cylinder, which is not well-suited to be described in terms of global coordinates. Yet, coordinates adapted to the underlying geometry might not be globally well defined, see Examples 7, 8 and 9.

2. At a dynamical level, the presence of a constraint complicates considerably the analysis of the motion even of point particles since one has to account in the Newton's equation the constraint force, see Equation (1).

In this section we start addressing the first problem introducing the first bits of what is commonly known as *Lagrangian formalism*. We shall refrain from working in the most general setting, in particular with reference to the regularity of all the functions that we shall consider. The reader is warned that the applicability of what follows is much wider and we content ourselves to encompass all scenarios which are of interest in mathematical and theoretical physics. As a starting point let us fix the **kinematic data** of the models that we shall analyze

- The *configuration space* is $\mathbb{R}^n$ endowed with a Cartesian reference system as per Equation (10), denoted by $(x_1, \dots, x_n)$ except if $n = 2, 3$ when we use the nomenclature (x, y) and (x, y, z) respectively,
- a collection of $N \geq 1$ point particles of mass m_i , $i = 1, \dots, N$.

A configuration of the system is the identification of the position of each particle either at a fixed instant of time $\bar{t}$ (instantaneous configuration) or for all $t \in \mathbb{R}$ and it will be denoted by

$$\mathbf{X}(t) = (\mathbf{x}_1(t), \dots, \mathbf{x}_N(t)),$$

where $\mathbf{x}_i(t) = (x_{1i}(t), \dots, x_{ni}(t))$ codifies the position/configuration of the i -th particle. In the language of calculus we say that

$$\mathbf{X} \in C^\infty(\mathbb{R}; \mathbb{R}^{nN}),$$

that is, for every instant of time $\mathbb{R} \ni t$, $\mathbf{X}(t) \in \mathbb{R}^{nN}$ is an admissible configuration of the system. In a more concise language we say that $C^\infty(\mathbb{R}; \mathbb{R}^{nN})$ is the space of **kinematically allowed configurations**, that is, a priori and without any assumption of the forces acting on the system, each point in $\mathbb{R}^{nN}$ is a potentially instantaneous configuration and every smooth curve in $\mathbb{R}^{nN}$ is a priori dynamically attainable.

Remark 3. *It is worth noticing that, in many textbooks, the word configuration of a system, especially if at a fixed instant of time $\bar{t}$, includes also the assignment of the velocity vector, namely $(\dot{x}_1(\bar{t}), \dots, \dot{x}_n(\bar{t}))$. We shall not follow this heresy and we shall refer to the assignment of both the position and velocity as the act of specifying the **state** of a system. If you think this is just nomenclature, wait one year and, we shall talk about it while you are taking the course of quantum mechanics.*

In other words we have introduced the standard arena of Newtonian mechanics, possibly with a slightly more formal language. Yet, we want to discuss what happens if one or more particles are subject to the presence of a constraint. As already discussed it is structurally different coping with a particle bouncing against a wall or moving along a sphere or on a plane. The following definitions serve to the scope of separating these completely different scenarios. For the time being we restrict the attention to a system with 1 particle, that is $N = 1$, although this is not necessary a priori and it is done only to simplify the presentation.

Definition 10. Let $\mathbb{R}^n$ be endowed with a Cartesian reference system $(x_1, \dots, x_n)$ and let $t \in \mathbb{R}$ be the time variable. We call **holonomic constraint** a collection of k equations ($1 \leq k < n$)

$$f_j(x_1, \dots, x_n, t) = 0, \quad j = 1, \dots, k \quad (20)$$

where each f_j is a real scalar function, that is $f_j \in C^\infty(\mathbb{R}^n \times \mathbb{R}; \mathbb{R})$. In addition the collection of all f_j are **functionally independent** for all $t \in \mathbb{R}$, namely the associated Jacobian matrix J whose entries are

$$J_{ij}(x_1, \dots, x_n, t) \doteq \frac{\partial f_j}{\partial x_i}(x_1, \dots, x_n, t), \quad i = 1, \dots, n, \text{ and } j = 1, \dots, k,$$

has rank k for all points $(\bar{x}_1, \dots, \bar{x}_n)$ such that $f_j(\bar{x}_1, \dots, \bar{x}_n, t) = 0$.

The reader, who is not used to mathematical statements applied to the analysis of physical system should neither be scared nor be pissed off by Definition 10. Mathematical physics is often denoted by MP which also means Military Police (Carabinieri) and the analogy is in a sense true. One of the two main goals of mathematical physics is that of protecting us from doing what it cannot be done. It sets the boundaries of a physical theory, pushing it to its limits and analyzing it in depths. Once these boundaries are established and we are aware of them, everything abiding by them is legit. For example, if you work with a physical system whose constraints fit with Definition 10, you are certain that all the results stemming from it apply. At the same time, going beyond the boundaries set by the mathematical formulation is not forbidden, it is actually necessary and vital. The second goal of mathematical physics is that of giving you the best possible tools to explore what lies beyond these boundaries and to establish new model and theories. Forgetting my runts, let us see two simple, yet instructive examples.

Example 11. Consider Definition 10 and let us set $n = 3$. If we want to bind a point particle to move on a sphere of fixed radius $R > 0$, we would impose the equation

$$f(x, y, z) = x^2 + y^2 + z^2 - R^2 = 0.$$

To establish whether this is a *holonomic constraint*, observe that $j = 1$ and that the Jacobian matrix is actually the vector

$$J_{i1}(x, y, z) = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \\ 2z \end{pmatrix}.$$

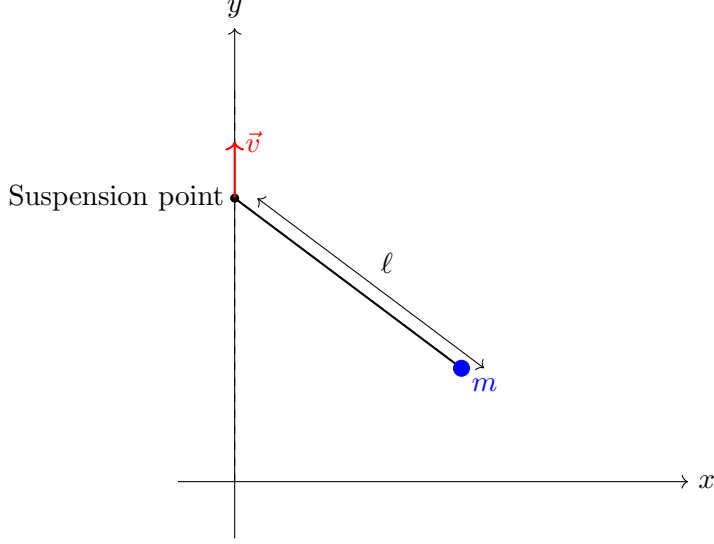


Figure 3: A point mass m constrained to move at the tip of a pendulum of length $\ell > 0$. The other end of the pendulum is constrained to move along the vertical axis with constant speed $v > 0$.

This is a rank 1 matrix for all collection of point $(\bar{x}, \bar{y}, \bar{z})$ such that $f(\bar{x}, \bar{y}, \bar{z}) = 0$. As a matter of fact $\text{rank}(J_{i1}) = 1$ unless $x = y = z = 0$, which is not a solution of the equation under scrutiny. Hence, this is a holonomic constraint, which is furthermore time-independent. This is the reason why we have omitted writing explicitly the time variable t .

Let us consider now an example where time dependence is explicit and we will start transitioning towards a language which is typical of problems in classical mechanics.

Example 12. We want to consider a point particle of mass $m > 0$ in $\mathbb{R}^2$ constrained to move at the tip of a pendulum of length $\ell > 0$ and of mass M so small that we can set it to 0. The other end of the pendulum is constrained to move along the vertical axis with constant speed $v > 0$ – See Figure 12 If we want to translate the constraint imposed by this problem on the position (x, y) of the mass m , we should consider the equation

$$f(x, y, t) = x^2 + (y - vt)^2 - \ell^2 = 0. \quad (21)$$

As in Example 11, in comparison with Definition 10, we set $j = 1$ and therefore the Jacobian matrix reduces to the following vector

$$J_{i1}(x, y, t) = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = \begin{pmatrix} 2x \\ 2(y - vt) \end{pmatrix}.$$

The rank is always 1 unless all components vanish, namely $x = 0$ and $y = vt$. Yet these points do not solve Equation 21 and, therefore, on account of Definition 10, we can conclude that we have identified an holonomic constraint.

As a last example we wish to investigate a case where we end up working with more than one equation. This will occur rarely in this course, but it is still instructive to have a look into it.

Example 13. Let us consider a point particle of mass $m > 0$ and let us suppose that we want to describe its motion constraining it to move on the equatorial plane of a sphere whose radius is exponentially growing with a law $R(t) = R_0 e^{Ht}$, with $H, R_0 > 0$. If we want to translate the constraint imposed by this problem on the position (x, y, z) of the mass m , we should consider the system of equations

$$\begin{cases} f_1(x, y, z, t) = x^2 + y^2 + z^2 - R^2(t) = 0 \\ f_2(x, y, z, t) = z = 0 \end{cases} \quad (22)$$

In this case the Jacobian is the following matrix

$$J(x, y, z, t) = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \end{bmatrix} = \begin{bmatrix} 2x & 0 & 2z \\ 0 & 0 & 1 \end{bmatrix}.$$

As one can infer the rank of this matrix is equal to 2 unless $x = y = 0$. In this case, regardless of the value of z the rank is 1. Yet there is no point of the form $(0, 0, z)$ which is a solution of Equation (22) since $R(t) \neq 0$ for all $t \in \mathbb{R}$. In other words, according to Definition 10, Equation (22) identifies an holonomic constraint.

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(Generalized) Holonomic Constraint

Definition 10 rests on the assumption that we are considering systems with only one particle or even systems with more than one particle, such that each of the underlying constraints refers to the position of only one of them. This means that we are forbidden to analyze with this language simple scenarios such as those involving two point particles constrained to keep their distance constant. This limitation can be bypassed with a rather straightforward generalization of the definition given.

Consider therefore $\mathbb{R}^n$ endowed with a Cartesian reference system $(x_1, \dots, x_n)$ and let $t \in \mathbb{R}$ be the time variable. Assume that we consider a system of N -point particles, $N > 1$, denoting a generic point of the underlying configuration space as $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_N)$, where $\mathbf{x}_i = (x_{1i}, \dots, x_{ni})$. We call **holonomic constraint** a collection of k equations ($1 \leq k < nN$)

$$f_j(\mathbf{X}, t) = f_j(\mathbf{x}_1, \dots, \mathbf{x}_N, t) = 0, \quad j = 1, \dots, k \quad (23)$$

where each f_j is a real scalar function, that is $f_j \in C^\infty(\mathbb{R}^{nN} \times \mathbb{R}; \mathbb{R})$. In addition the collection of all f_j are **functionally independent** for all $t \in \mathbb{R}$, namely the associated

Jacobian matrix J whose entries are

$$J_{ij}(x_{11}, x_{21}, \dots, x_{nN}, t) \doteq \frac{\partial f_j}{\partial x_i}(x_{11}, x_{21}, \dots, x_{nN}, t), \quad i = 1, \dots, nN, \text{ and } j = 1, \dots, k,$$

has rank k for all points $(\bar{x}_1, \dots, \bar{x}_n)$ such that $f_j(\bar{x}_1, \dots, \bar{x}_n, t) = 0$.

Exercise 14 (A Triangle in a Plane). Consider on $\mathbb{R}^2$ three point particles. Write down the equations constraining them to lie at the vertices of an equilateral triangle whose edges have length $\ell > 0$. Discuss whether this is an holonomic constraint.

[Hint: Observe that, in order to impose that the distance between two points is constant, it is sufficient to impose that it is constant its square.]

Remark 4. In the previous examples as well as in Definition 10 we have always considered systems where the underlying motion is constrained by means of equalities connecting the chosen coordinates. Yet, as pointed out at the beginning of the section, we might be interested in scenarios, such as a particle confined in a box, where the constraints can only be expressed in terms of inequalities. For example, considering $\mathbb{R}^2$, requiring that a point particle can move within a square of length 1 can be realized by imposing that, once the standard Cartesian coordinates (x, y) are chosen,

$$0 \leq x \leq 1 \quad \text{and} \quad 0 \leq y \leq 1.$$

In other words the constraints are expressed in terms of inequalities. In this instance we say that the constraint is **unilateral**. In all other cases it is called **bilateral**. Observe that holonomic constraint fall per definition in this class.

In the traditional literature of classical mechanics, given a holonomic constraint $f_j(x_1, \dots, x_n, t)$, $j = 1, \dots, k < n$ it is often convenient (not really, but you know... traditions are hard to die) to separate two cases:

1. an holonomic constraint is called **scleronomic** if it is *time-independent*, that is $\frac{\partial f_j}{\partial t} = 0$ for all $j = 1, \dots, k$. This is always written as $f_j(x_1, \dots, x_n) = 0$,
2. an holonomic constraint is called **rethonomic** if it is manifestly time-dependent, that is, there exists $j \in [1, k]$, $\bar{t} \in \mathbb{R}$ and $(\bar{x}_1, \dots, \bar{x}_n) \in \mathbb{R}^n$ such that $\frac{\partial f_j}{\partial t}(\bar{x}_1, \dots, \bar{x}_n, \bar{t}) \neq 0$.

Anholonomic Constraints

Definition 10 characterizes a class of constraints which set limitations to the kinematic configurations of a mechanical system. In some instance we are interested in reducing a priori the number of allowed dynamical configurations. One way to achieve this goal is by means of what are known as **anholonomic (non-holonomic) constraints**. We discuss their main realization for a single particle on $\mathbb{R}^n$ endowed with Cartesian coordinates

$(x_1, \dots, x_n)$ while $t \in \mathbb{R}$ is the time variable. Denoting by $\dot{x}_i(t)$, $i = 1, \dots, n$ the i -th component of the velocity vector of the trajectory of the particle, an anholonomic constraint is any algebraic equation of the form

$$F(x_1, \dots, x_n, \dot{x}_1, \dots, \dot{x}_n, t) = 0, \quad (24)$$

which depends explicitly at least on one of the velocity vector components and which cannot be integrated to an holonomic constraint.

The notion of anholonomic constraint is objectively very obscure. On the one hand this is natural since it is intrinsically defined as a complementary set. At the end of the day, a constraint which is not holonomic, but it is bilateral as per Remark 4, is anholonomic, although in Equation (24) we have adopted the pragmatic viewpoint that anholonomic constraints must involve explicitly the velocity components of a point particle. On the other hand, there is also the requirement that the constraint cannot be integrated to an holonomic one. This is a very reasonable request as one might infer thinking about this example. Take a point particle on $\mathbb{R}^2$ endowed with Cartesian coordinates (x, y) and assume that we impose the constraint $\dot{x} = 0$. This is tantamount to saying that $x(t) = x_0$, where x_0 is the initial position of the particle computed at any fixed time, say $t = 0$. This can also be realized by the holonomic constraint

$$f(x, y) = x - x_0 = 0.$$

In other words $\dot{x} = 0$ is not anholonomic. Let us now discuss a more subtle example. When I was a student this always represented a good source of headache and I see no reason not to share with you my memories. Actually we used to make fun of it since we thought it was unrealistic, but we were just shortsighted. We will be discussing a constraint which occurs for example if you end up with your car in muddy terrain. You try to move, but the wheels start rolling without the car moving any forward. This is the **no-slip constraint**.

Example 15 (No-Slip 1D – Holonomic). Consider an homogeneous disk of mass $M > 0$ in $\mathbb{R}^2$ which is free to rotate on the plane as per Figure 15. Although we have not yet discussed the motion of rigid bodies, from Newtonian mechanics, you can infer that, in absence of external forces, the motion of the disk can be split in that of C , the center of mass, and in the rotation component. The *1D no-slip* constraint aims at imposing that, at each instant of time $t \in \mathbb{R}$, the velocity of the contact point P between the disk and the horizontal x -axis vanishes. In addition the disk can move only horizontally. In other words, introducing the Cartesian coordinates (x, y) for the center of mass and the angular one θ , this is tantamount to imposing that

$$\dot{x} - R\dot{\theta} = 0 \quad \text{and} \quad y - R = 0,$$

where $R > 0$ is the radius of the disk. At first glance this is of the form of Equation (24), but, being R constant and assuming the initial value of the x - and of the θ -coordinates known, say (x_0, θ_0) , this can be integrated as

$$x(t) - R\theta(t) = x_0 - R\theta_0,$$

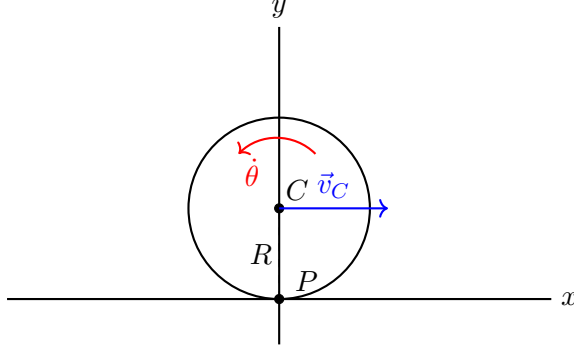


Figure 4: No-slip condition: $\vec{v}_P = \vec{v}_C + \dot{\theta}\mathbf{e}_z \times \vec{r} = 0$

which can be associated to the holonomic constraint

$$x - x_0 - R(\theta - \theta_0) = 0.$$

Example 16 (No-Slip 2D – Holonomic). Consider the same homogeneous disk of mass $M > 0$ in $\mathbb{R}^2$ as in Example 15. The 2D *no-slip* constraint aims at imposing that, at each instant of time $t \in \mathbb{R}$, the velocity of the contact point P between the disk and the horizontal x -axis vanishes. Yet, in this case, the disk is free to move in all directions and not only horizontally. To deal with this case the correct interpretation is the following:

- the disk is actually in $\mathbb{R}^3$ but constrained to move on the $z = 0$ plane,
- the position of the center of mass of the disk in Cartesian coordinates is $(x, y, 0)$, while, denoting by θ the angular coordinate encoding the rotation in the $(x-y)$ -plane, the angular velocity vector is $\vec{\omega} = \dot{\theta}\mathbf{e}_z$, where $\mathbf{e}_z = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

The no-slip condition entails that denoting by $\vec{v}_C = (\dot{x}, \dot{y}, 0)$ the velocity vector of the center of mass, it holds

$$\vec{v}_C + \vec{\omega} \times \vec{r},$$

where $\vec{r}$ here denotes the position of the contact point with respect to the center of mass. In other words

$$\vec{r} = (-R \sin \theta, R \cos \theta, 0).$$

This entails that the 2D no-slip constraint reads

$$\dot{x} - R\dot{\theta}\cos\theta = 0 \quad \text{and} \quad \dot{y} + R\dot{\theta}\sin\theta = 0. \quad (25)$$

Contrary to the 1D scenario, these two equations cannot be integrated out and therefore the 2D no-slip constraint is anholonomic.

From the previous discussion the reader should keep in mind that, in classical mechanics, holonomic constraints play a distinguished role. Yet, the question is why. On the one hand, requiring that there exists an algebraic equation written in terms of the underlying Cartesian coordinates is a reasonable hypothesis which includes many of the cases that we have in mind, save for those special instances where one might need unilateral or anholonomic constraints. Yet, a closer inspection of Definition 10 unveils that we have made the additional request of the Jacobian being of constant rank. This is a vital condition and it is at the heart of the whole Lagrangian approach to classical mechanics. In particular this hypothesis allows us to apply a key result from calculus, known as the **implicit function theorem**.

Implicit Function Theorem – smooth case (Advanced Topic)

Let $f \in C^\infty(\mathbb{R}^{n+m}; \mathbb{R}^m)$ where $m > 0$ while $n \geq 0$. Assume that there exists a point $(\bar{x}_1, \dots, \bar{x}_n, \bar{y}_1, \dots, \bar{y}_m)$ such that $f(\bar{x}_1, \dots, \bar{x}_n, \bar{y}_1, \dots, \bar{y}_m) = 0$ and assume also that denoting by f_j , $j = 1, \dots, m$ the components of f , the Jacobian matrix

$$J_f[\bar{x}_1, \dots, \bar{x}_n, \bar{y}_1, \dots, \bar{y}_m] = \frac{\partial f_j}{\partial y_i}(\bar{x}_1, \dots, \bar{x}_n, \bar{y}_1, \dots, \bar{y}_m),$$

is invertible. Then there exists an open neighbourhood $U \subseteq \mathbb{R}^n$ containing $(\bar{x}_1, \dots, \bar{x}_n)$ as well as $g \in C^\infty(U; \mathbb{R}^m)$ such that

$$f(x, g(x)) = 0, \quad \forall x \in U.$$

Exercise 17 (Only for Math-Oriented Readers (Challenge)). Assume to consider an holonomic constraint as per Definition 10 or, more generally, as per Equation (20). Show that it is possible to apply the Implicit Function Theorem to the collection of functions identifying the constraint.

In order to understand the relevance of the implicit function theorem combined with holonomic constraints, let us start from a simple showcase. Let us consider a point particle of mass $m > 0$ in $\mathbb{R}^3$ and let us assume that its kinematic configurations are subordinate to the holonomic constraint

$$f(x, y, z) = x^2 + y^2 + z^2 - R^2 = 0,$$

where (x, y, z) are Cartesian coordinates, while $R > 0$. This is an holonomic constraint as discussed in Example 11. Following Exercise 17 we can apply the Inverse Function Theorem. To see it explicitly, we can read the map as $f : \mathbb{R}^{2+1} \rightarrow \mathbb{R}$. In particular, let us consider z as the variable which we want to replace using the mentioned theorem and we observe that

$$J[f] = \frac{\partial f}{\partial z} = 2z,$$

which is always invertible unless $z = 0$. In particular we can choose the point $P = (0, 0, R)$ and we can infer that there exists a neighbourhood U of $(0, 0)$ as well as a function $g \in C^\infty(U; \mathbb{R})$

such that

$$f(x, y, g(x, y)) = 0, \quad \forall (x, y) \in U.$$

Yet, in this case, one can construct explicitly g from the constraint observing that

$$z = \pm \sqrt{R^2 - x^2 - y^2}.$$

Since we consider the point $(R, 0, 0)$, this means that $z(P) > 0$ and hence we discard the minus sign. In addition, since the Jacobian is always invertible unless $z = 0$, we can conclude that $U \equiv \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 < R^2\}$

$$f(x, y, \sqrt{R^2 - x^2 - y^2}) = 0, \quad \forall (x, y) \in U.$$

Heuristically speaking we are covering almost half of the sphere of radius R . From a physical viewpoint the consequences of combining the holonomic constraint with the implicit function theorem are far reaching. As a matter of fact we can now draw two immediate conclusions:

- In order to fully parameterize the collection of all kinematic configurations, only two variables/coordinates are necessary that is $3 - 1$, the number without any constraint to which we subtract the number of all holonomic constraints,
- The Cartesian coordinates (x, y) can be chosen as the independent coordinates to study the dynamics of the problem.

A reader might have the feeling that this was a long and tedious discussion, using ultra technowords to draw a rather intuitive conclusion which requires very basic math and a little bit of common sense. Yes, this is indeed the case and the reason lies in the fact that a system with one single particle and a very simple constraint, at least geometrically, is easy. This is the problem of introductory examples, since they might let you think that we are complicating easy problems. Well sometimes, but in many cases not, especially when the physical systems start to become complicated. As a matter of fact, let us look once more at the simple example above and let us make the following comment:

- The position of a particle in $\mathbb{R}^3$ constrained to move on a sphere of radius $R > 0$ centered at the origin is fully described by two independent coordinates which can be chosen as being Cartesian.

This is true, but nothing forces us to use Cartesian coordinates and actually the construction above seems to suggest that it might not be the unique or even the smartest choice. As a matter of facts following Example 7, we can switch from Cartesian to polar coordinates, that is

$$\begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta \end{cases} \quad (26)$$

In this respect the position of the point particle is fully codified by (r, θ, φ) rather than by (x, y, z) . In addition the holonomic constraint becomes very simple, namely $r = R$. This is tantamount to saying that the admissible kinematic configurations are (R, θ, φ) with no limitation on (θ, φ) barring their domain of definition.

The outcome of the previous digression can be generalized as follows: Suppose that

- on $\mathbb{R}^n$ we consider N point particles of mass $m_i > 0$, $i = 1, \dots, N$. Denoting by $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_N)$, the position of all N -particles where $\mathbf{x}_i = (x_{1i}, \dots, x_{ni})$,
- we consider k -holonomic constraints $f_j(\mathbf{x}_1, \dots, \mathbf{x}_N) = 0$ with $1 \leq k < N$.

Then we can conclude that, on account of the implicit function theorem, the collection of all *kinematically allowed configurations* has $nN - k$ **degrees of freedom**, namely one can introduce coordinates q_i , $i = 1, \dots, nN - k$, each with its own specific domain of definition, such that, for every instant of time $t \in \mathbb{R}$, the position of all particles abiding by the holonomic constraints can be written as

$$x_{ji} \equiv x_{ji}(q_1, \dots, q_{nN-k}, t), \quad i = 1, \dots, n \quad \text{and} \quad j = 1, \dots, N.$$

The collection $(q_1, \dots, q_{nN-k})$ are known as **Lagrangian (or generalized) coordinates**. Observe that, in addition of being non unique, there is no algorithmic way to construct them or to select a preferred choice. This shall be determined case by case depending on the underlying problem.

3 Lagrangian Mechanics

No diagrams, no figures – only equations.

Mécanique analytique by Lagrange

We are now in a position to use all the tools and results introduced in Section 2.1 to discuss the dynamics of one or more point particles in presence of constraints. Once and for all we recall the building blocks that will be assumed to be assigned in this section:

1. An ambient space $\mathbb{R}^n$ endowed with a reference frame, namely an origin O and an orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_n$ with associated **Cartesian coordinates** $(x_1, \dots, x_n)$,
2. A collection of $\mathbf{N} \geq \mathbf{1}$ **point particles** free to move on $\mathbb{R}^n$, $n \geq 1$, each of mass $m_i > 0$, $i = 1, \dots, N$. The position of each particle is denoted by $\mathbf{x}_j = (x_{1j}, \dots, x_{nj})$, $j = 1, \dots, N$,
3. a **time variable** $t \in \mathbb{R}$,
4. a collection of **k holonomic constraints** $f_j(\mathbf{x}_1, \dots, \mathbf{x}_N, t)$, $j = 1, \dots, k$, as per Definition 10 and Equation (24), where $1 \leq k < nN$.

Starting from these data, we can use the implicit function theorem, see Section 2.1, to draw the following two conclusions:

- a) there exists Σ_t , a **time-dependent open subset of $\mathbb{R}^{Nn-k}$** which codifies kinematically allowed configurations,
- b) a collection $(\mathbf{q}_1, \dots, \mathbf{q}_{Nn-k})$ of **Lagrangian coordinates**, such that, at each instant of time, the position of any among the point particles under investigation can be written as

$$\mathbf{x}_j(t) \equiv \mathbf{x}_j(q_1, \dots, q_{Nn-k}, t).$$

Observe that Σ_t might not be including all possible kinematic configurations compatible with the constraints since the inverse function theorem can be applied only in a neighbourhood of a point in $\mathbb{R}^{nN}$ which abides by all the underlying constraints. As the example of a particle moving on a sphere in $\mathbb{R}^3$ clarifies, it might be necessary to consider more than one region Σ_t to account for all the points which are compatible with the constraints assigned. In the following we shall not highlight this fact unless it is strictly necessary since all the constructions and results that we shall discuss are intrinsically local. This mean that one can work separately on each single set such as Σ_t .

Remark 5. *The mathematically inclined reader might feel very very unhappy by this lack of generality, claiming that there is always a lingering sense of imprecision in what we are constructing. Adapting a quote from Otto von Bismarck who used to say that politics is the art of compromise, be aware that giving a course at the 3rd semester of a BSc in physics is the art of compromise between mathematical rigor, theoretical intuition and experimental pragmatism. Hence, in order to discuss Lagrangian mechanics in full generality, one would need the full might of differential geometry. If you are keen on learning it, you are strongly encouraged to consider attending during your MSc studies the classes of General Relativity, Dynamical Systems and Mathematical Methods for Theoretical Physics.*

Having recalled the key ingredients which are available, we need two additional structures.

1. – Virtual Displacements

Assume that, for simplicity, we set $N = 1$, that is, we consider a single point particle and let us take two admissible configurations lying in Σ_t , that is $\mathbf{x}(t) = (x_1(t), \dots, x_n(t))$ and $\mathbf{x}'(t) = (x'_1(t), \dots, x'_n(t))$, where, for each $i = 1, \dots, n$,

$$x_i(t) \equiv x_i(q_1, \dots, q_{n-k}, t) \quad \text{and} \quad x'_i(t) \equiv x_i(q'_1, \dots, q'_{n-k}, t).$$

Here x'_i does not refer to a different Cartesian coordinate system, but to the choice of a second point lying in Σ_t . This is the reason why we move the prime from the x to the Lagrangian coordinates. Assume that t is arbitrary but fixed to the same values for both points. We can

introduce the displacement of $\mathbf{x}'$ relative to $\mathbf{x}$ as the vector $\Delta\mathbf{x} = \mathbf{x}' - \mathbf{x} \in \mathbb{R}^n$, whose generic i -th component reads

$$(\Delta\mathbf{x})_i = x_i(q'_1, \dots, q'_{n-k}, t) - x_i(q_1, \dots, q_{n-k}, t). \quad (27)$$

Recalling that, still as a consequence of the inverse function theorem, the Cartesian coordinates depend smoothly on the Lagrangian ones, we can consider a first order Taylor series expansion of $\mathbf{x}'$ centered at $(q_1 \dots q_{n-k})$:

$$(\Delta\mathbf{x})_i = \sum_{j=1}^{n-k} \left. \frac{\partial x_i}{\partial q^j} \right|_{(q_1, \dots, q_{n-k})} \delta q^j + \mathcal{O}((\delta q)^2), \quad (28)$$

where $\delta q^j = q'^j - q^j$. The first term in this expansion gives an expression of the displacement of two points abiding by the holonomic constraints and at an arbitrary but fixed time t . For this reason it is convenient to introduce what in the ninetieth century was known as **virtual displacement vector**

$$\delta\mathbf{x} \doteq \sum_{j=1}^{n-k} \left. \frac{\partial \mathbf{x}}{\partial q^j} \right|_{(q_1, \dots, q_{n-k})} \delta q^j = \left(\sum_{j=1}^{n-k} \left. \frac{\partial x_1}{\partial q^j} \right|_{(q_1, \dots, q_{n-k})} \delta q^j, \dots, \sum_{j=1}^{n-k} \left. \frac{\partial x_n}{\partial q^j} \right|_{(q_1, \dots, q_{n-k})} \delta q^j \right) \in \mathbb{R}^n. \quad (29)$$

The geometric interpretation of Equation (29) is that of a vector lying in $\mathbb{R}^n$ with base point $(q_1, \dots, q_{n-k})$ and tangent to $\Sigma_t \subseteq \mathbb{R}^{n-k} \subset \mathbb{R}^n$, the open subset of kinematically allowed configurations.

Exercise 18. Assume that on $\mathbb{R}^n$ we are considering $N > 1$ point particles and k holonomic constraints with $1 \leq k < Nn$. Construct the counterpart of Equation (29) in this scenario.

One might feel that the virtual displacement vector is strongly dependent on $(q_1, \dots, q_{n-k})$ but we shall prove the converse. We consider only the case with $N = 1$, but the results can be generalized mutatis mutandis to any $N > 1$.

Lemma 19. *Consider a point particle in $\mathbb{R}^n$ and k holonomic constraints with $1 \leq k < n$. Assume that $(q_1, \dots, q_{n-k})$ and $(\tilde{q}_1, \dots, \tilde{q}_{n-k})$ are two sets of Lagrangian coordinates for Σ_t , the chosen time-dependent open subset of $\mathbb{R}^{Nn-k}$ which codifies kinematically allowed configurations. Then Equation (29) does not depend on the choice of Lagrangian coordinates, namely*

$$\begin{aligned} \delta\mathbf{x} &\doteq \left(\sum_{j=1}^{n-k} \left. \frac{\partial x_1}{\partial q^j} \right|_{(q_1, \dots, q_{n-k})} (\delta q^j), \dots, \sum_{j=1}^{n-k} \left. \frac{\partial x_n}{\partial q^j} \right|_{(q_1, \dots, q_{n-k})} (\delta q^j) \right) = \\ &= \left(\sum_{j=1}^{n-k} \left. \frac{\partial x_1}{\partial \tilde{q}^j} \right|_{(q_1, \dots, q_{n-k})} (\delta \tilde{q}^j), \dots, \sum_{j=1}^{n-k} \left. \frac{\partial x_n}{\partial \tilde{q}^j} \right|_{(q_1, \dots, q_{n-k})} (\delta \tilde{q}^j) \right), \end{aligned}$$

where $|_{(q_1, \dots, q_{n-k})}$ denotes that the derivative are in both cases evaluated at the same point of Σ_t .

Proof. Considering the coordinates $(q_1, \dots, q_{n-k})$ and $(\tilde{q}_1, \dots, \tilde{q}_{n-k})$, standard results in calculus entail that, for all $i = 1, \dots, n-k$,

$$\frac{\partial}{\partial q^j} = \sum_{i=1}^{n-k} \frac{\partial \tilde{q}^i}{\partial q^j} \frac{\partial}{\partial \tilde{q}^i}.$$

Hence, for any $s = 1, \dots, n$, it holds that

$$\sum_{j=1}^{n-k} \frac{\partial x_s}{\partial q^j} \delta q^j = \sum_{j=1}^{n-k} \sum_{i=1}^{n-k} \frac{\partial \tilde{q}^i}{\partial q^j} \frac{\partial x_s}{\partial \tilde{q}^i} \delta q^j = \sum_{i=1}^{n-k} \frac{\partial x_s}{\partial \tilde{q}^i} \delta \tilde{q}^i,$$

where, in the last identity, we have used the relation

$$\delta \tilde{q}^i = \sum_{j=1}^{n-k} \frac{\partial \tilde{q}^i}{\partial q^j} \delta q^j.$$

This concludes the proof. $\square$

2. – Ideal Constraints

In Section 1 we have outlined a simple example which has highlighted the difficulty of solving Newton's equation in presence of constraints, due to the necessity of accounting also for constraint forces. Our goal is to look at this problem from a slightly more general perspective. Let us therefore assume to consider $\mathbb{R}^n$, a reference system therein and a collection of Cartesian coordinates $(x_1, \dots, x_n)$. Let us consider $N \geq 1$ point particles of mass $m_i > 0$ and let us denote by $\mathbf{x}_i$, $i = 1, \dots, n$ their generic position vector. Assume that there are k holonomic constraints $1 \leq k < N$ and denote by $\mathbf{F}_i(\mathbf{x}_1, \dots, \mathbf{x}_n)$ the active force on the i -th particle. Observe that, it would be possible to have $\mathbf{F}_i$ depending also on the velocities of the point particles, but we avoid going through this unnecessary complication.

As already mentioned, in Newtonian mechanics, each particle is compelled to abide by the underlying constraints by means of a suitable *reactive force*, say Φ_i , $i = 1, \dots, N$. This entails that the dynamics is ruled by the differential equations

$$m_i \mathbf{a}_i = \mathbf{F}_i + \Phi_i. \quad i = 1, \dots, N \quad (30)$$

Among all possible reactive forces, we single out a specific subclass.

Definition 20 (Ideal Constraints). Consider on $\mathbb{R}^n$ a reference system with Cartesian coordinates $(x_1, \dots, x_n)$, as well as $N \geq 1$ point particles and k holonomic constraints as per Definition 10 and Equation (20) with $1 \leq k < N$. Denote by $\Sigma_t \subset \mathbb{R}^{nN}$ the collection of all kinematically allowed configurations with $t \in \mathbb{R}$. We say the constraints are **ideal** if, for every $\mathbf{x}(t) = (\mathbf{x}_1(t), \dots, \mathbf{x}_N(t)) \in \Sigma_t$ the vector of the reactive forces

$$\Phi = (\Phi_1, \dots, \Phi_N),$$

once evaluated at $\mathbf{x}$ is perpendicular to Σ_t .

Remark 6. In comparison to [Mor20, Def. 8.11], we are considering a stronger notion of ideal constraints. In this reference it is required that the property of being perpendicular to Σ_t applies only to those point $\mathbf{x}(t) \in \mathbb{R}^{nN}$ which lie on Σ_t and they are solution of Equation (30). This is certainly the best abstract criterion although one has to keep in mind that the functional form for the reactive forces Φ_i is known only for ideal constraints and such form is obtained using this assumption. Hence, a priori, solving Equation (30) is rather challenging.

Remark 7. The notion of ideal constraints is relatively modern, since, in the nineteenth century, Definition 20 was replaced by the notion of **virtual work**. Under the same assumptions still of Definition 20 and given the vector of the reactive forces

$$\Phi = (\Phi_1, \dots, \Phi_N),$$

its virtual work is

$$\delta L \doteq \Phi \cdot \delta \mathbf{x},$$

where the symbol $\cdot$ denotes the scalar product in $\mathbb{R}^{nN}$, while $\delta \mathbf{x}$ is the virtual displacement vector as per Equation (29). The condition according to which, for an ideal constraint Φ is perpendicular to Σ_t is tantamount to saying that

$$\delta L = 0,$$

since $\delta \mathbf{x}$ is per construction tangent to Σ_t . In other words a constraint is ideal if and only if its associated virtual work vanishes. This is a generalization of the **principle of virtual work** which states that, if a mechanical system is in equilibrium, then the total virtual work done by all forces vanishes, that is

$$(\mathbf{F} + \Phi) \cdot \delta \mathbf{x} = 0,$$

where $\mathbf{F} = (\mathbf{F}_1, \dots, \mathbf{F}_N)$ includes all active forces on the N -particles.

Ideal Constraints

Definition 20 might feel reasonable at first glance, but at a closer look, it unveils a weak point. More precisely we have not discussed how to model the reactive forces. Hence, we are not in a position to verify concretely whether a constraint is ideal or not. A pragmatic viewpoint is the following: *All holonomic constraints are ideal unless there is a friction force with a tangential component along the hypersurface identified by the constraints.* The reason is quite simple: suppose to work on $\mathbb{R}^n$ with Cartesian coordinates $(x_1, \dots, x_n)$ and suppose that we consider a single point particle of mass $m > 0$

- if we have an holonomic constraint $f(x_1, \dots, x_n, t) = 0$, a model for the reactive force is

$$\Phi(x_1, \dots, x_n) \doteq \lambda \nabla f,$$

where ∇ is the gradient along the Cartesian coordinates $(x_1, \dots, x_n)$, while λ is a Lagrange multiplier, *cf.* Equation (1) signifying that we now the direction of the force but not its magnitude. At each instant of time $t \in \mathbb{R}$ this vector is per construcion perpendicular to the set of point such that $f(x_1, \dots, x_n) = 0$. Hence the constraint is ideal in this respect. Observe, that, physically, Φ serves its scope since it keeps the point particle anchored to the constraint hypersurface.

- if we want to think of a physically motivated model where the holonomic constraint as in the previous point is not ideal, we need to account for a tangential component of the reactive force which would therefore increase or reduce the component of the velocity tangent to the hypersurface $f(x_1, \dots, x_n, t) = 0$, see Equation (30). Reduction of the speed is indeed in physical models the effect of a friction force.

In a few words, at the end of the day, all holonomic constraints are perfectly fine!

3.1 The Euler-Lagrange Equations

The goal of this chapter is to develop a novel formalism to account for the classical dynamics of a system of N point particle in presence of ideal constraints. I will not lie to you. While reading the whole derivation, there will be moments when you can either make an act faith, or believe an heuristic and shallow explanation or study differential geometry. I will not ask you to do it, but I am just telling you that mathematical physics in the early nineteenth century was quite different from what we are used to now. Setting the complaints aside (for now), we will proceed in a systematic way step by step in the construction.

1 – Assumptions

We recall once and for all the assumptions at the heart of the following derivation, which are based on the content of Section 2.1. We shall list them all here and we will not repeat them until the end of the whole derivation. In the following

- a) We consider $N \geq 1$ point particles of mass $m_i > 0$, $i = 1, \dots, N$, free to move in $\mathbb{R}^n$, $n \geq 1$. Herein we fix a reference frame once and for all endowing it with Cartesian coordinates $(x_1, \dots, x_n)$.
- b) We denote by $t \in \mathbb{R}$ the time variable.
- c) We denote by $\mathbf{x}_i(t) = (x_{1i}(t), \dots, x_{ni}(t)) \in \mathbb{R}^n$ the position vector of each particle at an arbitrary instant of time t . If we want to highlight that also t can vary, then we write $\mathbf{x}_i \equiv \mathbf{x}_i(t) \in \mathbb{R}^{n+1}$. Furthermore we shall denote by

$$\mathbf{x}(t) = (\mathbf{x}_1(t), \dots, \mathbf{x}_N(t)) \in \mathbb{R}^{nN},$$

the position vector of all particles at an arbitrary instant of time t . If we want to highlight that also t can vary, then, also in this case, we write $\mathbf{x} \equiv \mathbf{x}(t) \in \mathbb{R}^{nN+1}$.

- d) We assume the existence of k ideal constraints, $1 \leq k < nN$, see Definition 20. In addition we denote by Σ_t , a time-dependent open subset of $\mathbb{R}^{nN}$ which codifies kinematically allowed configurations. Points of Σ_t are parameterized by $Nn-k$ Lagrangian coordinates $q^i \equiv q^i(t)$, $i = 1, \dots, Nn - k$.

To avoid burdening the reader with a heavy notation, we shall work with Σ_t without worrying that it represents just an open subset of the $(nN - k)$ -dimensional hypersurface in $\mathbb{R}^{nN}$ which is identified by the holonomic constraints. In addition, we shall not write the explicit dependence on time of the Lagrangian coordinates unless there might be a source of confusion.

2 – Kinematic Variables

In view of the above assumptions and of the analysis in Section 2.1, we are ready to translate all kinematic structures of our system of N point particles in terms of the Lagrange coordinates. To simplify a bit more the notation let us set

$$\tilde{N} \equiv \tilde{N}(N, n, k) = Nn - k.$$

Accordingly, we can write each position vector $\mathbf{x}_i \in \Sigma_t$ as

$$\mathbf{x}_i(t) \equiv \mathbf{x}_i(q^1, \dots, q^{\tilde{N}}, t), \quad i = 1, \dots, N$$

from which we derive that the associated velocity vector is

$$\mathbf{v}_i(t) \doteq \frac{d\mathbf{x}_i}{dt} = \frac{\partial \mathbf{x}_i}{\partial t} + \sum_{j=1}^{\tilde{N}} \frac{\partial \mathbf{x}_i}{\partial q^j} \frac{dq^j}{dt}, \quad (31)$$

where we have applied the chain rule and where

$$\frac{\partial \mathbf{x}_i}{\partial q^j} = \left(\frac{\partial x_{1i}}{\partial q^j}, \dots, \frac{\partial x_{ni}}{\partial q^j} \right).$$

Scalar Product in $\mathbb{R}^n$

Let us recall here one of the basic facts stemming from linear algebra. Consider two vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ whose components with respect to the standard basis $\mathbf{e}_i$, $i = 1, \dots, n$ read

$$\mathbf{v} \equiv (v_1, \dots, v_n) \quad \text{and} \quad \mathbf{w} \equiv (w_1, \dots, w_n),$$

namely $v = \sum_{i=1}^n v_i \mathbf{e}_i$ and $w = \sum_{j=1}^n w_j \mathbf{e}_j$. Then the *Euclidean scalar product* is a bilinear map $(,) : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$(\mathbf{v}, \mathbf{w}) \equiv \mathbf{v} \cdot \mathbf{w} = \sum_{i=1}^n v_i w_i = \begin{pmatrix} v_1 & \dots & v_n \end{pmatrix} \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix}. \quad (32)$$

Accordingly, for any vector $\mathbf{v} \in \mathbb{R}^n$ we define its *squared-length* as

$$\|\mathbf{v}\|^2 = (\mathbf{v}, \mathbf{v}) \equiv \mathbf{v} \cdot \mathbf{v} = \sum_{i=1}^n v_i^2.$$

In classical mechanics, the kinematic data are codified by the position and by the velocity of each point particle at each instant of time. Yet an additional, relevant quantity related to them which is conceptually independent from the forces which might act on the system is the **kinetic energy** which we rewrite in terms of the Lagrangian coordinates. Recall that, on $\mathbb{R}^n$, given N distinct particles

$$T = \sum_{i=1}^N T_i, \quad (33)$$

$$T_i = \frac{m_i}{2} \|\mathbf{v}_i\|^2 = \frac{m_i}{2} \mathbf{v}_i \cdot \mathbf{v}_i = \frac{m_i}{2} \sum_{j=1}^n v_{ji}^2,$$

Using herein Equation (31) we end up with

$$\begin{aligned} T_i &= \frac{m_i}{2} \left(\frac{\partial \mathbf{x}_i}{\partial t} + \sum_{j=1}^{\tilde{N}} \frac{\partial \mathbf{x}_i}{\partial q^j} \frac{dq^j}{dt} \right) \cdot \left(\frac{\partial \mathbf{x}_i}{\partial t} + \sum_{j=1}^{\tilde{N}} \frac{\partial \mathbf{x}_i}{\partial q^j} \frac{dq^j}{dt} \right) = \\ &= T_2 + T_1 + T_0, \end{aligned} \quad (34)$$

where

$$\begin{cases} T_2 = \frac{m_i}{2} \sum_{h,j=1}^{\tilde{N}} a_{hj} \frac{dq^h}{dt} \frac{dq^j}{dt}, & a_{hj} \equiv a_{hj}(q_1, \dots, q_{\tilde{N}}, t) \doteq \frac{\partial \mathbf{x}_i}{\partial q^h} \cdot \frac{\partial \mathbf{x}_i}{\partial q^j} \\ T_1 = \frac{m_i}{2} \sum_{j=1}^{\tilde{N}} b_j \frac{dq^j}{dt}, & b_j \equiv b_j(q_1, \dots, q_{\tilde{N}}, t) = \frac{\partial \mathbf{x}_i}{\partial q^j} \cdot \frac{\partial \mathbf{x}_i}{\partial t} \\ T_0 = \frac{\partial \mathbf{x}_i}{\partial t} \cdot \frac{\partial \mathbf{x}_i}{\partial t} \end{cases}. \quad (35)$$

Remark 8. Equation (35) tells us that a byproduct of working with Lagrangian coordinates is that, while the kinetic energy of a point particle is quadratic in the time derivatives of the

Cartesian coordinates, this is no longer true when working with the Lagrangian counterparts. This is not a big deal per se, but it highlights that simple expressions might get more complicated when working with $q_1, \dots, q_{\tilde{N}}$. A notable exception is the scenario where all constraints are scleronomic, hence time-independent. This entails that $\mathbf{x}_i(t) \equiv \mathbf{x}_i(q^1, \dots, q^{\tilde{N}}, t) = \mathbf{x}_i(q^1, \dots, q^{\tilde{N}})$ for all $i = 1, \dots, N$. Consequently in Equation (31)

$$\mathbf{v}_i(t) = \sum_{j=1}^{\tilde{N}} \frac{\partial \mathbf{x}_i}{\partial q^j} \frac{dq^j}{dt},$$

and, therefore, in Equation (35), $T_1 = T_0 = 0$. Hence, in this case, the kinetic energy is quadratic in the time-derivatives of the Lagrangian coordinates and, introducing the vector $\mathbf{q} = (q^1, \dots, q^{\tilde{N}})$, we can write

$$T_i = \frac{m}{2} \frac{d\mathbf{q}}{dt} \cdot_a \frac{d\mathbf{q}}{dt},$$

where, recalling Equation (35)

$$\frac{d\mathbf{q}}{dt} \cdot_a \frac{d\mathbf{q}}{dt} = \sum_{h,j=1}^n a_{hj} \frac{dq^h}{dt} \frac{dq^j}{dt} = \begin{pmatrix} \frac{dq^1}{dt} & \dots & \frac{dq^{\tilde{N}}}{dt} \end{pmatrix} \begin{pmatrix} a_{11} & \dots & a_{1\tilde{N}} \\ \vdots & \vdots & \vdots \\ a_{\tilde{N}1} & \dots & a_{\tilde{N}\tilde{N}} \end{pmatrix} \begin{pmatrix} \frac{dq^1}{dt} \\ \vdots \\ \frac{dq^{\tilde{N}}}{dt} \end{pmatrix}.$$

3 – Dealing with Newton’s Equations

We have introduced all the ingredients to deal with Newtonian dynamics with the due exception of one concept. It has far reaching consequences and at first glance, it is rather simple to state. Yet, if you think about it, it is by far not obvious and it is a sort of *paradigm shift* that from time to time occurs in physics. Other, even more radical examples, appear in special relativity, quantum mechanics and general relativity.

State Space – Paradigm Shift

Suppose to consider the simplest example of a single, massive point particle in $\mathbb{R}^3$. If I ask you how would you describe the state of the particle at each instant of time $t \in \mathbb{R}$, you would reply that we would need to specify three functions

$$\mathbf{x}(t) = (x(t), y(t), z(t)).$$

This is correct, but, if I would push asking how you select $\mathbf{x}(t)$ among the infinitely many curves in $\mathbb{R}^3$, you would answer that one needs to solve Newton’s equation. In absence of any constraint this is tantamount to say that $\mathbf{x}(t)$ is such that

$$\ddot{\mathbf{x}} = \frac{\mathbf{F}}{m}, \tag{36}$$

where $\mathbf{F}$ are the forces acting on the particle, which we could also assume to vanish for the sake of this explanation. In other words you would say that $\mathbf{x}(t)$ is admissible if it abides by Equation (36). If I would be wondering how many such curves exist, of course, I would conclude that there are infinitely many. If $\mathbf{F} = 0$, the answer is rather straightforward, but, in general, one could say that we can label the solutions of Equation (36) in terms of the admissible initial data, since, under very mild regularity conditions on $\mathbf{F}$, the initial value problem

$$\begin{cases} \ddot{\mathbf{x}} = \frac{\mathbf{F}}{m}, \\ \dot{\mathbf{x}}(\bar{t}) = (\dot{x}(\bar{t}), \dot{y}(\bar{t}), \dot{z}(\bar{t})) \text{ and } \mathbf{x}(\bar{t}) = (x(\bar{t}), y(\bar{t}), z(\bar{t})) \end{cases} \quad .$$

admits a unique solution and, in turn, every solution of Equation (36) is constructed in this way. Here $\bar{t} \in \mathbb{R}$ is arbitrary but fixed. If we read this reasoning carefully, we are stating that, regardless of our physical model for the forces acting on the system, the set of *admissible instantaneous configuration* is completely characterized by two vectors $\mathbf{x}(\bar{t}), \dot{\mathbf{x}}(\bar{t}) \in \mathbb{R}^3$ which are **independent from each other**. In the language of calculus, initial data are not constrained and they all can be chosen arbitrarily. To summarize

- At each instant of time, the set of all admissible states of a point particle is characterized by two independent vectors in $\mathbb{R}^3$ which we denote conventionally as $\mathbf{x}$ (kinematic configuration) and $\dot{\mathbf{x}}$. The collection of all these vectors is sometimes referred as the *state space*.
- Dynamics is encoded via Equation (36) and it selects those trajectories for which $\left. \frac{d\mathbf{x}(t)}{dt} \right|_{t=\bar{t}} = \dot{\mathbf{x}}(\bar{t})$.

Applying this reasoning to our analysis entails that, since Lagrangian coordinates serve the purpose to parameterize at each instant of time, the admissible configuration of the underlying particles, then it means that we need to consider each q_i as *independent* from $\dot{q}_i$ until we look for curves in Σ_t which are solutions to a specific set of Newton's equation. At that point we have to set $\dot{q}_i = \frac{dq_i}{dt}$.

Remark 9. *Let us be honest, it sounds very very convoluted and, indeed, this is the case. In addition, there is this lingering sense of uneasiness which let us wonder whether this is mathematically precise, physically sound etc... Yes, everything is perfectly sound. The problem is that we are not using the appropriate language to discuss classical mechanics. If we could use differential geometry, none of these issues would occur, but, as they say in Britain, this goes beyond the scope of these lectures. I recommend you the course of Elettrodinamica e Realtività for a first introduction to this very efficient language.*

Having in mind these caveats, let us consider our starting point, namely Equation (30). If we recall Definition 20 and Remark 7, we can get rid of the reactive forces by considering the

scalar product with a virtual displacement, namely

$$\sum_{i=1}^N (m_i \mathbf{a}_i - \mathbf{F}_i) \delta \mathbf{x}_i = 0 \implies \sum_{i=1}^N \sum_{j=1}^{\tilde{N}} (m_i \mathbf{a}_i - \mathbf{F}_i) \frac{\partial \mathbf{x}_i}{\partial q^j} \delta q^j = 0, \quad (37)$$

where $\delta \mathbf{x}_i$ is the i -th component of the vector of virtual displacements obtained combining Exercise 18 and Equation (29). Since we are considering arbitrary kinematic configurations we can omit writing down explicitly the point where $\frac{\partial \mathbf{x}_i}{\partial q^j}$ has been evaluated. In addition, since all δq^j are linearly independent per construction it turns out that, for all $j = 1, \dots, \tilde{N}$

$$\sum_{i=1}^N (m_i \mathbf{a}_i - \mathbf{F}_i) \frac{\partial \mathbf{x}_i}{\partial q^j} = 0. \quad (38)$$

We can now state one of the main results of these lecture notes.

Proposition 21. *Under the assumptions of Section 3.1 it holds that, for every trajectory solution of Equation (30), it holds that*

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{q}^j} - \frac{\partial T}{\partial q^j} = \sum_{i=1}^N m_i \mathbf{a}_i \frac{\partial \mathbf{x}_i}{\partial q^j}, \quad (39)$$

where T is the total kinetic energy as per Equation (33) and where we impose the identity $\dot{q}^j = \frac{dq^j}{dt}$ for all $j = 1, \dots, \tilde{N}$.

Proof. Since on the left hand side of Equation (39) only linear operators act on T , on account of Equation (33), we can consider

$$\frac{d}{dt} \frac{\partial T_i}{\partial \dot{q}^j} - \frac{\partial T_i}{\partial q^j},$$

where $i = 1, \dots, \tilde{N}$. Using the definition of T_i as $\frac{m_i}{2} \mathbf{v}_i \cdot \mathbf{v}_i$, and using the Leibnitz rule it descends that

$$\frac{d}{dt} \frac{\partial T_i}{\partial \dot{q}^j} = m_i \frac{d}{dt} \left(\mathbf{v}_i \cdot \frac{\partial \mathbf{v}_i}{\partial \dot{q}^j} \right) = m_i \left(\mathbf{a}_i \cdot \frac{\partial \mathbf{v}_i}{\partial \dot{q}^j} + \mathbf{v}_i \cdot \frac{d}{dt} \frac{\partial \mathbf{v}_i}{\partial \dot{q}^j} \right),$$

where we used implicitly that the scalar product is symmetric. At the same time and for the same reason, it holds that

$$\frac{\partial T_i}{\partial q^j} = m_i \mathbf{v}_i \cdot \frac{\partial \mathbf{v}_i}{\partial q^j}.$$

Putting together these two identities it descends that

$$\frac{d}{dt} \frac{\partial T_i}{\partial \dot{q}^j} - \frac{\partial T_i}{\partial q^j} = m_i \left(\mathbf{a}_i \cdot \frac{\partial \mathbf{v}_i}{\partial \dot{q}^j} + \mathbf{v}_i \cdot \left(\frac{d}{dt} \frac{\partial \mathbf{v}_i}{\partial \dot{q}^j} - \frac{\partial \mathbf{v}_i}{\partial q^j} \right) \right).$$

We can use Equation (31) to conclude that

$$\frac{\partial \mathbf{v}_i}{\partial \dot{q}^j} = \frac{\partial \mathbf{x}_i}{\partial q^j},$$

where we have claimed that $\frac{\partial^2 \mathbf{x}_i}{\partial \dot{q}^j \partial t} = 0$ since $\mathbf{x}_i = \mathbf{x}_i(q^1, \dots, q^{\tilde{N}}, t)$ and since the Lagrangian coordinates q^i and $\dot{q}^j$ are independent one from each other. At the same time, still using Equation (31), we can infer that

$$\begin{aligned} \frac{d}{dt} \frac{\partial \mathbf{v}_i}{\partial \dot{q}^j} - \frac{\partial \mathbf{v}_i}{\partial q^j} &= \frac{d}{dt} \frac{\partial}{\partial \dot{q}^j} \left(\frac{\partial \mathbf{x}_i}{\partial t} + \sum_{k=1}^{\tilde{N}} \frac{\partial \mathbf{x}_i}{\partial q^k} \dot{q}^k \right) - \frac{\partial}{\partial q^j} \left(\frac{\partial \mathbf{x}_i}{\partial t} + \sum_{k=1}^{\tilde{N}} \frac{\partial \mathbf{x}_i}{\partial q^k} \dot{q}^k \right) = \\ &= \frac{d}{dt} \frac{\partial \mathbf{x}_i}{\partial q^j} - \frac{\partial^2 \mathbf{x}_i}{\partial q^j \partial t} - \sum_{k=1}^{\tilde{N}} \frac{\partial^2 \mathbf{x}_i}{\partial q^j \partial q^k} \dot{q}^k = 0, \end{aligned}$$

where we have used once more that the Lagrangian coordinates q^i and $\dot{q}^j$ are independent one from each other and that we can write $\frac{d}{dt} = \frac{\partial}{\partial t} + \sum_{k=1}^{\tilde{N}} \dot{q}^k \frac{\partial}{\partial q^k}$ when acting on functions which depend only on the time variable and on the Lagrangian coordinates q^j , $j = 1, \dots, \tilde{N}$. Putting together all these data we have shown that

$$\frac{d}{dt} \frac{\partial T_i}{\partial \dot{q}^j} - \frac{\partial T_i}{\partial q^j} = m_i \mathbf{a}_i \cdot \frac{\partial \mathbf{x}_i}{\partial q^j},$$

which is the sought statement one we sum over all particles labeled by $i = 1, \dots, N$. $\square$

Combining Proposition 21 with Equation (38) we obtain a key result, the onset of **Lagrangian mechanics**

Theorem 22 (Euler-Lagrange Equations). *Under the assumptions of Section 3.1, the trajectory $\mathbf{x}(t) = (\mathbf{x}_1(t), \dots, \mathbf{x}_N(t))$ of N -point particles which abides by Newton's equations (30) is a solution of the **Euler-Lagrange equations***

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{q}^j} - \frac{\partial T}{\partial q^j} = \sum_{i=1}^N \mathbf{F}_i \frac{\partial \mathbf{x}_i}{\partial q^j}, \quad (40)$$

with the additional constraint that $\frac{dq^j}{dt} = \dot{q}^j$ for all $j = 1, \dots, \tilde{N}$. The right hand side is known as the Lagrangian component of the active forces.

4 – The Lagrange Function

Equation (40) is at this stage a result of dubious applicability since the left hand side can be computed in an algorithmic way once Lagrangian coordinates are chosen, but the right hand

seems more convoluted to be used efficiently. A further and spectacular simplification arises, if we assume that all the forces considered in the underlying physical system are **conservative**, that is, in 3 dimensions

$$\nabla \times \mathbf{F}_i = 0, \quad \forall i = 1, \dots, N$$

where $\nabla \times$ is the curl operator. A standard result from calculus entails that, if Σ_t is a simply connected domain, that is, all closed loops can be continuously contracted to a point, then there exists a *potential function* $V : \mathbb{R}^3 \rightarrow \mathbb{R}$ such that

$$\mathbf{F}_i = -\nabla_i V(x_{i1}, x_{i2}, x_{i3}), \quad (41)$$

where ∇_i entails that we are taking the gradient, only with respect to the Cartesian coordinates of the i -th particle. More generally, if we have a force in $\mathbb{R}^n$, that is $\mathbf{F}_i \in C^\infty(\mathbb{R}^{nN}; \mathbb{R}^n)$ we call it **conservative** if we can find a potential function $V : \mathbb{R}^{nN} \rightarrow \mathbb{R}$ such that Equation (41) holds true. Although, in practical applications it is hardly used, we can also allow V to depend explicitly on the time variable and, therefore, we shall write $V = V(\mathbf{x}_1, \dots, \mathbf{x}_N, t)$. As a consequence, when Equation (41) holds true, we can infer that the Lagrangian component of the active forces reads

$$\sum_{i=1}^N \mathbf{F}_i \frac{\partial \mathbf{x}_i}{\partial q^j} = - \sum_{i=1}^N \nabla_i V(\mathbf{x}_1, \dots, \mathbf{x}_N) \frac{\partial \mathbf{x}_i}{\partial q^j} = \frac{\partial V}{\partial q^j}, \quad (42)$$

where, in this expression, we have implicitly restricted V to Σ_t and therefore all the Cartesian coordinates are only dependent from the Lagrangian ones. In other words we get a modified and way more practical version of Theorem 22.

Theorem 23 (Lagrangian function). *Under the assumptions of Section 3.1 and assuming that all active forces on the system are conservative, the trajectory $\mathbf{x}(t) = (\mathbf{x}_1(t), \dots, \mathbf{x}_N(t))$ of N -point particles which abides by Newton's equations (30) is a solution of the **Euler-Lagrange equations***

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^j} - \frac{\partial \mathcal{L}}{\partial q^j} = 0 \quad (43)$$

with the additional constraint that $\frac{dq^j}{dt} = \dot{q}^j$ for all $j = 1, \dots, \tilde{N}$. Here

$$\begin{aligned} \mathcal{L} : \mathbb{R}^{\tilde{N}} \times \mathbb{R}^{\tilde{N}} \times \mathbb{R} &\rightarrow \mathbb{R}, \\ (q_1, \dots, q_{\tilde{N}}, \dot{q}_1, \dots, \dot{q}_{\tilde{N}}, t) &\mapsto \mathcal{L}[q_1, \dots, q_{\tilde{N}}, \dot{q}_1, \dots, \dot{q}_{\tilde{N}}, t] \doteq T - V, \end{aligned} \quad (44)$$

is known as **Lagrange function of the system** or, equivalently, **Lagrangian of the system**.

Remark 10. Observe that, in deriving Equation (43) from Equation (40), we have replaced the right hand side of the latter with Equation (42), but we have also used that

$$\frac{d}{dt} \frac{\partial V}{\partial \dot{q}^j} = 0. \quad \forall j = 1, \dots, \tilde{N}$$

This holds true if the potential function V does not depend on the generalized velocities $\dot{q}^j$, a statement which is true in most of the cases except a few notable physically interesting scenarios such as electromagnetism. In this case a generalization is needed and we have to resort to a more advanced notion of **generalized potentials**, see [Mor20, Sec. 10.2.2].

Remark 11. Observe that Equation (41) entails that, to a conservative force, we can associate a potential (function) which is not unique, since we have always the freedom to add to V any element in the kernel of ∇ . Yet, since $\mathbb{R}^n$, $n \geq 1$ is connected, this is nothing but a constant function. This translates at a mathematical level that we can always set the 0 of the potential arbitrarily and this has no physical consequence since, in a classical system, what matters is the difference between the total energy of the initial and of the final state. In this respect a constant, additive function is inconsequential. The same applies in Lagrangian mechanics as the Lagrange function $\mathcal{L}$ in Equation (44) does not carry per se a physical significance, but it serves as a source for the equations of motion, see Equation (43). Herein, only derivatives of $\mathcal{L}$ appear and therefore, shifting the potential V by a constant does not change them.

3.2 Examples

In order to grasp the deep significance of Theorem 23, especially from a pragmatic viewpoint, let us discuss two simple, yet very instructive examples which can also serve as the prototype of a written exam exercise.

A particle on a sphere

In $\mathbb{R}^3$ consider a Cartesian reference system as per Figure 5 and a sphere centered at the origin O of radius $R > 0$. A point particle of mass $m > 0$ is constrained to move on the sphere and it is subject to an external gravitational force here modeled as $\mathbf{F}_z = -mg\mathbf{e}_z$, where g is the standard gravitational acceleration while $\mathbf{e}_z$ is the unit vector pointing along the z -direction.

Solution: Observe that the constraint can be codified in the equation $f(x, y, z) = x^2 + y^2 + z^2 - R^2 = 0$ where (x, y, z) are the standard Cartesian coordinates in $\mathbb{R}^3$. This is an ideal constraint as per Definition 20 which is in addition time-independent (rethonomic). On top of this the only external force is conservative. As a matter of fact, recall that, given $\mathbf{F} = (F_x(x, y, z), F_y(x, y, z), F_z(x, y, z)) \in C^\infty(\mathbb{R}^3; \mathbb{R}^3)$, its **curl** reads

$$\nabla \times \mathbf{F} = \det \begin{bmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{bmatrix}$$

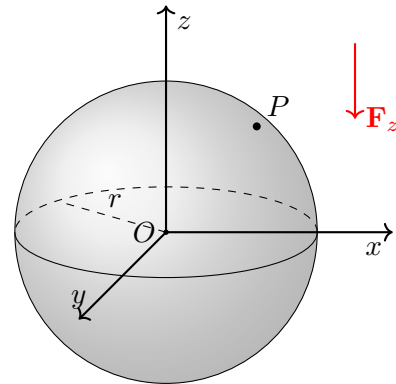


Figure 5: Point particle on a sphere

Expanding this determinant, we get:

$$\nabla \times \mathbf{F} = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \mathbf{i} + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right) \mathbf{j} + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right) \mathbf{k}.$$

A direct inspection shows that the force under scrutiny is conservative and it descends from a potential which can be obtained solving the equation $\mathbf{F} = -\nabla V$. This yields

$$V(x, y, z) = mgz + V_0,$$

where V_0 is a constant which we set to 0. This entails that we can apply Theorem 23. To this end, first of all, we observe that, since we have 1 ideal constraint, the number of *degrees of freedom (Lagrangian coordinates)* = 3-1 = 2. The geometry of the problem suggests how to choose these coordinates since the point particle is constrained to move on a sphere of a radius R , namely we employ spherical coordinates, see Example 7. Adapting them to the case in hand, we end up with

$$\begin{cases} x(t) = R \sin \theta(t) \cos \varphi(t) \\ y(t) = R \sin \theta(t) \sin \varphi(t) \\ z(t) = R \cos \theta(t) \end{cases} . \quad (45)$$

In other words we have two generalized Lagrangian coordinates (θ, φ) where $\theta \in [0, \pi)$ while $\varphi \in [0, 2\pi)$. The time dependence is here inserted to remind us that the position of the particle varies. As a matter of fact we can now compute the potential as a function of the new coordinates, namely

$$V \equiv V(\theta) = mgR \cos \theta(t),$$

as well as the instantaneous velocity of the particle by taking the time derivative of Equation 45:

$$\mathbf{v}(t) \equiv \dot{\mathbf{x}}(t) = \begin{cases} \dot{x}(t) = R \left(\dot{\theta}(t) \cos \theta(t) \cos \varphi(t) - \dot{\varphi}(t) \sin \theta(t) \sin \varphi(t) \right) \\ \dot{y}(t) = R \left(\dot{\theta}(t) \cos \theta(t) \sin \varphi(t) + \dot{\varphi}(t) \sin \theta(t) \cos \varphi(t) \right) \\ \dot{z}(t) = -R \dot{\theta}(t) \sin \theta(t) \end{cases} . \quad (46)$$

Observe that, in writing the velocity vector in this way, we have implemented the constraint $\dot{\mathbf{x}} = \frac{d\mathbf{x}}{dt}$. We can now write the kinetic energy using Equation (33), namely

$$T = \frac{m}{2} \mathbf{v}(t) \cdot \mathbf{v}(t) = \frac{m}{2} (\dot{x}^2(t) + \dot{y}^2(t) + \dot{z}^2(t)) = \frac{mR^2}{2} (\dot{\theta}^2(t) + \sin^2 \theta(t) \dot{\varphi}^2(t)) .$$

Putting together all these data we can construct the Lagrange function for the system under analysis using Equation (44):

$$\mathcal{L}(\theta, \dot{\theta}, \varphi, \dot{\varphi}) = \frac{mR^2}{2} (\dot{\theta}^2(t) + \sin^2 \theta(t) \dot{\varphi}^2(t)) - mgR \cos \theta(t). \quad (47)$$

The dynamics of the system is thus codified by the Euler-Lagrange equations (43):

$$\begin{cases} \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} - \frac{\partial \mathcal{L}}{\partial \theta} = mR^2 \left(\ddot{\theta}(t) - \sin \theta(t) \cos \theta(t) \dot{\varphi}^2(t) \right) - mgR \sin \theta(t) = 0 \\ \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\varphi}} - \frac{\partial \mathcal{L}}{\partial \varphi} = mR^2 \frac{d}{dt} (\sin^2 \theta(t) \dot{\varphi}(t)) = mR^2 \left(\sin^2 \theta(t) \ddot{\varphi}(t) + 2 \sin \theta(t) \cos \theta(t) \dot{\theta}(t) \dot{\varphi}(t) \right) = 0 \end{cases}, \quad (48)$$

which simplifies to

$$\begin{cases} \ddot{\theta}(t) - \sin \theta(t) \cos \theta(t) \dot{\varphi}^2(t) - \frac{g}{R} \sin \theta(t) = 0 \\ \ddot{\varphi}(t) + 2 \cot \theta(t) \dot{\theta}(t) \dot{\varphi}(t) = 0 \end{cases}, \quad (49)$$

Outcome: We have constructed the equations of motion in a rather simple and algorithmic way, but are we happy? Never, but this time in particular because we are left with the following open questions:

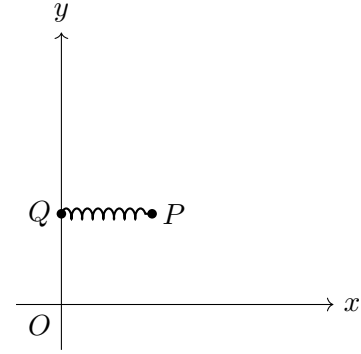
1. How does a solution of Equation (49) look like?
2. In Newtonian mechanics, we would have argued with the conservation of energy and, possibly of one of the components of the angular momentum, to simplify the study of the system. Have we done it?

The answer to these questions is a big NO!

It goes without saying that this is not acceptable and we have to take action to get out of this quandary. The second part of this course shall deal with the first question, while Section 4 with the second one.

Two particles connected by a spring

In $\mathbb{R}^2$ consider a Cartesian reference system as per Figure 6 and two point particles of same mass $m > 0$, denoted by P and Q . P abides to no constrain, while Q is forced to move along the vertical axis. They are subject to an external force here modeled by an elastic force attracting Q to P , that is $\mathbf{F}_{Q \rightarrow P} = \kappa(\mathbf{x}_Q - \mathbf{x}_P)$ where $\kappa > 0$ is an elastic constant while $\mathbf{x}_{P/Q}$ denote the position vector of P and Q respectively.



Solution: The system under investigation possesses a single constraint. If we denote by (x, y) the generic Cartesian coordinates on $\mathbb{R}^2$ and by $\mathbf{x}_P = (x_P, y_P)$, $\mathbf{x}_Q = (x_Q, y_Q)$ the position vectors of P and Q , the constraint reads $x_Q = 0$, which is ideal and time-independent (rethonomic) as per Definition 20. Consequently the system has 3 degrees of freedom and we can choose the Lagrangian coordinates to coincide this time with the Cartesian ones, since the geometry does not suggest a

Figure 6: Two point particles connected by a spring, one constrained on the vertical axis

smarter choice. Hence we choose (x_P, y_P, y_Q) . In addition the only active force on the system is conservative. Although in $\mathbb{R}^2$ one can find a way around, using the curl operator, to verify this statement, it is much easier to exhibit directly the potential function, which is our case reads

$$V(x_P, y_P, x_Q, y_Q) = \frac{\kappa}{2} \|\mathbf{x}_Q - \mathbf{x}_P\|^2.$$

A direct calculation shows that $\mathbf{F}_{Q \rightarrow P} = -\nabla_Q V$, where ∇_Q denotes the gradient with respect to (x_Q, y_Q) . Also in this case we have set the integration constant for V to 0, which, in many textbooks with a more engineering based mindset, is translated with the statement that we are considering Q and P connected by a spring with vanishing rest length. In other words we can apply Theorem 23. To summarize, implementing the constraint, it holds that the position vector of P and Q reads

$$\mathbf{x}_P(t) = (x_P(t), y_P(t)) \quad \text{and} \quad \mathbf{x}_Q(t) = (0, y_Q(t)),$$

where we have inserted the explicit time-dependence. Accordingly the potential reads

$$V = \frac{\kappa}{2} (x_P^2(t) + (y_Q(t) - y_P(t))^2),$$

while the velocity vector is

$$\dot{\mathbf{x}}_P(t) = (\dot{x}_P(t), \dot{y}_P(t)) \quad \text{and} \quad \dot{\mathbf{x}}_Q(t) = (0, \dot{y}_Q(t)).$$

Using once more Equation (34), we can compute the kinetic energy of the system as

$$T = \frac{m}{2} (\|\dot{\mathbf{x}}_P(t)\|^2 + \|\dot{\mathbf{x}}_Q(t)\|^2) = \frac{m}{2} (\dot{x}_P^2(t) + \dot{y}_P^2(t) + \dot{y}_Q^2(t)).$$

Putting together these data, we can construct the Lagrangian function (44):

$$\mathcal{L} = \frac{m}{2} (\dot{x}_P^2(t) + \dot{y}_P^2(t) + \dot{y}_Q^2(t)) + \frac{\kappa}{2} (x_P^2(t) + (y_Q(t) - y_P(t))^2), \quad (50)$$

from which we can derive the Euler-Lagrange equations (43)

$$\begin{cases} \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}_P} - \frac{\partial \mathcal{L}}{\partial x_P} = m\ddot{x}_P + \kappa x_P = 0 \\ \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{y}_P} - \frac{\partial \mathcal{L}}{\partial y_P} = m\ddot{y}_P - \kappa(y_Q - y_P) = 0 \\ \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{y}_Q} - \frac{\partial \mathcal{L}}{\partial y_Q} = m\ddot{y}_Q + \kappa(y_Q - y_P) = 0 \end{cases}, \quad (51)$$

Outcome: We have constructed the equations of motion in a rather simple and algorithmic way and, this time, they can be easily solved. The first one is decoupled from all other yielding and setting $\omega = \sqrt{\frac{\kappa}{m}}$

$$x_P(t) = x_P(0) \cos(\omega t) + \frac{\dot{x}_P(0)}{\omega} \sin(\omega t),$$

where $x_P(0)$ and $\dot{x}_P(0)$ are the initial position and velocity of the particle P along the x -direction at $t = 0$. The remaining two equations can be solved introducing the auxiliary coordinates known as center of mass mode and difference mode

$$z(t) = y_P(t) + y_Q(t) \quad \text{and} \quad w(t) = y_P(t) - y_Q(t).$$

The last two identities of Equation (51) become

$$\ddot{z} = 0 \quad \text{and} \quad \ddot{w} = -2\omega^2 w. \quad (52)$$

These admit the solutions

$$z(t) = z(0) + \dot{z}(0)t \quad \text{and} \quad w(t) = w_P(0) \cos(\sqrt{2}\omega t) + \frac{\dot{w}_P(0)}{\sqrt{2}\omega} \sin(\sqrt{2}\omega t),$$

where $(z(0), \dot{z}(0), w(0), \dot{w}(0))$ are the initial conditions associated to these variables at $t = 0$. To add a last question to those asked at the end of the previous example: could have we changed the Lagrangian coordinates from the very beginning adapting them in this case not only to the geometry but also to the special functional form of the potential? In other words

- How dependent is the Lagrange function and the associated equations of motion from the choice of Lagrangian coordinates. If we change them, does it suffice to change $\mathcal{L}$ accordingly?

We address this issue already in the next section.

3.3 Properties of the Euler-Lagrange Equations

In this section we investigate two noteworthy structural properties of the Euler-Lagrange equations. The first one is motivated by the discussion at the end of the previous section. It turned out that the choice of Lagrangian coordinates made at the beginning and subordinated to the underlying geometry was not optimal to analyze the dynamics. The solutions of the equations of motion were easy to find, changing coordinate system. While this operation is rather innocuous at the level of the Lagrangian as per Equation (44), since it is a scalar function and coordinate changes behave like we learned in calculus, it is not obvious a priori that this does not affect the solutions. We discuss that this is not the case following mainly [Mor20, Sec. 8.4.3].

Independence of the Euler-Lagrange Equations from the Lagrangian coordinates

We consider a system under the same hypotheses listed at the beginning of Section 3.1, with one caveat. Henceforth we assume that we have chosen two different systems of Lagrangian coordinates, both admissible on Σ_t :

$$\{q^i\}_{i=1,\dots,\tilde{N}} \quad \text{and} \quad \{q'^j\}_{j=1,\dots,\tilde{N}},$$

where $\tilde{N} = nN - k$ is the number of independent degrees of freedom of the configuration space. To start with since these coordinates are all valid in Σ_t , it is possible to express one system as a function of the other, namely

$$q'^j \equiv q'^j(q^1, \dots, q^{\tilde{N}}, t), \quad \forall j = 1, \dots, \tilde{N} \quad (53)$$

where we have also accounted for a possible explicit dependence on t in the coordinate change. Accordingly, the associated velocities obey

$$\dot{q}'^j = \frac{dq'^j}{dt} = \frac{\partial q'^j}{\partial t} + \sum_{k=1}^{\tilde{N}} \frac{\partial q'^j}{\partial q^k} \dot{q}^k,$$

which entails in turn the following identity

$$\frac{\partial \dot{q}'^j}{\partial \dot{q}^k} = \frac{\partial q'^j}{\partial q^k}, \quad (54)$$

where we assumed once more that the generalized velocities are independent from the Lagrangian coordinates. With reference to the generic form of the Euler-Lagrange equations as per Theorem 22, that is, under no assumption on the underlying active forces being conservative, we can establish the following structural result.

Proposition 24 (Independence of the Euler-Lagrange Equations). *Under the same hypotheses listed at the beginning of Section 3.1 and chosen two different systems of Lagrangian coordinates, both admissible on Σ_t :*

$$\{q^i\}_{i=1,\dots,\tilde{N}} \quad \text{and} \quad \{q'^j\}_{j=1,\dots,\tilde{N}},$$

it holds that, given any curve $\mathbf{x}(t) \equiv \mathbf{x}(q^1(t), \dots, q^{\tilde{N}}(t), t) = \mathbf{x}(q'^1(t), \dots, q'^{\tilde{N}}(t), t) \in \Sigma_t$, it holds that

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{q}'^j} - \frac{\partial T}{\partial q'^j} = \sum_{k=1}^{\tilde{N}} \frac{\partial q^k}{\partial q'^j} \left(\frac{d}{dt} \frac{\partial T}{\partial \dot{q}^k} - \frac{\partial T}{\partial q^k} \right),$$

where T is the kinetic energy as per Equation (34).

Proof. This is a brutal and exquisitely straightforward chain of identities. We start from the identity

$$\frac{\partial q^k}{\partial q'^j} \frac{d}{dt} \frac{\partial T}{\partial \dot{q}^k} = \frac{d}{dt} \left(\frac{\partial q^k}{\partial q'^j} \frac{\partial T}{\partial \dot{q}^k} \right) - \frac{d}{dt} \left(\frac{\partial q^k}{\partial q'^j} \right) \frac{\partial T}{\partial \dot{q}^k} = \frac{d}{dt} \left(\frac{\partial \dot{q}^k}{\partial \dot{q}'^j} \frac{\partial T}{\partial \dot{q}^k} \right) - \frac{d}{dt} \left(\frac{\partial q^k}{\partial q'^j} \right) \frac{\partial T}{\partial \dot{q}^k}, \quad (55)$$

where we used Equation (54). Recall that, in the Lagrangian setting, the collection of independent variables is $\{q^k, \dot{q}^k\}_{k=1,\dots,\tilde{N}}$ and $\{q'^j, \dot{q}'^j\}_{j=1,\dots,\tilde{N}}$ but we are assuming a change of coordinates of the form of Equation (53), this entails that

$$\frac{\partial}{\partial \dot{q}'^j} = \sum_{k=1}^{\tilde{N}} \left(\frac{\partial \dot{q}^k}{\partial \dot{q}'^j} \frac{\partial}{\partial \dot{q}^k} + \frac{\partial q^k}{\partial \dot{q}'^j} \frac{\partial}{\partial q^k} \right) = \sum_{k=1}^{\tilde{N}} \frac{\partial \dot{q}^k}{\partial \dot{q}'^j} \frac{\partial}{\partial \dot{q}^k},$$

where we have set $\frac{\partial q^k}{\partial \dot{q}'^j} = 0$ since we have assumed a coordinate transformation of the form $q^k = q^k(q'^1, \dots, q'^{\tilde{N}}, t)$ and viceversa. Hence, since we consider the coordinates $\dot{q}'^j$ as independent

when taking the derivatives, our result is justified. Reinstating the sum over k in Equation (55) and using this last identity, we end up with

$$\sum_{k=1}^{\tilde{N}} \frac{\partial q^k}{\partial q'^j} \frac{d}{dt} \frac{\partial T}{\partial \dot{q}^k} = \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}'^j} \right) - \sum_{k=1}^{\tilde{N}} \frac{d}{dt} \left(\frac{\partial q^k}{\partial q'^j} \right) \frac{\partial T}{\partial \dot{q}^k}.$$

This entails

$$\sum_{k=1}^{\tilde{N}} \frac{\partial q^k}{\partial q'^j} \left(\frac{d}{dt} \frac{\partial T}{\partial \dot{q}^k} - \frac{\partial T}{\partial q^k} \right) = \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}'^j} \right) - \sum_{k=1}^{\tilde{N}} \left(\frac{d}{dt} \left(\frac{\partial q^k}{\partial q'^j} \right) \frac{\partial T}{\partial \dot{q}^k} - \frac{\partial q^k}{\partial q'^j} \frac{\partial T}{\partial q^k} \right),$$

which, exchanging the derivatives $\frac{d}{dt}$ and $\frac{\partial}{\partial q'^j}$, yields

$$\sum_{k=1}^{\tilde{N}} \frac{\partial q^k}{\partial q'^j} \left(\frac{d}{dt} \frac{\partial T}{\partial \dot{q}^k} - \frac{\partial T}{\partial q^k} \right) = \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}'^j} \right) - \sum_{k=1}^{\tilde{N}} \left(\frac{\partial \dot{q}^k}{\partial q'^j} \frac{\partial T}{\partial \dot{q}^k} - \frac{\partial q^k}{\partial q'^j} \frac{\partial T}{\partial q^k} \right).$$

Observing that, for the most general transformation of coordinates, it holds always that

$$\sum_{k=1}^{\tilde{N}} \left(\frac{\partial \dot{q}^k}{\partial q'^j} \frac{\partial}{\partial \dot{q}^k} - \frac{\partial q^k}{\partial q'^j} \frac{\partial}{\partial q^k} \right) = \frac{\partial}{\partial q'^j},$$

we end up with

$$\sum_{k=1}^{\tilde{N}} \frac{\partial q^k}{\partial q'^j} \left(\frac{d}{dt} \frac{\partial T}{\partial \dot{q}^k} - \frac{\partial T}{\partial q^k} \right) = \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}'^j} \right) - \frac{\partial T}{\partial q'^j},$$

which concludes the proof. $\square$

From this result, we can infer a decisive corollary:

Corollary 25 (Independence of the Euler-Lagrange Equations - 2). *Under the same hypotheses listed at the beginning of Section 3.1 and chosen two different systems of Lagrangian coordinates, both admissible on Σ_t :*

$$\{q^i\}_{i=1,\dots,\tilde{N}} \quad \text{and} \quad \{q'^j\}_{j=1,\dots,\tilde{N}},$$

it holds that, given any curve $\mathbf{x}(t) \equiv \mathbf{x}(q^1(t), \dots, q^{\tilde{N}}(t), t)$ solution of the Euler-Lagrange equations as per Equation (40) with respect to the Lagrangian coordinates $\{q^i\}_{i=1,\dots,\tilde{N}}$, then it will also be solution of the same equations with respect to the Lagrangian coordinates $\{q'^j\}_{j=1,\dots,\tilde{N}}$.

Proof. Considering the right hand side of Equation (40), it holds that

$$\sum_{i=1}^N \mathbf{F}_i \frac{\partial \mathbf{x}_i}{\partial q'^j} = \sum_{k=1}^{\tilde{N}} \sum_{i=1}^N \mathbf{F}_i \frac{\partial \mathbf{x}_i}{\partial q^k} \frac{\partial q^k}{\partial q'^j},$$

where we have used implicitly that the position vector $\mathbf{x}_i$ cannot depend on the generalized velocities $\dot{q}^j$. Combining this result with Proposition 24, we end up with

$$\sum_{k=1}^{\tilde{N}} \frac{\partial q^k}{\partial q'^j} \left(\frac{d}{dt} \frac{\partial T}{\partial \dot{q}^k} - \frac{\partial T}{\partial q^k} - \sum_{i=1}^N \mathbf{F}_i \frac{\partial \mathbf{x}_i}{\partial q^k} \right) = \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}'^j} \right) - \frac{\partial T}{\partial q'^j} - \sum_{i=1}^N \mathbf{F}_i \frac{\partial \mathbf{x}_i}{\partial q'^j},$$

which entails the sought statement since, per definition, $\frac{\partial q^k}{\partial q'^j} \neq 0$. $\square$

(Non-)Uniqueness of the Lagrangian

In the derivation of the Lagrangian function in Theorem 23, one might have the feeling that $\mathcal{L}$ is essentially unique, barring the possibility of adding a constant to the potential function. Yet, we argued that this is irrelevant since it does not change the Euler-Lagrange equations (43). This reasoning opens actually several possibilities, namely, seeing Equation (43) as an unchangeable pillar of our theory, we would have the freedom of adding to $\mathcal{L}$ anything which does not alter the dynamics. A priori this sounds vague, but, in the following we will establish a class of universal functions which can be added to any Lagrangian without changing the dynamics. As far as the analysis of systems with finitely many degrees of freedom is concerned, this leeway plays no role and it does not carry a special physical significance. Yet, as soon as, one enters the realm of physical systems with infinitely many degrees of freedom, such as electromagnetism, or more generally any classical or quantum field theory, this has far reaching consequences to the point that it is really important to keep it in mind from the very beginning.

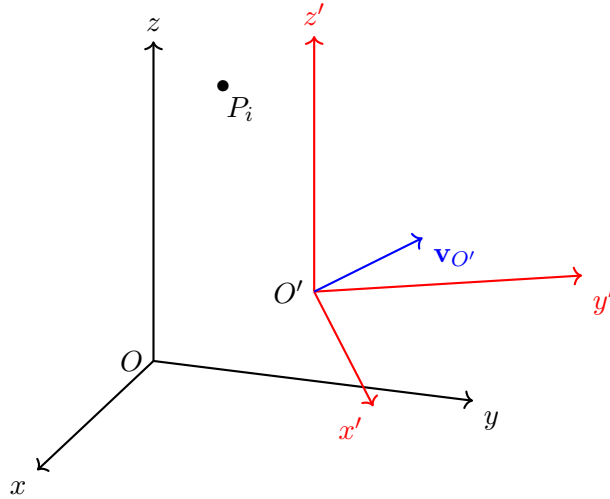


Figure 7: Two Cartesian reference systems which are inertial one to each other. The point O' is moving away from O with constant velocity $\mathbf{v}_{O'}$. The bullet denotes the position of the i -th particle labeled as P_i .

To this avail we discuss a special scenario. On the one hand, we consider a system of N -point particles abiding by all hypotheses detailed at the beginning of Section 3.1. On the other hand we consider a second reference system which is inertial with respect to the first one. This is pictorially depicted in 3D in Figure 7 and it consists of an origin labeled as O' as well as Cartesian coordinates $(x'_1, \dots, x'_n)$. In addition we assume that O' is moving relatively to O with constant velocity codified by the vector $\mathbf{v}_{O'}$. This entails, that, given any point particle, denoting by $\mathbf{v}_i$ and $\mathbf{v}'_i$ its velocity vector in the first and in the second reference frame respectively, it holds that

$$\mathbf{v}'_i = \mathbf{v}_i + \mathbf{v}_{O'}.$$

Let us discuss what are the changes the changes in the components of the Lagrangian when working in the two different reference systems under scrutiny:

Kinetic energy Focusing on Equation (34), we can consider each single component of the total kinetic energy and we denote T_i and T'_i the expression in the first and in the second reference frame respectively. Observing that the choice of Lagrangian coordinates $\{q^i\}_{i=1, \dots, \tilde{N}}$ does not depend on the underlying reference system, we can evaluate

$$\Delta T_i = T_i - T'_i = \frac{m_i}{2}(\|\mathbf{v}'_i\|^2 - \|\mathbf{v}_i\|^2) = \frac{m_i}{2}\|\mathbf{v}_{O'}\|^2 - m_i \mathbf{v}_i \cdot \mathbf{v}_{O'}, \quad (56)$$

where $\cdot$ is the scalar product on $\mathbb{R}^3$. At the same time, if $\mathbf{x}'_i \equiv \mathbf{x}'(q^1, \dots, q^{\tilde{N}}, t)$ is the position of the i -th particle in the reference system centered at O' seen as a function of time, it descends that

$$\mathbf{v}'_i = \frac{d\mathbf{x}'_i}{dt} = \frac{\partial \mathbf{x}'_i}{\partial t} + \sum_{k=1}^{\tilde{N}} \frac{\partial \mathbf{x}'_i}{\partial q^k} \dot{q}^k,$$

where, in this case, $\dot{q}^k$ is the derivative of q^k with respect to the time variable. This entails that Equation (56) can be read as

$$\begin{aligned} \Delta T_i &= \frac{d}{dt} g(t, q^1, \dots, q^{\tilde{N}}), \\ g(t, q^1, \dots, q^{\tilde{N}}) &= \frac{m_i}{2} \|\mathbf{v}_{O'}\|^2 t + m_i \mathbf{x}'_i(t, q^1, \dots, q^{\tilde{N}}), \end{aligned} \quad (57)$$

where we have used that $\|\mathbf{v}_{O'}\|^2$ is a constant since the relative speed between O and O' is constant.

Potential functions If the underlying system has only conservative, active forces, then we can work directly with the scalar function $V(q^1, \dots, q^{\tilde{N}}, t)$ as we assumed in Theorem 23. Since the Lagrangian coordinates do not depend on the underlying, inertial reference system, it turns out that V is the same both from the viewpoint of O and of that of O' . Since, in Remark 10, we have hinted to the possibility of considering generalized potentials which are also dependent on the underlying velocities, we emphasize that the conclusion drawn

holds true also in this case. As a matter of fact, if we consider Equation (40) and if we focus the attention to the right hand side, we can infer that

$$\sum_{i=1}^N \mathbf{F}_i \frac{\partial \mathbf{x}_i}{\partial q^j} = \sum_{i=1}^N \mathbf{F}_i \frac{\partial \mathbf{x}'_i}{\partial q^j},$$

where $\mathbf{F}_i = \mathbf{F}_i(q^1, \dots, q^{\tilde{N}}, \dot{q}^1, \dots, \dot{q}^{\tilde{N}}, t)$ and where we have used implicitly that $\mathbf{x}'_i = \mathbf{x}_i + \mathbf{v}_O t$ where $\mathbf{v}_O$ is a constant velocity vector. In other words the Lagrangian component of the active forces does not depend on the underlying reference system.

Putting together these observations, we can conclude that, denoting by $\mathcal{L}$ and by $\mathcal{L}'$ the Lagrangian of the system computed in the two underlying reference systems,

$$\Delta \mathcal{L} = \mathcal{L}' - \mathcal{L} = \sum_{i=1}^N \Delta T_i.$$

In order to infer the contribution to the dynamics of this difference, we can use the Euler-Lagrange equations (43) to infer that

$$\frac{d}{dt} \frac{\partial \Delta \mathcal{L}}{\partial \dot{q}^k} - \frac{\partial \Delta \mathcal{L}}{\partial q^k} = \sum_{i=1}^N \left(\frac{d}{dt} \frac{\partial \Delta T_i}{\partial \dot{q}^k} - \frac{\partial \Delta T_i}{\partial q^k} \right). \quad \forall k = 1, \dots, \tilde{N}$$

At the same time, rewriting Equation (57) as

$$\frac{d}{dt} g(q^1, \dots, q^{\tilde{N}}, t) = \frac{\partial g}{\partial t} + \sum_{j=1}^{\tilde{N}} \frac{\partial g}{\partial q^j} \dot{q}^j,$$

it holds that

$$\frac{d}{dt} \frac{\partial \Delta T_i}{\partial \dot{q}^k} - \frac{\partial \Delta T_i}{\partial q^k} = \frac{d}{dt} \frac{\partial g}{\partial q^k} - \frac{\partial^2 g}{\partial t \partial q^k} - \sum_{j=1}^{\tilde{N}} \frac{\partial^2 g}{\partial q^k \partial q^j} \dot{q}^j = 0.$$

In other words the difference in the functional form of the Lagrangian arising when considering two different inertial reference systems is immaterial at the level of dynamics and the reason stems from the fact that, adding to a Lagrangian the total time derivative of any function which depends only on time and on the underlying Lagrangian coordinates does not contribute to Equation (40) and hence also to Equation (43). We summarize this fact in the following definition which extends the range of validity of this fact beyond the case dealt above, which served the purpose of justifying the physical relevance of the following statement.

Definition 26 (Equivalence between Lagrange functions). *Under the assumption of Section 3.1 and of Theorem 23, we say that two Lagrange functions $\mathcal{L} \equiv \mathcal{L}(q^1, \dots, q^{\tilde{N}}, \dot{q}^1, \dots, \dot{q}^{\tilde{N}}, t)$ and $\mathcal{L}' \equiv \mathcal{L}'(q^1, \dots, q^{\tilde{N}}, \dot{q}^1, \dots, \dot{q}^{\tilde{N}}, t)$ are **dynamically equivalent** if they differ by the total*

time derivative of a function of time and of the Lagrange coordinates, namely there exists $g \in C^\infty(\mathbb{R}^{\tilde{N}+1}; \mathbb{R})$ where $g \equiv g(q^1, \dots, q^{\tilde{N}}, t)$ and

$$\mathcal{L}' = \mathcal{L} + \frac{dg}{dt}.$$

Up to now, what I personally find quite annoying and disturbing is this lingering uneasiness related to this ambiguous use of the symbol $\dot{q}$ sometimes as an independent coordinate, sometimes as the derivative of the Lagrangian coordinate. In the following section we approach this mess... ehm... this quandary discussing another largely underrated paradigm shift on Lagrangian mechanics and on the Euler-Lagrange equations.

3.4 Hamilton's Stationary Action Principle

Have you ever watched the 2016 sci-fi movie *Arrival*? It is quite good in my opinion and it deserved a few nominations at the Golden Globe and at the Academy Awards. Yet it pales in comparison to the novel on which it is based. Written by Ted Chiang with the title *The Story of Your Life* is from my viewpoint among the best sci-fi texts I have ever read, second only to masterpieces like the *Foundation Series*. Why I am telling you this? Because the story revolves [SPOILERS AHEAD, skip after the figure] around a group of scientists who needs to communicate with an alien species and they try the universal language of physics, but they fail until they realize a bias that we have when trying to model a system.

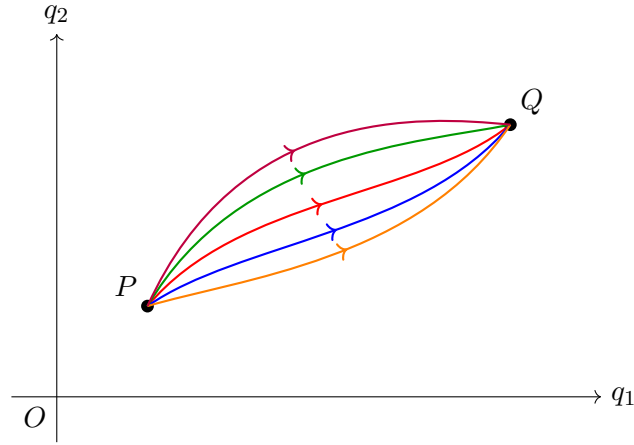


Figure 8: Pictorial representation of a system with two degrees of freedom and associated Lagrangian coordinates (q_1, q_2) and two points P, Q with different synchronous paths connecting them.

If you think closely about two standard courses of the first year of studies, Mechanics and Thermodynamics, there is a deep conceptual and structural difference on how they approach the investigation of a physical system:

- in mechanics, we always try to describe Nature as a collection of evolution laws. In other words, the idea is that we are able to measure and therefore know the constituents of a system at a given instant of time and our ultimate goal is to formulate *differential equations* which, resting upon this knowledge, allow us to predict the system trajectories, hence how all the components of the system evolve in time.
- in thermodynamics, on the contrary, once we individuate the macroscopic (extensive) variables of the system, we are not interested in establishing how they vary in time starting from a deterministic evolution equation, rather we consider all possible paths in the space of macroscopic variables with a fixed initial point, corresponding to our measurement at $t = t_0$ of these quantities and with a fixed end point corresponding to our target final configuration. Subsequently, among all possible paths, we establish a selection criterion which is subordinated to minimizing the change of a given quantity, such as energy, Gibbs free energy, entropy or enthalpy, depending on the system under scrutiny.

At first glance these two approaches to the analysis of a physical system might seem far apart from each other and quite detached both conceptually and from a practical viewpoint. Yet already at the beginning of the nineteenth century the studies of Hamilton proved that actually Lagrangian mechanics can be equivalently formulated in terms of a variational principle. This has far reaching consequences especially when dealing with systems with infinitely many degrees of freedom leading to the concept of *action* which is one of the building blocks of classical but, above all, of quantum field theory. Far more complicated is actually showing the converse, namely that a description in terms of a variational principle can be equivalently seen as stemming from evolution equations. In the realm of thermodynamics this question has been left open for decades until the pioneering work of Boltzmann and the onset of statistical mechanics, one of the pillars of modern theoretical physics. Here we will be only concerned with the “simple” implication. Yet, to this end, we need to introduce a few preliminary mathematical concepts and structures which go beyond those learned in calculus. Let us justify why:

- If we consider a function $f \in C^\infty(\mathbb{R}^n; \mathbb{R})$, $n \geq 1$, seeking its stationary points is tantamount to looking for

$$\bar{\mathbf{x}} \in \mathbb{R}^n \quad \text{such that} \quad \nabla f|_{\mathbf{x}=\bar{\mathbf{x}}} = 0,$$

where ∇ is the gradient.

- Under the assumptions at the beginning of Section 3.1, if we think of the Lagrangian in Equation (44) as the main structural ingredient of classical mechanics, we observe that

$$\mathcal{L} \equiv \mathcal{L}(q^1(t), \dots, q^{\tilde{N}}(t), \dot{q}^1(t), \dots, \dot{q}^{\tilde{N}}(t), t).$$

In the spirit of variational principles, we do not want to fix a priori a trajectory as a solution of a specific equations of motion, and therefore we are looking at

$$\mathcal{L} \equiv \mathcal{L}(\gamma(t), t),$$

where $\gamma \in C^\infty(\mathbb{R}; \mathbb{R}^{2\tilde{N}})$ is a curve such that, for each $t \in \mathbb{R}$,

$$\gamma(t) = (q^1(t), \dots, q^{\tilde{N}}(t), \dot{q}^1(t), \dots, \dot{q}^{\tilde{N}}(t)).$$

This creates a problem if we look for stationary points, as the gradient is meaningful only on functions with a finite dimensional domain.

The solution to this problem is quite simple and very pragmatic. We report it here adapting the example above although we will be using it in a slight more general context where the following definition can be adapted slavishly. The general theory of these structures requires a tool in functional analysis, known as theory of distributions. Be relieved or be sad, most likely, you will never hear about it. I am sad.

Definition 27 (Gateaux derivative for paths). *Let $I = [a, b] \subseteq \mathbb{R}$ be an interval and let $C^\infty(I; \mathbb{R}^n)$, $n \geq 1$ be the space of all smooth paths from I to $\mathbb{R}^n$. Given a functional $F : C^\infty(I; \mathbb{R}^n) \rightarrow \mathbb{R}$ we say that it is Fréchet derivable at $\gamma \in C^\infty(I; \mathbb{R}^n)$ along the direction $u \in C^\infty(I; \mathbb{R}^n)$ if there exists*

$$DF[\gamma; u] \doteq \left. \frac{dF[\gamma + \alpha u]}{d\alpha} \right|_{\alpha=0}. \quad (58)$$

*In addition we say that γ is a **stationary point** for F if*

$$\left. \frac{dF[\gamma + \alpha u]}{d\alpha} \right|_{\alpha=0} = 0,$$

for all $u \in C^\infty(I; \mathbb{R}^n)$.

We apply these concepts to a classical system abiding by the hypotheses of Section 3.1 and in particular of Theorem 23. Hence we can assume the existence of a Lagrange function $\mathcal{L}[\gamma(t), t] = \mathcal{L}(q^1(t), \dots, q^{\tilde{N}}(t), \dot{q}^1(t), \dots, \dot{q}^{\tilde{N}}(t), t)$. Following the rationale sketched above, we do not want to look for all possible admissible paths, but we want to fix the initial and final configurations. Heuristically speaking we know where all particles of the system start from and where we want them to end up, regardless of their velocities. In other words, fixing $Q_i, Q_f \in \Sigma_t \subseteq \mathbb{R}^{\tilde{N}}$, we look for the following class of paths, historically known as *synchronous trajectories*:

$$\Gamma_{if} \doteq \{\tilde{\gamma} \in C^\infty([a, b]; \Sigma_t) \mid \tilde{\gamma}(a) = Q_i \text{ and } \tilde{\gamma}(b) = Q_f\}. \quad (59)$$

Observe that the image is $\Sigma_t \subseteq \mathbb{R}^{\tilde{N}}$ since we consider the simple case that all trajectories lie within the domain of definition of the Lagrangian coordinates. This hypothesis could be removed, though it would need advanced tools of differential geometry and we omit going in this direction. Furthermore we are also imposing that each trajectory is covered in a span of time equal to $b - a$. This is not a restriction since we have always the freedom to perform an affine transformation of the time variable, *i.e.*, $t \mapsto t' = \lambda t + t_0$, $\lambda, t_0 \in \mathbb{R}$. At last we introduce

a functional, which is the first instance of one of the most important structures in modern theoretical physics, the **(Hamilton) action functional**

$$S_{if} : \Gamma_{if} \rightarrow \mathbb{R}, \quad \tilde{\gamma} \mapsto S_{if}[\tilde{\gamma}] \doteq \int_a^b \mathcal{L}[\tilde{\gamma}(t), t] dt = \int_a^b \mathcal{L}(q^1(t), \dots, q^{\tilde{N}}(t), \dot{q}^1(t), \dots, \dot{q}^{\tilde{N}}(t), t) dt, \quad (60)$$

where the subscripts i and f highlight the dependence on the initial and final configuration, while that on the Lagrangian is left implicit. The following result is here discussed as in [Mor20, Thm. 10.2].

Theorem 28 (Hamilton's Stationary Action Principle). *Given a Hamilton action functional as per Equation (60), a curve $\tilde{\gamma} \in \Gamma_{if}$, see Equation (59), is a solution of the Euler-Lagrange equations (43) if and only if it is a stationary point for S_{if} in the sense of Definition 27.*

Proof. On account of Equation (58), we consider for $\eta \in C^\infty([a, b]; \mathbb{R}^{\tilde{N}})$

$$\frac{d}{d\alpha} S_{if}[\tilde{\gamma} + \alpha\eta] = \frac{d}{d\alpha} \int_a^b \mathcal{L}[\tilde{\gamma}(t) + \alpha\eta(t), t] dt = \int_a^b \partial_\alpha \mathcal{L}[\tilde{\gamma}(t) + \alpha\eta(t)] dt,$$

where in the last equality we have used Leibnitz integral rule, while we have implicitly assumed that η is such that $\tilde{\gamma} + \alpha\eta \in \Gamma_{if}$. Denoting $\eta(t) = (\eta^1(t), \dots, \eta^{\tilde{N}}(t))$, we can write

$$S_{if}[\tilde{\gamma} + \alpha\eta] = \int_a^b \mathcal{L}(q^1(t) + \alpha\eta^1(t), \dots, q^{\tilde{N}}(t) + \alpha\eta^{\tilde{N}}(t), \dot{q}^1(t) + \alpha\dot{\eta}^1(t), \dots, \dot{q}^{\tilde{N}}(t) + \alpha\dot{\eta}^{\tilde{N}}(t), t) dt,$$

from which it descends that

$$\begin{aligned} \frac{d}{d\alpha} S_{if}[\tilde{\gamma} + \alpha\eta] &= \\ &= \int_a^b \partial_\alpha \mathcal{L}(q^1(t) + \alpha\eta^1(t), \dots, q^{\tilde{N}}(t) + \alpha\eta^{\tilde{N}}(t), \dot{q}^1(t) + \alpha\dot{\eta}^1(t), \dots, \dot{q}^{\tilde{N}}(t) + \alpha\dot{\eta}^{\tilde{N}}(t), t) dt, \end{aligned}$$

which, upon taking the evaluation as $\alpha = 0$, yields

$$\left. \frac{d}{d\alpha} S_{if}[\tilde{\gamma} + \alpha\eta] \right|_\alpha = \int_a^b \sum_{i=1}^{\tilde{N}} \left(\eta^i(t) \frac{\partial \mathcal{L}}{\partial q^i} + \dot{\eta}^i(t) \frac{\partial \mathcal{L}}{\partial \dot{q}^i} \right) dt,$$

where we have left implicit that $\mathcal{L}$ is a lone function of $(q^1, \dots, q^{\tilde{N}}, \dot{q}^1, \dots, \dot{q}^{\tilde{N}})$ after evaluation

at $\alpha = 0$. At this point we observe that

$$\begin{aligned} \int_a^b \dot{\eta}^i \frac{\partial \mathcal{L}}{\partial \dot{q}^i} dt &= \int_a^b \left(\frac{d}{dt} \left(\eta^i \frac{\partial \mathcal{L}}{\partial \dot{q}^i} \right) - \eta^i \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^i} \right) dt = \\ \eta^i(b) \frac{\partial \mathcal{L}}{\partial \dot{q}^i}(b) - \eta^i(a) \frac{\partial \mathcal{L}}{\partial \dot{q}^i}(a) - \int_a^b \eta^i \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^i} dt &= - \int_a^b \eta^i \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^i} dt, \end{aligned}$$

where, in the last identity, we have used that the condition $\tilde{\gamma} + \alpha\eta \in \Gamma_{if}$ entails that $\eta(a) = \eta(b) = 0$. Putting together our results, we end up with

$$\left. \frac{d}{d\alpha} S_{if}[\tilde{\gamma} + \alpha\eta] \right|_{\alpha} = \int_a^b \sum_{i=1}^{\tilde{N}} \eta^i(t) \left(\frac{\partial \mathcal{L}}{\partial q^i} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^i} \right) dt.$$

From this identity we, infer that, if $\tilde{\gamma}$ is a trajectory, solution of the Euler-Lagrange equations (43), then the left hand side is stationary for any choice of the components η^i . Hence it is a stationary point for the action functional. Conversely, if we assume $\tilde{\gamma}$ to be a stationary point, it means that the right hand side vanishes for all choices of $\eta^i(t)$. The constraint that $\tilde{\gamma} + \alpha\eta \in \Gamma_{if}$ does not allow us to draw the sought conclusion by invoking linear independence of the coefficients η^i . Yet, we can get to the sought result reasoning by contradiction. Let us assume that $\tilde{\gamma}(t)$ is a stationary point, but the Euler-Lagrange equations (43) are not satisfied. Then there must exist at least an index $\bar{i}$ in the allowed range such that

$$\frac{\partial \mathcal{L}}{\partial q^{\bar{i}}} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^{\bar{i}}} \neq 0.$$

Hence, we can assume that this quantity is of definite sign, say positive. If it changes sign, we can argue as follows: Take any $\bar{t} \in I$ where it is positive. Since it is a continuous function, there must exist a closed interval $[c, d] \ni \bar{t}$ where it is strictly positive. In addition if a curve is stationary on a compact interval $[a, b]$, it will be such also on all compact intervals $[c, d] \subseteq [a, b]$. In other words we can choose c, d such that there exists $C > 0$ for which

$$\frac{\partial \mathcal{L}}{\partial q^{\bar{i}}} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^{\bar{i}}} > C, \quad \forall t \in [c, d]$$

which is tantamount to saying that, on $[c, d]$

$$\left. \frac{d}{d\alpha} S_{if}[\tilde{\gamma} + \alpha\eta] \right|_{\alpha} > C(d - c),$$

which is a contradiction since it should vanish. If there is more than one component not abiding by the Euler-Lagrange equations, we can repeat iteratively the same reasoning for a finite number of steps, at most $\tilde{N}$. $\square$

Observe that Hamilton's stationary action principle theorem only guarantees that, working with a solution of the Euler-Lagrange equations in a given interval is tantamount to considering a stationary configuration of the action principle. This does not entail that such configuration exists nor that it is unique. This pertains to the realm of calculus and of existence and solutions of systems of second order ordinary differential equations with smooth coefficients, the topic of the second part of this course.

Leibnitz integral rule – compact case (Advanced Topic)

Let $K \subset \mathbb{R}^n$ be a compact set and, given $I = (a, b) \subset \mathbb{R}$, let $f : I \times K \rightarrow \mathbb{R}$ be such that

1. $f \in C^0(I \times K; \mathbb{R})$,
2. $\frac{\partial f(t, x)}{\partial t}$ exists and it is a continuous function on $I \times K$.

Then, for every $t_0 \in I$, it holds that $F(t) \doteq \int_K f(t, x) d^n x$ is a differentiable function on I such that

$$\left. \frac{d}{dt} \right|_{t=t_0} \int_K f(t, x) d^n x = \int_K \left. \frac{\partial f(t, x)}{\partial t} \right|_{t=t_0} d^n x.$$

Observe that, although in the statement of the theorem, the interval $I = (a, b)$, we can replace it with the closed counterpart $[a, b]$.

4 Conservation Laws

You mustn't be afraid to dream a little bigger, darling.

Inception

In the previous section we have discussed the building blocks of Lagrangian mechanics, particularly the Euler-Lagrange equations (43). Yet, in the first of the examples that we have discussed in Section 3.2, we ended up with a set of highly non linear, coupled, second order, ordinary differential equations, *cf.* Equation (49). This entails that, despite having established the laws of dynamics, finding an actual solution even after assigning initial conditions is highly non trivial. One may tempted to resort immediately to numerical analyses, which is certainly a viable solution as you will realize in the second half of this course. Yet, one should not be in a hurry. If we try to think back at Newtonian mechanics, this situation was a routine, but, to reduce the *complexity* of the set of differential equations, the first task when studying a physical system was to investigate its symmetries and its conserved quantities. Among these we recall the three main classes:

- **momenta** conservation,

- **angular momentum** conservation, be it total or the component along a direction,
- **energy** conservation.

Each of these carried tremendous physical significance, but, in addition, it helped in simplifying the complexity of the equations of motion, a feat which is desirable even if we have ultimately to resort to numerical schemes. Hence the goal of this whole chapter will be to investigate how symmetries and conserved quantities enter the game of Lagrangian mechanics.

4.1 Cyclic coordinates

We address the question raised in the previous discussion starting from setting the necessary language:

Nomenclature 2. We call **Lagrangian system** any system abiding by the hypotheses of Section 3.1. Unless otherwise specified, we assume that we are working on $\mathbb{R}^n$ with N point particles and $\tilde{N}$ Lagrangian coordinates.

Having established our playground, we make precise a rather intuitive concept.

Definition 29. Given a Lagrangian system, we call **first integral of motion/constant of motion/conserved quantity** any function

$$F : \Sigma_t \times \mathbb{R}^{\tilde{N}+1} \rightarrow \mathbb{R}, \quad (q^1, \dots, q^{\tilde{N}}, \dot{q}^1, \dots, \dot{q}^{\tilde{N}}, t) \mapsto F(q^1, \dots, q^{\tilde{N}}, \dot{q}^1, \dots, \dot{q}^{\tilde{N}}, t),$$

such that $\frac{dF}{dt} = 0$ on every smooth curve $\gamma : I \subseteq \mathbb{R} \rightarrow \Sigma_t$, solution of the Euler-Lagrange equation (43).

The different nomenclatures come from historical reasons and nowadays they are used mostly as synonyms although, in physics, the wording “first integral of motion” is almost unknown since it serves the purpose to highlight that the function F in Definition 29 helps in integrating out the equations of motion. Establishing sufficient conditions for the existence of conserved quantities is the main goal of our analysis, but, to this end we the Euler-Lagrange equations (43) help us in individuating a distinguished and physically relevant class.

Definition 30. Given a Lagrangian system and denoting by $\mathcal{L}$ the associated Lagrangian as per Equation (44), we say that a generalized coordinate q^k is **cyclic** if $\mathcal{L} \neq \mathcal{L}(q^k)$, that is if $\mathcal{L}$ does not depend explicitly on q^k , or, equivalently, if

$$\frac{\partial \mathcal{L}}{\partial q^k} = 0.$$

A direct investigation of Equation (43) unveils that

$$\frac{\partial \mathcal{L}}{\partial q^k} = 0 \iff \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^k} = 0, \tag{61}$$

which entails that we have individuated the first instance of conserved quantity. This

Definition 31. Given a Lagrangian system and denoting by $\mathcal{L}$ the associated Lagrangian as per Equation (44), we call **conjugate momentum** to the Lagrangian coordinate q^k the function $p^k : \Sigma_t \times \mathbb{R}^{\tilde{N}+1} \rightarrow \mathbb{R}$

$$p_k \equiv p^k(q^1, \dots, q^{\tilde{N}}, \dot{q}^1, \dots, \dot{q}^{\tilde{N}}, t) = \frac{\partial \mathcal{L}}{\partial \dot{q}^k}.$$

Observe that, Equation (61) can be equivalently rewritten as

$$\frac{\partial \mathcal{L}}{\partial q^k} = 0 \iff \frac{dp_k}{dt} = 0, \quad (62)$$

where, under the assumptions of Theorem 23, in particular that the potential depends only on the Lagrangian coordinates $\{q^k\}_{k=1, \dots, \tilde{N}}$ and not on the generalized velocities, it turns out that

$$p_k = \frac{\partial \mathcal{L}}{\partial \dot{q}^k} = \frac{\partial T}{\partial \dot{q}^k} = \sum_{i=1}^N \sum_{j=1}^{\tilde{N}} m_i \left(\frac{\mathbf{x}_i}{\partial q^j} \dot{q}^j \right) \cdot \frac{\partial \mathbf{x}_i}{\partial q^k} = \sum_{i=1}^N m_i \mathbf{v}_i \cdot \frac{\partial \mathbf{x}_i}{\partial q^k}, \quad (63)$$

where we have used Equation (35) first and Equation (31) in the last identity. In other words we have individuated a conserved quantity along the solutions of the Euler-Lagrange equations whenever a Lagrangian coordinate is cyclic. Equation (63) highlights that the conjugate momentum is somehow different from what we are used to from Newtonian mechanics, since it accounts also for a projection along the constraint hypersurface Σ_t due to the scalar product along $\frac{\partial \mathbf{x}_i}{\partial q^k}$. Hence the physical interpretation of p^k is not at all straightforward and it can depend on the case under scrutiny, particularly on the nature of the Lagrangian coordinate q^k . In the following we discuss the two main scenarios which can occur in physical model.

Translational Coordinate and Conservation of Linear Momentum

Let us now assume that we consider a Lagrangian system with a cyclic coordinate q^k and potential functions which do not depend on the generalized velocities. A first interesting scenario is when q^k is **translational**, namely it describes a translation along a direction in $\mathbb{R}^n$ individuated by a fixed vector $\mathbf{n}$. This is tantamount to saying that

$$\mathbf{x}_i(q^1, \dots, q^k + \Delta q^k, \dots, q^{\tilde{N}}, t) = \mathbf{x}_i(q^1, \dots, q^k, \dots, q^{\tilde{N}}, t) + \Delta q^k \mathbf{n}, \quad \forall i = 1, \dots, N$$

where $\delta q^k \in \mathbb{R}$ quantifies an arbitrary increment of the coordinate q^k . Accordingly, Equation (63) becomes

$$p_k = \sum_{i=1}^N m_i \mathbf{v}_i \cdot \frac{\partial \mathbf{x}_i}{\partial q^k} = \sum_{i=1}^N m_i \mathbf{v}_i \cdot \mathbf{n} = \mathbf{P} \cdot \mathbf{n},$$

where $\mathbf{P} \doteq \sum_{i=1}^N m_i \mathbf{v}_i$ is the total momentum of the whole system of n -particles. This entails that p_k coincides in this instance with *the component along $\mathbf{n}$ of the total momentum*.

Remark 12. The identification of translational coordinates allows to recover one of the main results that you learned in Newtonian mechanics, **the center of mass theorem**. As a matter of fact, suppose that we are working with a Lagrangian system and, to make contact, with standard mechanics, we assume $n = 3$, although this is completely irrelevant for the following reasoning. In addition we assume that the system is subject to the action only of internal and conservative forces, namely the associated potential function V depends only on $\mathbf{x}_i - \mathbf{x}_j$, $i, j = 1, \dots, N$, that is the difference between the position of two particles.

Under these assumptions, we can choose the Lagrangian coordinates to coincide with the Cartesian counterparts, individuating the position of each of the particles in $\mathbb{R}^3$. Yet, a smarter choice is the following: We fix (q^1, q^2, q^3) to coincide with the Cartesian coordinates of $\mathbf{x}_C$, the center of mass of the system, so that

$$\mathbf{x}_C = (x_C^1, x_C^2, x_C^3) = \sum_{j=1}^3 q^j \mathbf{e}_j,$$

where $\mathbf{e}_j = \frac{x_C^j}{\|\mathbf{x}_C\|}$. The remaining $3N - 3$ can be chosen arbitrarily, but, for definiteness, we let them coincide with the components of the vectors $\mathbf{x}_i - \mathbf{x}_1$, $i = 2, \dots, N$ and we denote them by $\{(q_i^1, q_i^2, q_i^3)\}_{i=2, \dots, N}$. Here the subscript 1 identifies an arbitrary point particle, chosen randomly among the N in the system. With these choices we can write the position vectors of each particle as follows

$$\mathbf{x}_1 = \mathbf{x}_1 - \mathbf{x}_C + \mathbf{x}_C \quad \text{and} \quad \mathbf{x}_i = \mathbf{x}_i - \mathbf{x}_1 + \mathbf{x}_1.$$

Since the position of the center of mass can be written as

$$\mathbf{x}_C = \sum_{i=1}^n \frac{m_i \mathbf{x}_i}{M}, \quad M = \sum_{j=1}^n m_j,$$

it descends that

$$\mathbf{x}_C - \mathbf{x}_1 = \sum_{i=1}^n \frac{m_i (\mathbf{x}_i - \mathbf{x}_1)}{M} = \sum_{i=2}^n \frac{m_i (\mathbf{x}_i - \mathbf{x}_1)}{M} = \sum_{i=2}^N \frac{m_i q_i^k}{M} \mathbf{e}_{k,i},$$

where $\mathbf{e}_{k,i} = \frac{x_i^k - x_1^k}{\|\mathbf{x}_i - \mathbf{x}_1\|}$ having denoted $\mathbf{x}_i = (x_i^1, x_i^2, x_i^3)$. In other words

$$\mathbf{x}_1 = \sum_{i=2}^N \frac{m_i q_i^k}{M} \mathbf{e}_{k,i} + \sum_{j=1}^3 q^j \mathbf{e}_j \quad \text{and} \quad \mathbf{x}_i = \sum_{j=2}^N \sum_{k=1}^3 \frac{m_j q_j^k}{M} \mathbf{e}_{k,j} + \sum_{j=1}^3 q^j \mathbf{e}_j + \sum_{k=1}^3 q_i^k \mathbf{e}_{k,i}. \quad (64)$$

By direct inspection, we can infer that a translation of one of the coordinates of the center of mass, e.g., $q^j \rightarrow q^j + \Delta q^j$, entails that

$$\mathbf{x}_i \mapsto \mathbf{x}_i + \Delta q^j \mathbf{e}_j.$$

In other words we have individuated translational coordinates. Since the potential does not depend on q^j , $j = 1, 2, 3$, per hypothesis and since the kinetic energy does not depend explicitly on the Lagrangian coordinates as one can infer directly from the time derivative of Equation (64), we can conclude that

$$\frac{\partial \mathcal{L}}{\partial q^j} = 0 \implies \frac{dp_j}{dt} = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}^j} \right) = 0, \quad j = 1, 2, 3.$$

Here $p_j = \sum_{k=1}^N m_k \dot{x}_k^j$ is the component of the total momentum of the system along the j -th direction. Since this is conserved for all possible choices of j , we conclude once more that the total momentum is conserved. This justifies also in the framework of Lagrangian mechanics that, in presence of a system with conservative, internal forces, it is convenient to choose a reference system with the origin coincident with the center of mass, since this is moving inertially with respect to the original Cartesian reference system.

Despite the relevance of Remark 12, cyclic coordinates are powerful tools to simplify the complexity of a concrete problem. Let us see it by means of another explicit example, as it could occur in any written exam.

Exercise: In $\mathbb{R}^2$ consider a Cartesian reference system as per Figure 9 and two point particles of same mass $m > 0$, denoted by P and Q . P is constrained to move along the horizontal axis. At P it is attached a rod of length $\ell > 0$ and negligible mass. The second particle Q is fixed to the other end of the rod. They are subject to an external gravitational force here modeled by $\mathbf{F}_z = mg\mathbf{e}_z$ where g is the standard gravitational acceleration while $\mathbf{e}_z$ is the unit vector pointing along the z -direction.

Solution: The system is Lagrangian since we have 2 point particles whose position vectors with respect to the standard Cartesian coordinates can be written as $\mathbf{x}_P = (x_P, y_P)$ and $\mathbf{x}_Q = (x_Q, y_Q)$. There are two ideal constraints as per Definition 20, the first being $y_P = 0$, while the second being $x_Q^2 + y_Q^2 - \ell^2 = 0$. This entails that the system has 2 degrees of freedom. One Lagrangian coordinate can be chosen to coincide with $x \equiv x_P$, while the geometry of the problem suggests to consider $\theta \in [0, 2\pi)$, the angle between the horizontal axis and the rod as the second Lagrangian coordinate. Accordingly

$$\mathbf{x}_P = (x, 0) \quad \text{and} \quad \mathbf{x}_Q = (x + \ell \cos \theta, \ell \sin \theta).$$

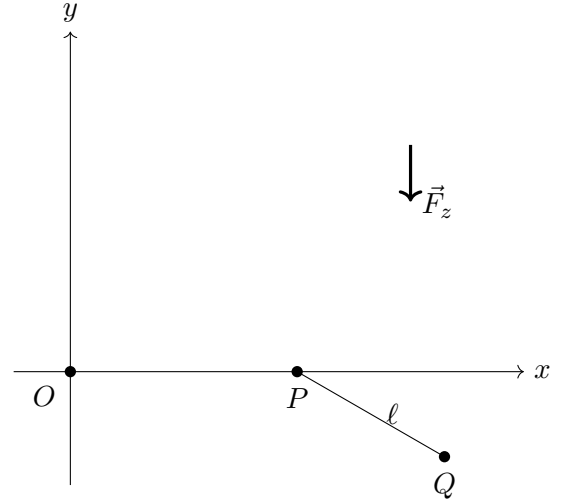


Figure 9: Two point particles connected by a rod of negligible mass and length ℓ , one constrained on the horizontal axis.

The velocity vectors become

$$\dot{\mathbf{x}}_P = (\dot{x}, 0) \quad \text{and} \quad \dot{\mathbf{x}}_Q(\dot{x} - \ell\dot{\theta} \sin \theta, \ell\dot{\theta} \cos \theta).$$

Accordingly the kinetic energy reads

$$T = \frac{m}{2} (\|\dot{\mathbf{x}}_P\|^2 + \|\dot{\mathbf{x}}_Q\|^2) = \frac{m}{2} (2\dot{x}^2 + \ell^2\dot{\theta}^2 - 2\ell\dot{x}\dot{\theta} \sin \theta),$$

while, following what already discussed in Section 3.2, $\vec{F}_z$ is a conservative force whose associated potential reads

$$V = mg(y_P + y_Q) = mg\ell \sin \theta.$$

In other words the Lagrangian is

$$\mathcal{L} = \frac{m}{2} (2\dot{x}^2 + \ell^2\dot{\theta}^2 - 2\ell\dot{x}\dot{\theta} \sin \theta) - mg\ell \sin \theta.$$

Observe that $\mathcal{L}$ does not depend explicitly on x and, hence, using Definition 30, we can infer that x is a cyclic coordinate and the associated momentum is conserved, namely

$$p_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = 2m\dot{x} - m\ell\dot{\theta} \sin \theta = p_x(0),$$

where $p_x(0)$ is a constant computed once initial condition on the position and on the velocities of P and Q are assigned. Observe that x is a *translational coordinate* but the associated momentum is not just $m\dot{x}$. It is the component of the total momentum along the x -direction as one could compute by looking at the momentum of the system at the center of mass. Going back to solving the exercise, if we also write down the Euler-Lagrange equation for θ , see Equation (43), we end up with

$$\begin{cases} 2m\dot{x} - m\ell\dot{\theta} \sin \theta = p_x(0) \\ m\ell^2 \ddot{\theta} - m\ell \ddot{x}_P \sin \theta - mg\ell \cos \theta = 0 \end{cases} ,$$

where we have replaced the equation of motion for x with the equivalent conservation of its associated momentum. Observe that, if we take the time derivative of the first equation, we end up with

$$2m \ddot{x}_P - m\ell \ddot{\theta} \sin \theta - m\ell \dot{\theta}^2 \cos \theta = 0,$$

which can be used in the second equation to get rid of $\ddot{x}$, hence obtaining the final system:

$$\begin{cases} \dot{x} = \frac{\ell}{2} \dot{\theta} \sin \theta - \frac{p_x(0)}{2} \\ \ddot{\theta} = \frac{\cos \theta (2g + \ell \dot{\theta}^2 \sin \theta)}{\ell (1 + \cos^2 \theta)} \end{cases} .$$

Hence the use of a conserved quantity has greatly reduced the complexity of the dynamical system, since we are left with solving one single second order ordinary differential equation, that for θ . Once this function is known explicitly, $x(t)$ can be reconstructed integrating the first one with respect to the time variable. Alas, we are still stuck with the big problem that the ensuing ordinary differential equation is highly non-linear and we are not aware at the moment of any method to construct its solutions.

Rotational Coordinate and Conservation of Angular Momentum

Let us now assume that we consider a Lagrangian system with a cyclic coordinate q^k and potential functions which do not depend on the generalized velocities. A second interesting scenario is when q^k is **rotational**, namely it describes a rotation along a given axis in $\mathbb{R}^n$ individuated by a fixed vector $\mathbf{n}$. Working once more with $n = 3$ for simplicity, though the discussion extends to generic values of $n \geq 2$, this entails that

$$\mathbf{x}_i(q^1, \dots, q^k + \Delta q^k, \dots, q^{\tilde{N}}, t) = R_{\mathbf{n}}(\Delta q^k) \mathbf{x}_i(q^1, \dots, q^k, \dots, q^{\tilde{N}}, t), \quad \forall i = 1, \dots, N$$

where $R_{\mathbf{n}}(\Delta q^k)$ is a rotation matrix along the $\mathbf{n}$ -axis of an angle equal to Δq^k . Observe that this is a *rigid translation* in the sense that all particles have been rotated simultaneously of the same angle along the same direction. Using Equation (63), we can infer that

$$p_k = \sum_{i=1}^N m_i \mathbf{v}_i \cdot \frac{\partial \mathbf{x}_i}{\partial q^k} = \sum_{i=1}^N m_i \mathbf{v}_i \cdot (\mathbf{n} \wedge \mathbf{x}_i),$$

where $\wedge$ denotes the vector product, where we have used the results of Appendix A according to which

$$\frac{d}{d\theta} (R_{\mathbf{n}}(\theta) \mathbf{x}) = \mathbf{n} \wedge (R_{\mathbf{n}}(\theta) \mathbf{x}) \quad \text{and} \quad R_{\mathbf{n}}(0) = \mathbb{I},$$

$\mathbb{I}$ being the identity matrix acting on $\mathbb{R}^3$. Recalling the *scalar triple product* according to which, given $\mathbf{v}, \mathbf{w}, \mathbf{z} \in \mathbb{R}^3$, it holds that

$$\mathbf{v} \cdot (\mathbf{w} \wedge \mathbf{z}) = \mathbf{w} \cdot (\mathbf{z} \wedge \mathbf{v}) = \mathbf{z} \cdot (\mathbf{v} \wedge \mathbf{w}), \quad (65)$$

we can rewrite the conserved generalized momentum as

$$p_k = \mathbf{n} \cdot \sum_{i=1}^N m_i (\mathbf{x}_i \wedge \mathbf{v}_i) = \mathbf{n} \cdot \mathbf{L}, \quad (66)$$

where $\mathbf{L}$ is the total angular momentum of the system. In other words, if we have a *cyclic rotational coordinate* codifying rigid rotations along an axis $\mathbf{n}$, then the associated conjugate momentum expresses the *conservation of the component of the angular momentum along $\mathbf{n}$* .

Example 32. We have already encountered an instance of a rotational cyclic coordinate. Let us give a fresher look at the first example in Section 3.2. In particular the system had 2 degrees of freedom and the Lagrangian in Equation (47) read

$$\mathcal{L}(\theta, \dot{\theta}, \varphi, \dot{\varphi}) = \frac{mR^2}{2} \left(\dot{\theta}^2(t) + \sin^2 \theta(t) \dot{\varphi}^2(t) \right) - mgR \cos \theta(t).$$

We can now observe that φ is a cyclic coordinate and it goes without saying that it is rotational since a shift of φ by an angle φ_0 is tantamount to a rotation by that angle around the z -axis. It also means that we can use Equation (61) to infer that

$$p_{\varphi} = mR^2 \sin^2 \theta \dot{\varphi} = p_0,$$

where p_0 is a constant to be computed once initial conditions on the position and on the generalized velocities of the Lagrangian coordinates are assigned. Hence we can rewrite the equations of motion ruling the dynamics as

$$\begin{cases} \ddot{\theta} - \sin \theta \cos \theta \dot{\varphi}^2 - \frac{g}{R} \sin \theta = 0 \\ mR^2 \sin^2 \theta \dot{\varphi} = p_0 \end{cases}, \quad (67)$$

This system can therefore be simplified eliminating the dependence on $\dot{\varphi}$ on the first equation, namely

$$\begin{cases} \ddot{\theta} - \frac{p_0^2 \cos \theta}{mR^3 \sin^3 \theta} - \frac{g}{R} \sin \theta = 0 \\ mR^2 \sin^2 \theta \dot{\varphi} = p_0 \end{cases}, \quad (68)$$

Once more we used a conserved quantity to reduce the complexity of a dynamical system. We are left with solving a second order, ordinary differential equation for θ and, once done, we can recover $\varphi(t)$ integrating the second equation with respect to time. Yet, we are back with the same old problem, namely the differential equation is highly nonlinear and, at the moment, we have no tools to solve it.

Remark 13. *The reader should keep in mind that, while most of the results in Lagrangian mechanics are independent from the chosen coordinates, see, e.g., Proposition 24, this is not the case when discussing cyclic coordinates. As a matter of fact, if a given Lagrangian is not independent from any of the coordinates $\{q^i\}_{i=1,\dots,\tilde{N}}$, this does not mean that this statement applies to any other choice $\{q'^k\}_{k=1,\dots,\tilde{N}}$.*

To clarify the content of this remark, which might be overlooked as a triviality, let us consider a concrete example which we have already partly investigated.

Example 33. Consider the second example discussed in Section 3.2, namely two particles connected by a spring. We recall that with the natural choice of Lagrange coordinates, coinciding with the Cartesian ones, the Lagrangian in Equation (50) is

$$\mathcal{L} = \frac{m}{2} (\dot{x}_P^2(t) + \dot{y}_P^2(t) + \dot{y}_Q^2(t)) + \frac{\kappa}{2} (x_P^2(t) + (y_Q(t) - y_P(t))^2).$$

None among the coordinates is cyclic since the Lagrangian is manifestly dependent on all of them. Yet, consider the following change of coordinates:

$$(x_P, y_P, y_Q) \rightarrow (x, w, z) \text{ where } \begin{cases} x = x_P \\ w = y_Q - y_P \\ z = y_Q + y_P \end{cases} \iff \begin{cases} x_P = x \\ y_P = \frac{z-w}{2} \\ y_Q = \frac{z+w}{2} \end{cases}.$$

We can write the Lagrangian in the new coordinates observing first of all that the last identity entails that

$$\begin{cases} \dot{x}_P = \dot{x} \\ \dot{y}_P = \frac{\dot{z}-\dot{w}}{2} \\ \dot{y}_Q = \frac{\dot{z}+\dot{w}}{2} \end{cases}.$$

Hence we end up with

$$\mathcal{L} = \frac{m}{2} \left(\dot{x}^2 + \frac{\dot{z}^2}{2} + \frac{\dot{w}^2}{2} \right) + \frac{\kappa}{2} (x^2 + w^2),$$

which is independent on z . Therefore this is a cyclic coordinate, translational in nature, whose momentum

$$p_z = m\dot{z} = p_{z,0},$$

is constant along the trajectories solution of the Euler-Lagrange equations (43). This is in perfect agreement with Equation (52) and it has a clear physical interpretation comparing this system with Remark 12. More precisely, the center of mass is not moving inertially with respect to the origin of our reference system O since the conservative forces are not internal along all directions. More precisely the component along the y -direction of the elastic force is internal in the sense that it depends only on $y_P - y_Q$, while, that along the x -direction, not since only x_P appears. Hence we can expect a weaker version of the center of mass theorem to hold true, namely only the y -component of the center of mass is moving with constant speed along the vertical direction. This is exactly the information that the cyclic coordinate z is encoding.

Lesson Learned: In a Lagrangian system, all generalized coordinates are equally valid, in the sense that all of them can be used to construct the Lagrange function and the equations of motion. Yet, not all of them, are a smart choice, some might better than other since they reduce manifestly the complexity of the problem. Alas, there is no algorithmic way to find them and, even the abstract question whether a distinguished choice of such a system of coordinates exist, is open at this point. This goes beyond the immediate scopes of these notes but let me tell you that a positive answer to this last question exists and it goes under the name of **Hamilton-Jacobi** theory which is of interest also for its deep rooted connections with Fermi's argument to derive the Schrödinger equation in quantum mechanics. An interested reader might wish to learn it from [FM06, Chap. 11].

4.2 Noether's Theorem

The outcome of the analysis in Section 4.1 is quite satisfactory since it lead to the identification of a notion of conserved quantities in Lagrangian mechanics, associating to it cyclic coordinates, whose physical interpretation become clear once the details of the model under investigation are given. Yet, we cannot be really satisfied with the outcome of our analysis since at least two, apparently very different, open questions are left open:

1. Barring the question of finding the most appropriate set of generalized coordinates, conserved quantities are up to now associated to translational or rotational cyclic coordinates. These are the transliteration in Lagrangian mechanics of the well-known conserved quantities, well established in Newton's theory: the components of linear and angular momenta. Yet one is left wondering whether there might exist other conserved quantities which are eluding our grasp, leading to a further reduction in the complexity of the dynamics of the systems under investigation.

2. When discussing both translational or rotational cyclic coordinates, there is always an underlying dichotomy which is left implicit. On the one hand, the notion of being cyclic is encoded in a differential equation of the form $\frac{\partial \mathcal{L}}{\partial q^k} = 0$, where $\mathcal{L} = \mathcal{L}(q^1, \dots, q^{\tilde{N}}, \dot{q}^1, \dots, \dot{q}^{\tilde{N}}, t)$. This is a local identity in the sense that, to establish whether at a given point the derivative of a function, either total or partial, vanishes, only the knowledge of the function in an arbitrarily small neighbourhood of that point is necessary. This is a *local statement* and it encompasses what in physics is known as a **local symmetry** of the system¹. On the other hand, being cyclic is also equivalent to saying that

$$\mathcal{L}(q^1, \dots, q^k + \Delta q^k, \dots, q^{\tilde{N}}, \dot{q}^1, \dots, \dot{q}^{\tilde{N}}, t) = \mathcal{L}(q^1, \dots, q^k, \dots, q^{\tilde{N}}, \dot{q}^1, \dots, \dot{q}^{\tilde{N}}, t),$$

where Δq^k is arbitrary and only constrained to the requirement that $q^k + \Delta q^k$ lies in the domain of definition of the Lagrangian coordinate. This is a *global statement* and it encompasses what in physics is known as a **global symmetry** of the system.

While symmetries and conserved quantities were known in physics since the early days of Lagrangian and even of Newtonian mechanics, a systematic study of their connection had to wait the first two decades on the twentieth century, culminating in the work of Emmy Noether. The results that we shall discuss in the following see their application in Lagrangian mechanics, but their true power appears only when one applies them to systems with infinitely many degrees of freedom. What is known as Noether's theorem is one of the pillars of modern, classical and quantum field theory, including among these general relativity. For this reason, it is without fear of being proven wrong that I claim that this is the most important topic of the whole course.

Let us consider a Lagrangian system – see Nomenclature 2 – and let us introduce a more compact notation for the underlying Lagrangian coordinates and for their associated velocities:

$$(q, \dot{q}, t) \equiv (q^1, \dots, q^{\tilde{N}}, \dot{q}^1, \dots, \dot{q}^{\tilde{N}}, t),$$

where the subscripts will be always henceforth avoided unless possible confusion might arise or it becomes necessary to highlight the role of one or more specific coordinates/velocities. We introduce in the following a distinguished class of coordinate transformations which are well suited to codify the idea of symmetry of a system. In order to catch heuristically the idea behind what is coming next, let us think for a second about two systems which we expect to abide by our intuitive idea of being symmetric:

1. a point particle in $\mathbb{R}^3$ constrained to move on the surface of fixed radius $R > 0$,
2. a point particle in $\mathbb{R}^3$ constrained to move on a plane, which we take without loss of generality as the locus $z = 0$.

¹The reader is warned that, when switching from systems with finite degrees of freedom to those with an infinite number, such as electrodynamics, the notion of local symmetry is generalized to include a larger class of symmetries known as *gauge symmetries*, which exceed the scopes of this course, but are well rooted in classical mechanics.

In absence of external forces or, equivalently, if we think of the kinematically allowed configurations, we expect the first system to be invariant with respect to rotations of $\mathbb{S}^2$, while the second with respect to translations tangent to the plane $z = 0$. In other words, given a kinematically allowed configuration every rotation on $\mathbb{S}^2$ (*resp.* translation along the plane) yields another admissible configuration for the first (*resp.* second) system. Even more strongly given one single kinematically allowed configuration, all others can be reached by one of these transformations. These two viewpoints are not equivalent, but their difference shall not play a role in our discussion. We just warn the reader that a subtle difference exists. This geometric perspective changes when external forces are present, since these the associated potentials which enter the Lagrangian function might not be compatible with them. For instance, if we consider $V(z) = mgz$ as in many of the examples discussed in these notes, we realize by direct inspection that a rotation along $\mathbb{S}^2$ does not leave V invariant, hence affecting the dynamics, while a translation along the plane $z = 0$ leaves $V(z)$ fixed. In the first case we do not expect the dynamics encoded in the Euler-Lagrange equations to be invariant under rotations, while, in the second case, we expect the opposite. If we try to extract from these model scenarios the mathematical building blocks, we recognize a key ingredient:

- a collection of transformations of the configurations of the underlying system which admits the structure of a group, *e.g.*, translations or rotations. This encodes the idea of repeatability, namely the action of two consecutive admissible transformations is itself a single admissible transformation (*composition law* of group elements) and every admissible transformation can be reverted back by an opposite transformation (*existence of an inverse element*).

As already mentioned a few times in these notes, we do not want to delve into establishing the minimal class of regularity admissible for all the functions appearing in these notes and therefore we consider them to be always smooth. Hence, in many textbooks, especially [Mor20], a more detailed analysis can be found, but, being for many readers the first time that they are exposed to these concepts, we prefer to avoid too many technicalities (yup, trust me, this is by far non technical).

Definition 34. *Given a Lagrangian system and the coordinate $(q, t) \equiv (q^1, \dots, q^{\tilde{N}}, t)$ on Σ_t , we call **one-parameter group of diffeomorphisms/transformations** any smooth change of coordinates $q'_\epsilon \equiv (q'^1_\epsilon, \dots, q'^{\tilde{N}}_\epsilon)$ such that*

$$q'^k_\epsilon \equiv q'^k(\epsilon, q, t) \equiv q'^k(\epsilon, q^1, \dots, q^{\tilde{N}}, t), \quad \forall k = 1, \dots, \tilde{N}$$

where $\epsilon \in \mathbb{R}$ and where, for all $k = 1, \dots, \tilde{N}$

$$\begin{cases} q'^k_0 = q^k \\ q'^k(\epsilon, q'^1_{\epsilon'}, \dots, q'^{\tilde{N}}_{\epsilon'}, t) = q'^k_{\epsilon+\epsilon'} \end{cases} \quad (69)$$

Observe that, in some textbooks, *e.g.* [Mor20], ϵ and consequently ϵ' are required only to exist as elements lying in an interval $I_\alpha \doteq (-\alpha, \alpha)$, $\alpha > 0$. This is sometimes referred as a *local*

one-parameter group of diffeomorphism and it is especially useful when working with non trivial geometries different from $\mathbb{R}^n$ or where the domain of the underlying Lagrangian coordinates is the whole real axis. Although we loose the structure of group with respect to the addition since $\epsilon + \epsilon'$ might not lie in I_α , we gain in generality and the following discussions will not be affected by these limitations on the range of ϵ . We can now clarify what do we mean by a Lagrangian system invariant under a (geometric symmetry).

Definition 35. *Given a Lagrangian system and a Lagrange function $\mathcal{L}$, we call it **invariant** under a one-parameter group of transformations if*

$$\left. \frac{\partial \mathcal{L}(q'_\epsilon, \dot{q}'_\epsilon, t)}{\partial \epsilon} \right|_{\epsilon=0} = 0, \quad (70)$$

where $\dot{q}'_\epsilon \equiv (\dot{q}'_\epsilon^1, \dots, \dot{q}'_\epsilon^{\tilde{N}})$ and, for all $k = 1, \dots, \tilde{N}$,

$$\dot{q}'_\epsilon^k = \frac{\partial q'^k}{\partial t}(\epsilon, q, t) + \sum_{j=1}^{\tilde{N}} \frac{\partial q'^k}{\partial q^j}(\epsilon, q, t) \dot{q}^j.$$

In this case we say that the coordinate transformation q'_ϵ is a **symmetry group** for the Lagrangian system.

A slightly weaker form of this definition plays a distinguished role in what is coming next.

Definition 36. *Given a Lagrangian system and a Lagrange function $\mathcal{L}$, we call it **weakly invariant** under a one-parameter group of transformations if there exists a function $G \in C^\infty(\Sigma_t \times \mathbb{R}^{\tilde{N}+1}; \mathbb{R})$ such that, for all $(q, \dot{q}, t) \in \Sigma_t \times \mathbb{R}^{\tilde{N}+1}$,*

$$\left. \frac{\partial \mathcal{L}(q'_\epsilon, \dot{q}'_\epsilon, t)}{\partial \epsilon} \right|_{\epsilon=0} = G(q, \dot{q}, t), \quad (71)$$

where there exists $g \in C^\infty(\Sigma_t \times \mathbb{R})$ such that, for all $(q, t) \in \Sigma_t \times \mathbb{R}$

$$G(q, \dot{q}, t) = \frac{\partial g}{\partial t}(q, t) + \sum_{k=1}^{\tilde{N}} \frac{\partial g}{\partial q^k}(q, t) \dot{q}^k.$$

In this case we say that the coordinate transformation q'_ϵ is a **weak symmetry group** for the Lagrangian system.

We are now in a position to state and to prove the celebrated Noether theorem which connects symmetries of a Lagrangian system to conserved quantities.

Theorem 37 (Noether's Theorem). *Given a Lagrangian system, which is invariant under the action of a (local) one-parameter group of diffeomorphism as per Definition 34, then*

1. if the Lagrangian satisfies Equation (70) for all configurations $(q, \dot{q}, t)$ which are solutions of the Euler-Lagrange equation (43), then the function

$$I(q, \dot{q}, t) = \sum_{k=1}^{\tilde{N}} \frac{\partial \mathcal{L}}{\partial \dot{q}^k} \frac{\partial q_\epsilon'^k}{\partial \epsilon} \Big|_{\epsilon=0} (t, q, \dot{q}), \quad (72)$$

is a constant of motion known as **Noether invariant**.

2. if the Lagrangian satisfies Equation (71) for all configurations $(q, \dot{q}, t)$ which are solutions of the Euler-Lagrange equation (43), then the function

$$I_g(q, \dot{q}, t) = \sum_{k=1}^{\tilde{N}} \frac{\partial \mathcal{L}}{\partial \dot{q}^k} \frac{\partial q_\epsilon'^k}{\partial \epsilon} \Big|_{\epsilon=0} (t, q, \dot{q}) - g(t, q), \quad (73)$$

is a constant of motion.

Proof. We prove only item 1., the second case being analogous. Under the given assumptions, it holds that

$$\frac{\partial \mathcal{L}(q'_\epsilon, \dot{q}'_\epsilon, t)}{\partial \epsilon} = \sum_{k=1}^{\tilde{N}} \left(\frac{\partial \mathcal{L}(q'_\epsilon, \dot{q}'_\epsilon, t)}{\partial q'^k} \frac{\partial q'^k}{\partial \epsilon} + \frac{\partial \mathcal{L}(q'_\epsilon, \dot{q}'_\epsilon, t)}{\partial \dot{q}'^k} \frac{\partial \dot{q}'^k}{\partial \epsilon} \right).$$

Evaluating at $\epsilon = 0$ and using that, according to Equation (69), $q'_0 = q$, it turns out that

$$\frac{\partial \mathcal{L}(t, q'_\epsilon, \dot{q}'_\epsilon)}{\partial \epsilon} \Big|_{\epsilon=0} = \sum_{k=1}^{\tilde{N}} \left(\frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial q^k} \frac{\partial q'^k}{\partial \epsilon} \Big|_{\epsilon=0} + \frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial \dot{q}^k} \frac{\partial \dot{q}'^k}{\partial \epsilon} \Big|_{\epsilon=0} \right).$$

Focusing on the last term we observe that, using a variation of Schwarz theorem,

$$\frac{\partial \dot{q}'^k}{\partial \epsilon} \Big|_{\epsilon=0} = \frac{d}{dt} \frac{\partial q'^k}{\partial \epsilon} \Big|_{\epsilon=0},$$

which can be inserted in the preceding expression to obtain

$$\frac{\partial \mathcal{L}(t, q'_\epsilon, \dot{q}'_\epsilon)}{\partial \epsilon} \Big|_{\epsilon=0} = \sum_{k=1}^{\tilde{N}} \left(\frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial q^k} \frac{\partial q'^k}{\partial \epsilon} \Big|_{\epsilon=0} + \frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial \dot{q}^k} \frac{d}{dt} \frac{\partial q'^k}{\partial \epsilon} \Big|_{\epsilon=0} \right).$$

Recalling that we consider only configurations which are solution of the Euler-Lagrange equation (43), it turns out that

$$\frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial q^k} = \frac{d}{dt} \frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial \dot{q}^k},$$

which entails that the previous identity can be rewritten as

$$\begin{aligned} \left. \frac{\partial \mathcal{L}(t, q'_\epsilon, \dot{q}'_\epsilon)}{\partial \epsilon} \right|_{\epsilon=0} &= \sum_{k=1}^{\tilde{N}} \left(\left. \frac{d}{dt} \frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial \dot{q}^k} \frac{\partial q'^k}{\partial \epsilon} \right|_{\epsilon=0} + \left. \frac{\partial \mathcal{L}(q, \dot{q}, t)}{\partial \dot{q}^k} \frac{d}{dt} \frac{\partial q'^k}{\partial \epsilon} \right|_{\epsilon=0} \right) = \\ &= \frac{d}{dt} \left(\sum_{k=1}^{\tilde{N}} \left. \frac{\partial \mathcal{L}}{\partial \dot{q}^k} \frac{\partial q'^k}{\partial \epsilon} \right|_{\epsilon=0} (t, q, \dot{q}) \right) = \frac{dI(q, \dot{q}, t)}{dt}. \end{aligned}$$

In other words, if the left hand side vanishes, then $I(q, \dot{q}, t)$ is constant in time, provided that q is a solution of the Euler-Lagrange equations. $\square$

Remark 14. *An immediate consequence of this result is that the conservation of the conjugate momentum of a cyclic coordinate is a special case of Noether's Theorem 37. As a matter of fact suppose that we have a Lagrangian system such that the coordinate q^k is cyclic. We can define a one-parameter group of diffeomorphisms q'_ϵ where $q'^k_\epsilon = q^k + \epsilon$, where $\epsilon \in I \subset \mathbb{R}$, while $q'^j_\epsilon = q^j$ for all $\epsilon \in I$ and for all $j \neq k$. Here I is chosen so to include 0 and so that $q^k + \epsilon$ lies in the domain of definition of q^k depending on the case under scrutiny.*

In addition, being q^k cyclic, it turns out that we have identifies a symmetry group for the underlying Lagrangian since $\dot{q}' = \dot{q}$ and therefore, per construction Equation (70) holds true. As a consequence, we can apply Noether's theorem 37 which entails that it is a constant of motion

$$I(q, \dot{q}, t) = \sum_{j=1}^{\tilde{N}} \left. \frac{\partial \mathcal{L}}{\partial \dot{q}^j} \frac{\partial q'^j}{\partial \epsilon} \right|_{\epsilon=0} (t, q, \dot{q}) = \frac{\partial \mathcal{L}(t, q, \dot{q})}{\partial \dot{q}^j} = p_j.$$

This is nothing but the conjugate momentum as per Definition 31.

Before commenting further on the usefulness of Noether's theorem beyond its mere conceptual aspects, we highlight one point. At the beginning of Section 4, we have commented that, in Newtonian mechanics, reducing the complexity of the dynamics is often achieved using conservation laws. Among those we listed, energy plays a prominent role, but the reader should keep in mind what follows. As far as I am concerned, claiming the opposite is a very serious mistake:

Energy conservation in Lagrangian mechanics is not a consequence of Noether's theorem!

4.3 Jacobi Integral and Conservation of Energy

It is quite remarkable that energy conservation in the framework of Lagrangian mechanics pre-dates Noether's theorem by almost a century. Before commenting how is it possible and why I am so strict on ascribing this result to a completely different conceptual realm, let us state it and prove it, since this will also set the ground for our later comments. Even in the following we shall stick to Nomenclature 2.

Theorem 38 (Jacobi's Theorem). *Given a Lagrangian system, we call **Hamiltonian (function)** of the system*

$$\mathcal{H} \doteq \sum_{k=1}^{\tilde{N}} \dot{q}^k \frac{\partial \mathcal{L}}{\partial \dot{q}^k} - \mathcal{L}. \quad (74)$$

In addition, it holds that

1. *for every curve $\gamma(t) = (q^1(t), \dots, q^{\tilde{N}}(t), \dot{q}^1(t), \dots, \dot{q}^{\tilde{N}}(t))$ which is a solution of the Euler-Lagrange equations (43), thereon it holds that*

$$\frac{d\mathcal{H}}{dt} = -\frac{\partial \mathcal{L}}{\partial t}.$$

2. *if, in addition, $\mathcal{L} \neq \mathcal{L}(t)$, that is the Lagrangian is not explicitly dependent on time, then $\mathcal{H}$ is a conserved quantity, also known as **Jacobi integral**².*

Proof. The proof is by brute force, namely, starting from Equation (74), we compute using the Leibnitz rule

$$\frac{d\mathcal{H}}{dt} = \sum_{k=1}^{\tilde{N}} \left(\frac{d\dot{q}^k}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^k} + \dot{q}^k \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}^k} \right) \right) - \frac{d\mathcal{L}}{dt} = \sum_{k=1}^{\tilde{N}} \left(\frac{d\dot{q}^k}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^k} + \dot{q}^k \frac{\partial \mathcal{L}}{\partial q^k} \right) - \frac{d\mathcal{L}}{dt}$$

Since $\mathcal{L} = \mathcal{L}(q^1, \dots, q^{\tilde{N}}, \dot{q}^1, \dots, \dot{q}^{\tilde{N}}, t)$, it turns out that

$$\frac{d\mathcal{L}}{dt} = \sum_{k=1}^{\tilde{N}} \left(\frac{d\dot{q}^k}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}^k} + \dot{q}^k \frac{\partial \mathcal{L}}{\partial q^k} \right) + \frac{\partial \mathcal{L}}{\partial t}.$$

This entails the sought identity and item 1.. The second point is a direct consequence of the first one. $\square$

Theorem 38 seems to suggest that we have found a conserved quantity which is different from the one we were looking for, but actually this is not the case. It is a generalization as this lemma clarifies once and for all.

Lemma 39. *Under the assumptions of Theorem 38, if all the underlying constraints are time independent and, if we are working with Lagrangian coordinates subordinated to the underlying inertial reference system, then the Hamiltonian function coincides with the **total mechanic energy** $\mathcal{E}(t)$ of the system, namely*

$$\mathcal{H}(t) = \mathcal{E}(t) = T(q, \dot{q}) + V(q, \dot{q}),$$

where here we use the same notation as in Section 4.2 for denoting the collection of all Lagrangian coordinates and of all the associated velocities.

²Here we make contact with the nomenclature of [Mor20], although the reader should be aware that this terminology is applied in most of the literature to the unique conserved quantity of the reduced three-body problem. In the general setting one refers to the Hamiltonian as a conserved quantity.

Proof. Under the given assumption, Equation (35) entails that, for each particle of the system the kinetic energy

$$T = T_2 = \sum_{i=1}^N \frac{m_i}{2} \sum_{h,j=1}^{\tilde{N}} a_{hj} \frac{dq^h}{dt} \frac{dq^j}{dt},$$

where

$$a_{hj} \equiv a_{hj}(q_1, \dots, q_{\tilde{N}}, t) \doteq \frac{\partial \mathbf{x}_i}{\partial q^h} \cdot \frac{\partial \mathbf{x}_i}{\partial q^j}.$$

As a consequence

$$\mathcal{L} = \sum_{i=1}^N \sum_{h,j=1}^{\tilde{N}} \frac{m_i}{2} a_{hj} \frac{dq^h}{dt} \frac{dq^j}{dt} - V(q^1, \dots, q^{\tilde{N}}).$$

Replacing this expression in Equation (74) yields

$$\mathcal{H} = \sum_{i=1}^N \sum_{h,j=1}^{\tilde{N}} \frac{m_i}{2} a_{hj} \frac{dq^h}{dt} \frac{dq^j}{dt} + V(q^1, \dots, q^{\tilde{N}}),$$

which is exactly the sought conclusion. $\square$

Time vs Lagrangian Coordinates

As already mentioned, the Jacobi integral and Noether invariants are conceptually quite different. Although they are both conservation laws, the second one address only symmetries of the system with respect to changes of the underlying Lagrangian coordinates as codified by Definition 35. At the same time Theorem 38 can be seen as a by product of the system to be invariant under time translations $t \mapsto t' = t + a$, $a \in \mathbb{R}$.

From this viewpoint one might say that there is no difference whatsoever between the two cases. This is not at all true and the origin of the discrepancy lies in the very foundations of classical Newtonian mechanics. Most notably, time is not considered as a coordinate on par with the others, say $x_1, \dots, x_n$, subordinated to the choice of a reference frame. Actually, to be precise, time is just an **affine parameter**. This is not a mere mathematical statement, but, it tell us that, in the setting of Lagrangian mechanics, our only freedom to redefine time is via a transformation of the form

$$t \mapsto t' = \lambda t + a, \quad \lambda \in \mathbb{R} \setminus \{0\} \text{ and } a \in \mathbb{R}.$$

On the contrary all other coordinates do not have this privileged and separate role. We are free to change them as we see fit, namely

$$(x_1, \dots, x_n) \mapsto (x'_1, \dots, x'_n),$$

where $x'_k = x'_k(x_1, \dots, x_n)$ for all $k = 1, \dots, n$. The transformation needs to be smooth and invertible, but, a part from this, it carries no additional limitation, and, actually,

most notably we can also make it time-dependent. This is not the case for t , which is the reason why it should be best referred as a parameter rather than a coordinate, although we refrain from doing so. The reason is that, ultimately, this asymmetric behaviour is incorrect and this is probably the main lesson of *special relativity*. Yet, to really get to this conclusion one needs an extra ingredient that it is absent in Lagrangian mechanics: the speed of light being finite and its value being independent from the chosen reference frame. In relativistic mechanics time and Lagrangian coordinates do share the same role and, in that instance, one can formulate a relativistic version of Noether's theorem which also includes energy conservation.

Applications and Examples –

We are now ready to discuss a few applications and examples of Noether's and especially of Jacobi's theorem. We start from a result which basically justifies why in many textbooks, *e.g.* [GPS14], it is often stated that 1D problems are trivial

Remark 15 (Conservation of Energy with 1 degree of freedom). *Let us consider a Lagrangian system with one degree of freedom, abiding to the hypotheses of Theorem 23. Denoting by q the underlying Lagrangian coordinate and recalling Equations (34) and (35), the Lagrange function reads*

$$\mathcal{L}(q, \dot{q}, t) = \frac{m}{2} a(q, t) \dot{q}^2 - V(q, t),$$

where $m > 0$ plays the role of a mass term and, if we are dealing with a single point particle, it coincides with its mass. Let us further assume that $a(q, t) = a(q)$ and that $V(q, t) = V(q)$. We can therefor apply Theorem 38 to conclude that the Hamiltonian function is constant. Hence Equation (74) entails that

$$\mathcal{H}_0 = \frac{m}{2} a(q) \dot{q}^2 + V(q),$$

where $\mathcal{H}_0$ is the constant value of the Hamiltonian which depends on the initial conditions of the underlying system. It is also worth noticing that there is no necessity for $\mathcal{H}_0$ to coincide with the mechanic energy of the system. In addition this last identity entails

$$\dot{q} = \pm \sqrt{\frac{2(\mathcal{H}_0 - V(q))}{ma(q)}} \implies t = \pm \int_{q_0}^{q(t)} \sqrt{\frac{ma(x)}{2(\mathcal{H}_0 - V(x))}} dx, \quad (75)$$

where the $\pm$ -sign has to be chosen depending on the given initial conditions and where $q_0 = q(0)$. Observe that, we have solved the problem by integration and we have constructed the exact solution of the equation of motion but implicitly. Depending on the form of $a(x)$ and of $V(x)$ it might be very complicated, if not utterly impossible, doing the integral. Even more extreme is trying to invert the result from $t = t(q)$ into a normal form $q = q(t)$.

An attentive reader might not be fully convinced by Equation (75) since it contains two potential problem sources. The first is the square root which has to be positive. On the one hand $a(q)$ comes from the kinetic energy which is per construction a positive quantity and, therefore, having a single degree of freedom, $a(q) > 0$. On the other hand $\mathcal{H}_0 - V(x) \geq 0$ since $\mathcal{H} = T + V$. Hence the only problem occurs when $V(x) = \mathcal{H}_0$ which is tantamount to saying that we have assigned as initial conditions $\dot{q}(0) = 0$ and $q(0) = q_0$. In this case the square root vanishes and thus there is a priori the possibility that the right hand side of Equation (75) is non integrable. Yet this never happens as one can expect on physical grounds.

Exercise 40 (Hard). Prove that the right hand side of Equation (75) is always integrable. [Hint: Focus on the case $\dot{q}(0) = 0$ and $q(0) = q_0$. Consider then two cases, either $\left. \frac{dV}{dq} \right|_{q=q_0} = 0$ or $\left. \frac{dV}{dq} \right|_{q=q_0} \neq 0$, and use a Taylor series expansion.]

A second class of problems which can lead to finding a solution by integration occurs when there are more than 1 degree of freedom, but the presence of cyclic coordinates helps us via Noether's theorem 37 to reduce the complexity of the underlying set of ordinary differential equations. All the examples that we have seen up to now purposely fell in this category and we analyze one of the them in detail, leaving to the readers repeating the same procedure in all other cases.

Example 41 (Particle on a Sphere – Jacobi's and Noether's Integrals). In Section 3.2 we have considered the motion of a point particle in $\mathbb{R}^3$ constrained to move on a sphere of fixed radius $R > 0$ under the action of a constant force directed along the z -axis. We have shown that this is a Lagrangian system and in Example 32, we have used the existence of a cyclic coordinate to reduce the complexity of the underlying equation of motion to solving a nonlinear second order, ordinary differential equation.

We review this problem using the Jacobi integral. As a matter of fact we recall that the underlying Lagrangian is

$$\mathcal{L}(\theta, \dot{\theta}, \varphi, \dot{\varphi}) = \frac{mR^2}{2} \left(\dot{\theta}^2(t) + \sin^2 \theta(t) \dot{\varphi}^2(t) \right) - mgR \cos \theta(t).$$

In addition, since φ is a cyclic coordinate, the associated conjugate momentum is conserved:

$$p_\varphi = mR^2 \sin^2 \theta \dot{\varphi} = p_0.$$

By direct inspection $\mathcal{L} \neq \mathcal{L}(t)$ and, therefore, we can apply Theorem 38 and Lemma 39 to conclude that the Hamiltonian function is conserved and, furthermore, it coincides with the total mechanic energy. Using Equation (74) and rewriting for convenience the conservation law for p_φ , we obtain the following system of conserved quantities:

$$\begin{cases} \frac{mR^2}{2} \left(\dot{\theta}^2 + \sin^2 \theta \dot{\varphi}^2 \right) + mgR \cos \theta = \mathcal{H}_0 \\ mR^2 \sin^2 \theta \dot{\varphi} = p_0 \end{cases},$$

where the values of $\mathcal{H}_0$ and of p_0 depend on the assigned initial data. Consequently, inserting the second equation in the first one, we obtain

$$\frac{mR^2}{2}\dot{\theta}^2 + \frac{p_0^2}{2mR^2\sin^2\theta} + mgR\cos\theta = \mathcal{H}_0 \implies \dot{\theta} = \pm \sqrt{\frac{2}{mR^2}(\mathcal{H}_0 - mgR\cos\theta) - \frac{p_0^2}{m^2R^4\sin^2\theta}},$$

where the sign $\pm$ is chosen according to the underlying initial conditions. We can solve this ordinary differential equation by separation of variables, namely

$$t = \pm \int_{\theta_0}^{\theta(t)} \frac{1}{\sqrt{\frac{2}{mR^2}(\mathcal{H}_0 - mgR\cos x) - \frac{p_0^2}{m^2R^4\sin^2 x}}} dx, \quad (76)$$

which is once more a highly non trivial integral which yields $t = t[\theta]$. One should bear in mind that we need to have the solution in the form $\theta = \theta(t)$ since this is necessary also to integrate the equation defining $\dot{\varphi}$. Also in this case we have a highly implicit analytic problem.

Solution via Special Functions

It is often perceived that integrals such as those in Equation (76) cannot be solved at all and they are destined to be left in integral form. Actually this is not the case. Inspired by classical and quantum mechanics, this class of integrals and the ordinary differential equations from which they originate have been thoroughly studied in the second half of the nineteenth century and in the first half of the twentieth, leading to introducing several classes of *special functions*. In the course of the bachelor in physics, the most famous examples are Hermite polynomials and spherical harmonics which will appear during the courses of introduction to modern physics and of quantum mechanics. Yet, we feel interesting showcasing the application of this rather important and practical part of analysis by investigating in detail Equation (76). Hence let us rewrite Equation (76) with the following change of coordinates and parameters:

$$z := \cos\theta, \quad d\theta = -\frac{dz}{\sqrt{1-z^2}}, \quad \varepsilon := \frac{2\mathcal{H}_0}{mR^2}, \quad \beta := \frac{p_0^2}{m^2R^4}.$$

so that, setting $z_0 = z(0)$, the integral can be rewritten as

$$t = \mp \int_{z_0}^z \frac{dz}{\sqrt{N(z)}}, \quad N(z) := (1-z^2)\left(\varepsilon - \frac{2g}{R}z\right) - \beta.$$

We rewrite this function as

$$N(z) = a_3 z^3 + a_2 z^2 + a_1 z + a_0, \quad a_3 = \frac{2g}{R}, \quad a_2 = -\varepsilon, \quad a_1 = -\frac{2g}{R}, \quad a_0 = \varepsilon - \beta,$$

assuming in addition that it has three real roots $z_1 > z_2 > z_3$. The case where only one exists can be dealt with in a similar way but we shall not focus on it since our purpose

is not to be exhaustive, but to outline a case study. We introduce a class of functions known as **incomplete elliptic (or Legendre) integrals** of first kind [NIST, §19.2(ii)]:

$$F(\varphi, k) \doteq \int_0^\varphi \frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}} = \int_0^{\sin \varphi} \frac{dt}{\sqrt{1 - t^2} \sqrt{1 - k^2 t^2}}, \quad (77)$$

where in the second equality we have given an equivalent representation via the substitution $t = \sin \theta$. Here k is any complex number such that $1 - \sin^2 \varphi \in \mathbb{C} \setminus \{(-\infty, 0]\}$ and $1 - k^2 \sin^2 \varphi \in \mathbb{C} \setminus \{(-\infty, 0]\}$. This integral can be expressed as a power series, namely

$$F(\varphi, k) = \sum_{n=0}^{\infty} \frac{(2n)!}{2^{2n}(n!)^2} k^{2n} \cdot \frac{1}{2} B_{\sin^2 \varphi} \left(n + \frac{1}{2}, \frac{1}{2} \right), \quad (78)$$

where $B_x(a, b)$ denotes the incomplete Beta function, see [NIST, §8.17]:

$$B_x(a, b) = \int_0^x t^{(a-1)} (1-t)^{(b-1)} dt.$$

In the case under scrutiny this can be computed recursively as

$$I_n(\varphi) := B_{\sin^2 \varphi} \left(n + \frac{1}{2}, \frac{1}{2} \right) = \int_0^{\sin^2 \varphi} t^{n-\frac{1}{2}} (1-t)^{-\frac{1}{2}} dt,$$

and

$$I_n(\varphi) = \frac{2n-1}{2n} I_{n-1}(\varphi) - \frac{\sin^{2n-1} \varphi \cos \varphi}{2n}, \quad I_0(\varphi) = 2\varphi.$$

More importantly observe that Equation (78) is absolutely convergent for all $\varphi \in \mathbb{R}$ and $|k| < 1$. Going back to our specific problem, if we decompose

$$N(z) = a_3(z - z_1)(z - z_2)(z - z_3),$$

and if we use the substitution

$$\sin \varphi = \sqrt{\frac{z_1 - z}{z_1 - z_2}}, \quad k^2 = \frac{z_1 - z_2}{z_1 - z_3} \quad (0 < k^2 < 1),$$

the sought definite integral reads

$$t = \pm \int_{z_0}^z \frac{dz}{\sqrt{N(z)}} = \pm \frac{2}{\sqrt{a_3(z_1 - z_3)}} [F(\varphi(z), k) - F(\varphi_0, k)], \quad (79)$$

where $F(\varphi, k)$ is as per Equation (78) while φ_0 corresponds to $z_0 = \cos \theta_0$. Since scientists are never happy, they looked for an inverse of this lovely expression, which, with a little bit of irony, is based on the Jacobi elliptic function sn , see [NIST, §22.2]:

$$\text{sn}(x, k) = \sum_{n=0}^{\infty} a_n x^{2n+1}, \quad a_0 = 1,$$

where the coefficients a_n are defined recursively as

$$a_0 = 1, \quad a_n = \frac{1}{(2n+1)(2n)} \cdot \left[\sum_{j=0}^{n-1} (2j+1)(2(n-j)-1)a_j a_{n-j-1} - k^2 \sum_{i+j+l=n-1} (2i+1)a_i (2j+1)a_j (2l+1)a_l \right], \quad n \geq 1.$$

Also in this case the series is uniformly convergent for all $x \in \mathbb{R}$ and for all k such that $|k| < 1$. The Jacobi elliptic function is closely related to F^{-1} , the inverse of the truncated elliptic function via the identity [NIST, §22.15]:

$$F^{-1}(x, k) = \arcsin(\text{sn}(x, k)).$$

Applying this property to Equation (79), after a few algebraic manipulations, we end up with

$$z(t) = z_1 - (z_1 - z_2) \text{sn}^2\left(\frac{\sqrt{a_3(z_1 - z_3)}}{2}(t - t_*), k\right)$$

where t_* is a constant fixed by the initial condition and

$$k^2 = \frac{z_1 - z_2}{z_1 - z_3}, \quad t_* = \pm \frac{2}{\sqrt{a_3(z_1 - z_3)}} F(\varphi_0, k),$$

with φ_0 given by $\sin \varphi_0 = \sqrt{\frac{z_1 - z_0}{z_1 - z_2}}$. To conclude,

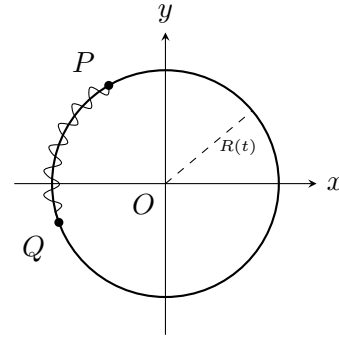
$$\theta(t) = \arccos(z(t)).$$

From the view point of applied analysis, this is a remarkable result, but, if the reader feels discouraged, this was part of the plan. Solving even the simplest among the problems in Lagrangian mechanics requires advanced technical tools. They are not at all useless or an overkill, but they can get us just so far. Even, at the end of this lengthy calculation, we know the solution in the form of a convergent power series, which is more than we can do in most of the cases. Yet, even in this situation, there is no way out than to resort to numerical techniques to get to numerical values. Should have we started directly with a

numerical analysis? Many claim so, I do not. It is a balance and a compromise. Pushing as far as we can the analytic tools allows for more reliable, precise and fast numerical answers, but, sometimes it is also fair to say that the effort in doing so compared to the actual gain is disputable and one might wish to start at an earlier stage this process. Where is the right point? I have no idea, I lean to the extreme side of pushing the analytic work as much as I can, but I reckon that each of you should have her/his own opinion on this issue.

The next example serves the purpose of highlighting that the condition $\frac{\partial \mathcal{L}}{\partial t}$ in Theorem 38 is not equivalent to requiring that all constraints are time-independent. Rather this is a special and relevant subclass.

Exercise: In $\mathbb{R}^2$ consider a Cartesian reference system as per Figure 10 and two point particles of the same mass $m > 0$, denoted by P and Q and constrained to move along the circumference of a circle of radius $R(t) = R_0 e^{Ht}$, with $T_0, H > 0$. Between them there is an attracting force descending from a potential $V = \frac{\kappa R^2(t)}{2}(\theta_P - \theta_Q)^2$ where θ_P, θ_Q are respectively the angles between P or Q and the horizontal axis:



1. Write the equations of motion,
2. Establish whether there exists a Jacobi integral,
3. Solve the equations of motion.
4. repeat the first two points assuming $V = \frac{\kappa R^2(t)}{2}(\theta_P - \theta_Q)^2 + mH^2 R^2(t)$.

Figure 10: Two point particles on an expanding circle connected by a spring running along the circumference.

Solution: The system is Lagrangian since we have 2 point particles which are constrained to move on a circle, namely $x_{P/Q}^2(t) + y_{P/Q}^2(t) - R^2(t) = 0$. These are 2 rehomonic, ideal constraints as per Definition 20. Hence, overall, we have two degrees of freedom. Accordingly it is convenient to choose as Lagrangian coordinates the angle θ between P and the horizontal axis as well the angle φ between P and Q . The position of P and Q reads

$$\mathbf{x}_P(t) = (R(t) \sin \theta(t), R(t) \cos \theta(t)) \quad \text{and} \quad \mathbf{x}_Q(t) = (R(t) \sin(\theta(t) + \varphi(t)), R(t) \cos(\theta(t) + \varphi(t))).$$

Observing that $\dot{R}(t) = HR(t)$ and omitting henceforth the explicit dependence on t to simplify the notation, the velocity vector becomes

$$\dot{\mathbf{x}}_P(t) = (\dot{R} \sin \theta + R\dot{\theta} \cos \theta, \dot{R} \cos \theta - R\dot{\theta} \sin \theta) = (HR \sin \theta + R\dot{\theta} \cos \theta, HR \cos \theta - R\dot{\theta} \sin \theta),$$

and, similarly

$$\dot{\mathbf{x}}_Q(t) = (HR \sin(\theta + \varphi) + R(\dot{\theta} + \dot{\varphi}) \cos(\theta + \varphi), HR \cos(\theta + \varphi) - R(\dot{\theta} + \dot{\varphi}) \sin(\theta + \varphi)).$$

Hence, using Equation (35), the total kinetic energy becomes

$$T = mR^2 \left(\dot{\theta}^2 + \dot{\theta}\dot{\varphi} + \frac{\dot{\varphi}^2}{2} \right) + mH^2 R^2,$$

while the potential energy reads

$$V = \frac{\kappa}{2} R^2 \varphi^2 + \alpha m H^2 R^2,$$

where $\alpha = 0, 1$ serves as a bookkeeping device to analyze both cases under scrutiny at the same time. The Lagrangian reads:

$$\mathcal{L} = mR^2 \left(\dot{\theta}^2 + \dot{\theta}\dot{\varphi} + \frac{\dot{\varphi}^2}{2} \right) - \frac{\kappa}{2} R^2 \varphi^2 + (1 - \alpha) m H^2 R^2, \quad (80)$$

from which we can derive that θ is a cyclic coordinate. Hence we have a conserved quantity according to Theorem 37:

$$p_\theta = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = mR^2 (2\dot{\theta} + \dot{\varphi}). \quad (81)$$

This is equivalent to writing down the Euler-Lagrange equation (43) relative to θ , while that related to φ reads

$$mR^2 \ddot{\varphi} + mR^2 \ddot{\theta} + 2m\dot{R}R(\dot{\varphi} + \dot{\theta}) + \kappa R^2 \varphi = 0. \quad (82)$$

We can use Equation (81) in Equation (82) as well as its time derivative to get rid of both $\dot{\theta}$ and $\ddot{\theta}$, obtaining at the end of the day

$$\ddot{\varphi} + 2 \frac{\dot{R}R}{\varphi} + \frac{2\kappa}{m} \varphi = 0$$

Using that $R(t) = R_0 e^{Ht}$ we end up with

$$\ddot{\varphi} + 2H\dot{\varphi} + \frac{2\kappa}{m} \varphi = 0, \quad (83)$$

which is the differential equation of a damped harmonic oscillator, whose generic solution splits in the renown following three cases:

1. **Underdamped** ($H^2 < 2\kappa/m$):

$$\varphi(t) = e^{-Ht} [A \cos(\omega t) + B \sin(\omega t)], \quad \omega = \sqrt{\frac{2\kappa}{m} - H^2}. \quad (84)$$

2. **Critically damped** ($H^2 = 2\kappa/m$):

$$\varphi(t) = (A + Bt) e^{-Ht}. \quad (85)$$

3. **Overdamped** ($H^2 > 2\kappa/m$):

$$\varphi(t) = A e^{(-H + \sqrt{H^2 - 2\kappa/m})t} + B e^{(-H - \sqrt{H^2 - 2\kappa/m})t}. \quad (86)$$

To conclude we observe that Equation (81) can be rewritten as

$$\dot{\theta} = -\frac{\dot{\varphi}}{2} + \frac{p_\theta}{2mR_0^2} e^{-2Ht},$$

which can be integrated as

$$\theta(t) = \theta_0 + \frac{1}{2}(\varphi_0 - \varphi(t)) + \frac{p_\theta}{4mHR_0^2} (1 - e^{-2Ht}),$$

where the initial conditions are here assigned as (θ_0, φ_0) . Observe that the solution of the equation of motion is independent from the choice of α and therefore we can conclude the following

- If the Lagrangian in Equation (80) is with $\alpha = 0$ then it is manifestly time-dependent and therefore according to Theorem 38 the Hamiltonian is not conserved,
- if we set $\alpha = 1$, then the Lagrangian is time independent and the Hamiltonian is conserved but Lemma 39 tells us that it does not coincide with the total mechanic energy.

One might think that this is absurd and actually, at first glance, the first interpretation is the one we expect. Equation (83) is telling us that φ is behaving like a damped oscillator, a feature that in Newtonian mechanics is often associated to friction or to non conservative systems. Yet the dynamics is not altered by adding a time dependent function to the Lagrangian and, in that case, we can construct a conserved quantity. The reason is that, if we look at the Hamiltonian in both cases, it reads that

$$\mathcal{H}_{\alpha=0} = \mathcal{H}_{\alpha=1} + mH^2 R^2(t).$$

In other words we have extracted from the Hamiltonian in the first scenario one bit, which is the conserved part $\mathcal{H}_{\alpha=1}$ and a second bit which is responsible for destroying this symmetry, namely $mH^2 R^2(t)$. Heuristically speaking it is accounting for the work necessary to inflate continuously and eternally the circle exponentially.

Remark 16. *A reader might think that the last example is an ad hoc construction, which carries no connection to the physical world. This is only partly true, because, suppose I want to look at the celestial sphere and I want to look at two galaxies which are far apart and of comparable mass. With respect to the scale of the Universe they are like points. Suppose that the Universe is*

undergoing a phase of exponential expansion and model the force between the galaxies as in the problem above. We have considered a 2D reduction of this scenario and Equation (83) occurs in cosmology in theories known as inflationary models. The only weak point is that the force used in the exercise seems far even from the Newtonian potential, even worse if we think of general relativity. This is only partly true and we shall address the reason in Section 5.

Let us now discuss one last example which addresses the problem of apparent forces and it highlights why in Lemma 39 we emphasized the necessity of working with Lagrangian coordinates subordinated to the underlying inertial reference system.

Exercise: In $\mathbb{R}^2$ consider a Cartesian reference system as per Figure 11 and an horizontal rail which is modeled as a rectangle of height $L > 0$, length $2L$ which is constrained to move anchored to the horizontal axis with a uniform acceleration along the x -direction $a > 0$. The initial speed is set to $v_0 = 0$. On top of the rail there is a point particle of mass $m > 0$ which is constrained to move on its horizontal edge and there is an elastic force of constant $\kappa > 0$, connecting the particle to the end of the edge. The rest length is 0. Compute the Lagrangian and the equations of motion both in the inertial frame and in that comoving with the rail assuming that the system is studied when the particle is far from the edges of the rail. Discuss whether the Hamiltonian function is conserved in the two scenarios.

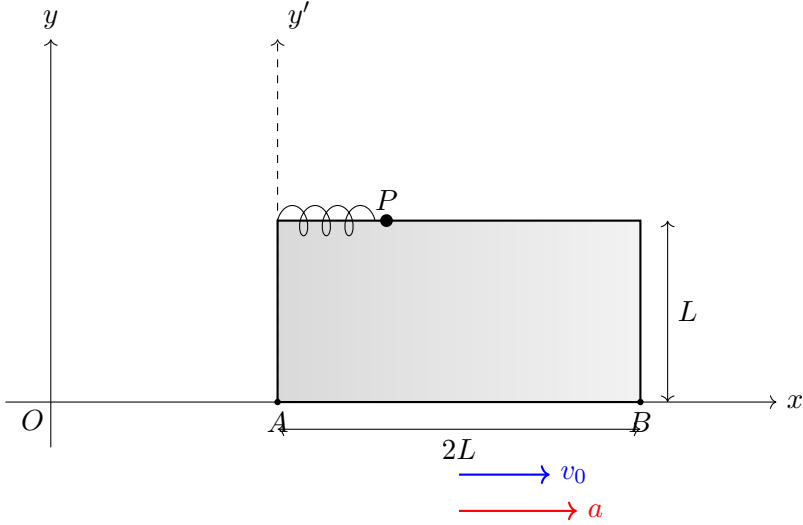


Figure 11: An horizontal rail uniformly accelerated along the x -axis with a point particle on top it attached to a spring.

Solution – Inertial Frame: The system is Lagrangian since the only constraint is $y_P - L = 0$, where the position of the particle is denoted by $\mathbf{x}_P = (x_P, y_P)$. Hence this is ideal as per Definition 20. With reference to Figure 11 we can choose as Lagrangian coordinate x , the horizontal distance between P and A . In other words

$$\mathbf{x} = (x + x_A(t), L),$$

where the given hypotheses entail that $x_A(t) = x_0 + \frac{a}{2}t^2$, x_0 being the initial position of the point A at $t = 0$. Hence the velocity vector reads

$$\dot{\mathbf{x}} = (\dot{x} + at, 0),$$

from which we can construct the kinetic energy as $T = \frac{m}{2}(\dot{x} + at)^2$. Following the second example in Section 3.2, the potential energy of the particle is $V = \frac{\kappa}{2}x^2$, from which it descends that the

Lagrangian as per Equation (44) reads

$$\mathcal{L} = \frac{m}{2}(\dot{x} + at)^2 - \frac{\kappa}{2}x^2, \quad (87)$$

whose associated Euler-Lagrange equation (43) is

$$m(\ddot{x} + a) + \kappa x = 0 \implies \ddot{x} = -\frac{\kappa}{m}x - a.$$

This is an ordinary differential equation with a source term and we know from calculus that the general solution is the linear combination of a particular solution $x_p(t)$ and of a solution of the homogeneous equation, which, in this case, is an harmonic oscillator. Being the source constant in time one can infer that the sought particular solution is $x_p(t) = -\frac{ma}{\kappa}$ and hence

$$x(t) = A \cos \omega(t) + \frac{B}{\omega} \sin \omega(t) - \frac{ma}{\kappa},$$

where $A = x(0) + \frac{ma}{\kappa}$ while $B = \dot{x}(0)$. To conclude it suffices to observe that Equation (87) is manifestly time-dependent and, therefore, according to Theorem 38 the Hamiltonian is not conserved.

Solution – Comoving Frame: Consider now a frame comoving with the horizontal rail. The system is Lagrangian since the only constraint is still $y_P - L = 0$, where the position of the particle is denoted by $\mathbf{x}_P = (x_P, y_P)$. Hence this is ideal as per Definition 20. With reference to Figure 11 we can choose as Lagrangian coordinate q , the horizontal distance between P and A . We denote it by the letter q to differentiate this from the previous case. Hence the position and velocity vectors of the particle are simply

$$\mathbf{x} = (q, L) \implies \dot{\mathbf{x}} = (\dot{q}, L),$$

from which it descends that the kinetic energy is $T = \frac{m}{2}\dot{q}^2$. In this case one has to be careful in constructing the potential since one has to account also for the apparent force which, in this case, reads as $\mathbf{F}_{app} = -ma\mathbf{e}_x$, where $\mathbf{e}_x = (1, 0) \in \mathbb{R}^2$. This allows for the use of the Lagrangian formalism since it descends from a potential $V_{app} = maq$. Hence, overall the Lagrange function in the comoving frame reads

$$\mathcal{L} = \frac{m}{2}\dot{q}^2 - maq - \frac{\kappa}{2}q^2, \quad (88)$$

where we have also accounted for the elastic potential. The Euler Lagrange equation (43) reads

$$m\ddot{q} + ma + \kappa q = 0,$$

which is, as expected, the same equation of motion as before. Yet, in this reference system, not only the Lagrangian is time independent and hence we can apply Theorem 38, but there is no time dependent constraint and, therefore, also Lemma 39 is valid. In other words the Hamiltonian function is conserved and it coincides with the total mechanical energy of the system:

$$\mathcal{H} = \frac{m}{2}\dot{q}^2 + maq + \frac{\kappa}{2}q^2.$$

The physical interpretation of the result is twofold. From the viewpoint of the inertial system, the Hamiltonian cannot be conserved since one needs to account for the moving rail. From that of the accelerated observer, on the contrary, energy is conserved because he/she is able to account for the motion of the rail in the form of an apparent force. This holds true only because the rail is uniformly accelerating, hence this reasoning would fail since the apparent force would not descend from a potential function. This phenomenon has deep rooted consequences in modern physics both in our understanding of the concept of inertial frames and in one of the most iconic consequences of quantum field theory on curved spacetimes: the Unruh effect. But this is a story to be told in a few years.

4.4 Central Force Problem – Hidden Symmetries

In this last section of this chapter, we revise the most iconic application of Newtonian gravity: the Kepler problem. The reader might wonder why since, using standard techniques, it could be solved exactly. Hence one might not see any benefit in using the Lagrange formalism and the whole enterprise might look moot. My goal is to convince you that this is not the case since this is the prototypical example of a system which one might study using Noether's symmetries and the Jacobi integral, but, in doing so, it is missing one conserved quantity, known as the *Runge-Lenz vector*. This is the prime example of a whole bunch of hidden symmetries which do not descend directly from Theorems 37 and 38, signaling that they need to be extended as they cannot cover the whole underlying story. We also emphasize that the investigation on the central force problem is vast and whole books have been written about it. Here we shall only scratch the surface focusing mainly on the connection with the theory of conserved quantities, also because we assume that the reader is already familiar with the Newtonian two body problem and with the reduced three body counterpart from the course of mechanics. A reader, who is willing to learn more, is encouraged to start from [GPS14, Chap. 3].

Two two body problem

Let us consider $\mathbb{R}^3$ with a standard reference frame to which we associate the canonical Cartesian coordinates (x, y, z) . Herein two point particles of mass m_1 and m_2 are free to move constrained only by a conservative internal force descending by a potential $V = V(\mathbf{r}_1 - \mathbf{r}_2)$ where $\mathbf{r}_{1/2}$ denotes the position vector of the two particles, see Figure 12. Since there are no constraints and the forces are all internal, the system is manifestly Lagrangian and it possesses 6 degrees of freedom. Following Remark 12 we can infer that the center of mass is moving inertially with respect to O . Therefore it is convenient to choose as Lagrangian coordinates (x_R, y_R, z_R) , the components of the position vector $\mathbf{R}$ of the center of mass and (x', y', z') the components of $\mathbf{r} \doteq \mathbf{r}_1 - \mathbf{r}_2$. A direct computation shows that the kinetic energy can be rewritten as per Equation (34):

$$T = \frac{m_1}{2} \|\dot{\mathbf{r}}_1\|^2 + \frac{m_2}{2} \|\dot{\mathbf{r}}_2\|^2 = \frac{M}{2} \|\dot{\mathbf{R}}\|^2 + \frac{\mu}{2} \|\dot{\mathbf{r}}\|^2,$$

where $M = m_1 + m_2$, $\mu = \frac{m_1 m_2}{m_1 + m_2}$ is the reduced mass, while $\dot{\mathbf{R}}$ and $\dot{\mathbf{r}}$ are the velocity vectors associated to $\mathbf{R}$ and $\mathbf{r}$ respectively. Hence, in view of the assumptions, the total Lagrangian

reads

$$\mathcal{L}_{Tot} = \frac{M}{2} \|\dot{\mathbf{R}}\|^2 + \frac{\mu}{2} \|\dot{\mathbf{r}}\|^2 - V(\mathbf{r}). \quad (89)$$

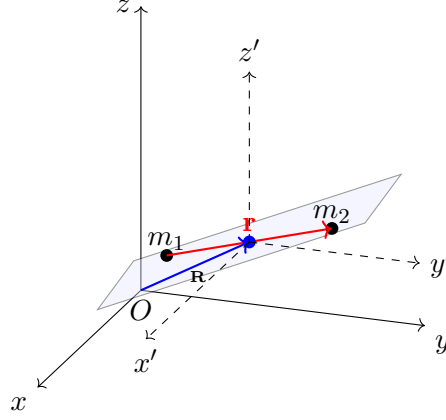


Figure 12: A not so original picture of two point particles of mass m_1 and m_2 moving in $\mathbb{R}^3$. The shaded region is the orbital plane

As discussed in Remark 12, since the center of mass is inertial, it turns out that each momentum $p_{x_R}, p_{y_R}, p_{z_R}$ is conserved. This entails that the contribution to the kinetic energy coming from $\frac{M}{2} \|\dot{\mathbf{R}}\|^2$ is a constant determined by the initial conditions. In other words, from a dynamical viewpoint we can discard completely this term from Equation (89). This is tantamount to saying that we can study the motion from a reference frame having its origin at the center of mass. Herein the system has three degrees of freedom and we can use once more (x', y', z') as Lagrangian coordinates, see Figure 12. The Lagrange function in the center of mass system reads

$$\mathcal{L} = \frac{\mu}{2} \|\dot{\mathbf{r}}\|^2 - V(\mathbf{r}). \quad (90)$$

We make the following additional assumption:

- the potential $V(\mathbf{r})$ is central, namely it depends only on the distance $\|\mathbf{r}\|$. That is $V \equiv V(\|\mathbf{r}\|)$.

As far as the two particles do not collide (*collisional singularity*), we can assume that $\|\mathbf{r}\| > 0$ and therefore it is convenient to switch to new Lagrangian, spherical coordinates,

$$(x', y', z') \rightarrow (r, \theta, \varphi),$$

where the transformation laws are those of Equation (14). In these coordinates the Lagrangian reads

$$\mathcal{L} = \frac{\mu}{2} \left(\dot{r}^2 + r^2 \left(\dot{\theta}^2 + \sin^2 \theta \dot{\varphi}^2 \right) \right) - V(r). \quad (91)$$

Following Equation (66) and the related analysis, we can infer that the components of the angular momentum along any direction is conserved. This has two notable consequences:

1. the first is nothing but the observation that φ is a cyclic coordinate and therefore its associated conjugate momentum is conserved:

$$p_\varphi = \frac{\partial \mathcal{L}}{\partial \dot{\varphi}} = \mu r^2 \sin^2 \theta \dot{\varphi} = \ell, \quad (92)$$

where ℓ is determined from the initial conditions. The letter ℓ is chosen since it is the magnitude of the angular momentum.

2. since $\mathbf{L} \cdot \mathbf{n}$ is conserved along any direction $\mathbf{n}$, it turns that also the vector $\mathbf{L}$ is conserved. In particular this entails that its direction does not change in time. If we recall that, in Cartesian coordinates, $\mathbf{L} = \mathbf{r} \wedge \mathbf{p}$ where $\mathbf{p} = m\dot{\mathbf{r}}$ and if we recall that, in the system under investigation there are no constraints, this entails that, no matter the set of Lagrangian coordinates used, $\mathbf{L}$ is always perpendicular to the plane spanned by $\mathbf{r}$ and $\mathbf{p}$ unless they are parallel.
3. If $\mathbf{r}$ is parallel to $\mathbf{p}$ then $\mathbf{L} = 0$. This descend to a purely radial motion. As a matter of fact Equation (92) boils down to

$$\mu r^2 \sin^2 \theta \dot{\varphi} = 0,$$

which entails either $\theta = 0, \pi$ or $\dot{\varphi} = 0$. The second option implies also that $\varphi(t) = \varphi_0$, this value being determined by the initial condition. As far as θ is concerned we can look at this associated Euler-Lagrange equation starting from Equation (91):

$$\mu r^2 \left(\ddot{\theta} - \sin \theta \cos \theta \dot{\varphi}^2 \right) = 0.$$

This allows for two options:

- (a) $\theta(0) = \theta_0 \neq 0, \pi$ in which case we impose that $\dot{\theta}(0) = \theta_0$, $\varphi(0) = \varphi_0$ and $\dot{\varphi}(0) = 0$,
- (b) $\theta(0) = 0, \pi$, in which case we need to impose that $\dot{\theta}(0) = 0$.

In the first case the motion is radial since the angular coordinates are fixed. In the second one, the discrepancy is only apparent, since the motion is constrained along the z -axis which is still a radial motion. Yet the conclusion is *incorrect* because, in Example 7, we have proven that spherical coordinates are ill defined on the z -axis. Therefore the case $\theta(0) = 0, \pi$ cannot be considered because these values are not within the range of validity of these coordinates. Hence case (b) does not exist and, if our initial conditions force us to work on the z -axis, then one needs to change coordinates.

To summarize, unless we are interested in a purely radial motion, we have that the angular momentum vector $\mathbf{L}$ is for all values of the time variable $t \in \mathbb{R}$ perpendicular to the plane

generated by $\mathbf{r}$ and $\mathbf{p}$. In other words, by a rigid rotation of the reference frame labeled by the axis (x', y', z') we can choose a third system (x'', y'', z'') where the z'' -axis is parallel to $\mathbf{L}$. This tantamount to *setting* $\theta = \frac{\pi}{2}$. All these data can be combined together observing that, since there is no explicit time dependence in Equation (91) and there are no constraints, we can apply Lemma 39 to conclude that *the Hamiltonian function is conserved and it coincides with the mechanical energy*:

$$\mathcal{H} = \mathcal{E} = \frac{\mu}{2} \left(\dot{r}^2 + r^2 \left(\dot{\theta}^2 + \sin^2 \theta \dot{\varphi}^2 \right) \right) + V(r), \quad (93)$$

which further simplifies, setting $\theta = \frac{\pi}{2}$ and using Equation (92), to

$$\mathcal{E} = \frac{\mu}{2} \left(\dot{r}^2 + \frac{\ell^2}{\mu^2 r^2} \right) + V(r). \quad (94)$$

In other words we are in the same situation of Remark 15, namely we can solve implicitly the dynamics by integrating out this equation as

$$t = \pm \int_{r_0}^{r(t)} \frac{dx}{\sqrt{\frac{2\mathcal{E}}{\mu} - \frac{\ell^2}{\mu^2 x^2} - V(x)}} \quad (95)$$

where $r(0) = r_0$ and where the sign in front of the integral is chosen depending on the initial conditions.

Remark 17. *It is worth revisiting Equation (92) under the further assumption that $\mathbf{L} \neq 0$. As mentioned, this entails that we can set $\theta = \frac{\pi}{2}$ which implies in turn that $\mu r^2 \dot{\varphi}$ is constant in time. Equivalently, being μ a number*

$$\frac{d}{dt} \left(\frac{r^2}{2} \dot{\varphi} \right) = 0,$$

where the factor $\frac{1}{2}$ has been added since the integrand is now the **areal velocity**, namely the area swept by the radius vector per unit time. We have recovered Kepler's second law, though with any radial potential.

Two body problem with the Newtonian potential

We consider now the special case where the radial potential is Newtonian, namely

$$V(r) = -\frac{\kappa}{r},$$

where $\kappa > 0$ is here a constant depending on the parameters of the problem under scrutiny, but of no significance for our discussion. We report Equations (92) and (94):

$$\begin{cases} \mu r^2 \dot{\varphi} = \ell \\ \mathcal{E} = \frac{\mu}{2} \left(\dot{r}^2 + \frac{\ell^2}{\mu^2 r^2} \right) - \frac{\kappa}{r} \end{cases} .$$

We are interested in individuating the orbits followed by a particle under the additional assumption that $\ell \neq 0$. From the first equation we can infer that

$$\frac{d}{dt} = \frac{d\varphi}{dt} \frac{d}{d\varphi} = \frac{\ell}{\mu r^2} \frac{d}{d\varphi},$$

which entails that the conservation of energy becomes

$$\mathcal{E} = \frac{\ell^2}{2\mu r^4} \left(\frac{dr}{d\varphi} \right)^2 + \frac{\ell^2}{2\mu r^2} - \frac{\kappa}{r}.$$

Once more, by separation of variables, we end up with

$$\varphi - \varphi_0 = \int_{r_0}^{r(\varphi)} \frac{dx}{x^2 \sqrt{\frac{2\mu\mathcal{E}}{\ell^2} + \frac{2\mu\kappa}{\ell^2 x} - \frac{1}{x^2}}},$$

where $\varphi(0) = \varphi_0$ and $r(0) = r_0$. By the substitution $r \mapsto u \doteq r^{-1}$, we end up with

$$\varphi - \varphi_0 = \int_{u_0}^{u(\varphi)} \frac{dy}{\sqrt{\frac{2\mu\mathcal{E}}{\ell^2} + \frac{2\mu\kappa y}{\ell^2} - y^2}}, \quad (96)$$

which can be integrated out as

$$\varphi - \varphi'_0 = -\arccos \frac{\frac{\ell^2 u}{\mu\kappa} - 1}{\sqrt{1 + \frac{2\mathcal{E}\ell^2}{\mu\kappa^2}}},$$

where

$$\varphi'_0 = \varphi_0 + \arccos \frac{\frac{\ell^2 u_0}{\mu\kappa} - 1}{\sqrt{1 + \frac{2\mathcal{E}\ell^2}{\mu\kappa^2}}}.$$

By restoring the radius r and inverting this expression, we obtain the final form for the orbit, namely

$$\frac{1}{r} = \frac{\mu\kappa}{\ell^2} \left(1 + \sqrt{1 + \frac{2\mathcal{E}\ell^2}{\mu\kappa^2}} \cos(\varphi - \varphi') \right) = \frac{\mu\kappa}{\ell^2} (1 + e \cos(\varphi - \varphi')), \quad (97)$$

which is the general form of the equation of a conic with **eccentricity**

$$e = \sqrt{1 + \frac{2\mathcal{E}\ell^2}{\mu\kappa^2}}. \quad (98)$$

For the sake of completeness we recall that there exist 4 possible orbits depending on the value of e , see [GPS14, Chap. 3]:

1. if $e > 1$, then $\mathcal{E} > 0 \implies$ hyperbolic motion,
2. if $e = 1$, then $\mathcal{E} = 0 \implies$ parabolic motion,
3. if $0 < e < 1$, then $\mathcal{E} \in \left(-\frac{\mu\kappa^2}{2\ell^2}, 0\right) \implies$ elliptic motion,
4. if $e = 0$, then $\mathcal{E} = -\frac{\mu\kappa^2}{2\ell^2} \implies$ circular motion.

Conic curve

We recall here the notion of conic curve. Historically, it can be introduced in two different, albeit equivalent ways. In the first one a conic curve is the intersection between a cone in $\mathbb{R}^3$ and plane. We shall not discuss this definition and we focus instead on the second one, intrinsically two-dimensional, according to which, given

1. a plane diffeomorphic to $\mathbb{R}^2$ and a distinguished element therein, F called *focal point*,
2. a line γ lying in $\mathbb{R}^2$, known as the *directrix*,

we call **conic curve** of *eccentricity* $e > 0$, the set of points $P \in \mathbb{R}^2$ such that

$$\frac{\|\mathbf{x}_{PF}\|}{d(P, \gamma)} = e, \quad (99)$$

where $\mathbf{x}_{PF}$ is the vector connecting P to F while $d(P, \gamma)$ denotes the distance from P to the directrix. In order to obtain a parameterization of Equation (99) in terms of coordinates, we consider a reference frame in $\mathbb{R}^2$ centered at F with standard Cartesian coordinates (x, y) . Accordingly the position vector of P can be written in polar coordinates as

$$\mathbf{x}_P = \mathbf{x}_{PF} = (r \cos \theta, r \sin \theta) \implies \|\mathbf{x}_{PF}\| = r.$$

Denoting by d_0 the distance between the origin and its closest point $Q \in \gamma$, it turns out that every $(x_d, y_d) \in \gamma$ abides by the equation

$$x_d \cos \theta_0 + y_d \sin \theta_0 = d_0,$$

where θ_0 is the angle between the outward pointing normal to γ at Q . If we recall that, given a generic point P of coordinate $(x, y) \in \mathbb{R}^2$ and a line of equation $ax' + by' = c$, the distance from P to the line is $\frac{|ax+by-c|}{\sqrt{a^2+b^2}}$, we can conclude that

$$d(\mathbf{x}_P, \gamma) = |\cos \theta_0 x + \sin \theta_0 y - d_0| = |r \cos \theta \cos \theta_0 + r \sin \theta \sin \theta_0 - d_0| = |r \cos(\theta - \theta_0) - d_0|.$$

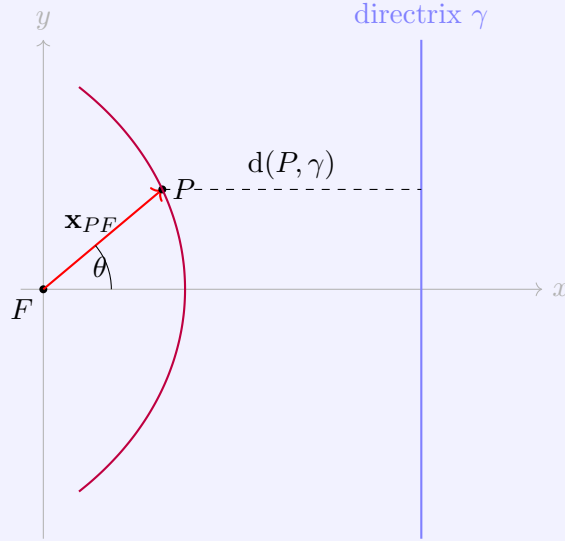
In other words Equation (99) becomes

$$\frac{r}{|r \cos(\theta - \theta_0) - d_0|} = e, \quad (100)$$

In this equation, the absolute value entails that we have two options, but it is customary to consider those points for which $|r \cos(\theta - \theta_0) - d_0| = d_0 - r \cos(\theta - \theta_0)$ which means, at a geometric level, that we consider only those points lying on the same side of the focal F with respect to the directrix. In this way we get exactly to the same form appearing in Equation (97), namely

$$r = \frac{ed_0}{1 + e \cos(\theta - \theta_0)}, \quad (101)$$

where $p \doteq ed_0$ is the *semilatus rectum*, that is half of the length of the latus rectum which is a line segment parallel to the directrix and intersecting the focal point.



In order to obtain the cozier and more familiar form of a conic curve, consider the reference system having the focal point at the origin such that the y -axis is parallel to the directrix, namely we set $\theta_0 = 0$ in Equation (101):

$$p = r(1 + e \cos \theta) \implies \sqrt{x^2 + y^2} = p - ex,$$

where we have reinstated the Cartesian coordinates $x = r \cos \theta$ and $y = r \sin \theta$. By squaring both sides of this identity we obtain

$$x^2 + y^2 = p^2 + e^2 x^2 - 2epx,$$

which leads to the equation of a generic conic of eccentricity e and semilatus rectum p :

$$(1 - e^2)x^2 + y^2 + 2epx - p^2 = 0. \quad (102)$$

The reader can now replace the value of $e > 0$ to recognize the different possibilities sketched above. Notice also that we have excluded the case $e = 0$ since it is degenerate

with respect to Equation (99) and, absurdly, the conic reduces to the lone focal point. As a matter of fact, this case is seen as a degenerate instance of an ellipsis $0 < e < 1$. In this case observe that Equation (102) reduces to

$$\frac{(x - x_0)^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a = \frac{p}{1 - e^2}, \quad b = \frac{p}{\sqrt{1 - e^2}}, \quad x_0 = -ea$$

If we send $e \rightarrow 0$ we obtain a contradiction since also $p \rightarrow 0$ being $p = d_0 e$. Hence, in order to obtain a circle one needs both to let the eccentricity vanish but also to let d_0 diverge, namely sending the directrix at infinity, though keeping the product $d_0 e$ constant. In this way as $e \rightarrow 0$ Equation (102) reduces to that of a circle.

Hidden Symmetry: the Laplace-Runge-Lenz vector

It is really hard to fully appreciate the richness of the Kepler problem, already in the simpler scenario with two point particles. As a matter of fact, this is the first instance of a model where we learn a hard truth, namely that in many Lagrangian systems, one can find *hidden symmetries*. First of all let us discuss it for the case under investigation, commenting on its origin and significance at the end. Also in this case we recommend to consult [GPS14, Sec. 3.9] for additional details.

Consider therefore the two body problem in an inertial reference system comoving with the center of mass. The underlying Lagrangian has been already derived in Equation (90) for a generic central potential. In the gravitational scenario where we set $V(r) = -\kappa r^{-1}$, $\kappa > 0$, it reads

$$\mathcal{L} = \frac{\mu}{2} \|\dot{\mathbf{r}}\|^2 + \frac{\kappa}{r},$$

where $\mathbf{r} = (x', y', z')$ are the components of the relative position vector between the two bodies. The associated conjugate momenta are

$$p_{x'} = \frac{\partial \mathcal{L}}{\partial \dot{x}'} = \mu \dot{x}', \quad p_{y'} = \frac{\partial \mathcal{L}}{\partial \dot{y}'} = \mu \dot{y}', \quad p_{z'} = \frac{\partial \mathcal{L}}{\partial \dot{z}'} = \mu \dot{z}',$$

or, in a more compact form,

$$\mathbf{p} = \mu \dot{\mathbf{r}}.$$

From the Euler-Lagrange equations or, equivalently from Newton's equations, being the system without constraints, we can infer that

$$\mu \ddot{\mathbf{r}} = -\nabla V = -\frac{\kappa}{r^3} \mathbf{r} \implies \dot{\mathbf{p}} = \frac{\kappa}{r^3} \mathbf{r}. \quad (103)$$

Considering the angular momentum vector $\mathbf{L} = \mu \mathbf{r} \wedge \dot{\mathbf{r}}$, we can construct

$$\dot{\mathbf{p}} \wedge \mathbf{L} = \frac{\mu \kappa}{r^3} (\mathbf{r} \wedge (\mathbf{r} \wedge \dot{\mathbf{r}})) = \frac{\mu \kappa}{r^3} (\mathbf{r}(\mathbf{r} \cdot \dot{\mathbf{r}}) - r^2 \dot{\mathbf{r}}), \quad (104)$$

where in the last identity we used that *triple vector product rule*:

$$\mathbf{v} \wedge \mathbf{w} \wedge \mathbf{z} = \mathbf{w}(\mathbf{v} \cdot \mathbf{z}) - \mathbf{z}(\mathbf{v} \cdot \mathbf{w}),$$

and we $\mathbf{r} \cdot \mathbf{r} = r^2$. Using the identity $2\mathbf{r} \cdot \dot{\mathbf{r}} = \frac{d}{dt}(\mathbf{r} \cdot \mathbf{r}) = \frac{dr^2}{dt} = r\dot{r}$, where r and $\dot{r}$ are the radial coordinate and its time derivative, we can rewrite Equation (104) as

$$\dot{\mathbf{p}} \wedge \mathbf{L} = \frac{\mu\kappa}{r^3} (\mathbf{r}(r\dot{r}) - r^2\dot{\mathbf{r}}) = \mu\kappa \left(\frac{\mathbf{r}\dot{r}}{r^2} - \frac{\dot{\mathbf{r}}}{r} \right) \Rightarrow \frac{d}{dt}(\mathbf{p} \wedge \mathbf{L}) = \mu\kappa \frac{d}{dt} \left(\frac{\mathbf{r}}{r} \right),$$

where, in the last implication, we employed implicitly that the angular momentum vector is conserved, *i.e.*, $\dot{\mathbf{L}} = 0$. This last identity unveils a novel conserved quantity, known as **Laplace-Runge-Lenz vector**

$$\mathbf{A} \doteq \mathbf{p} \wedge \mathbf{L} - \frac{\mu\kappa\mathbf{r}}{r} \Rightarrow \frac{d\mathbf{A}}{dt} = 0. \quad (105)$$

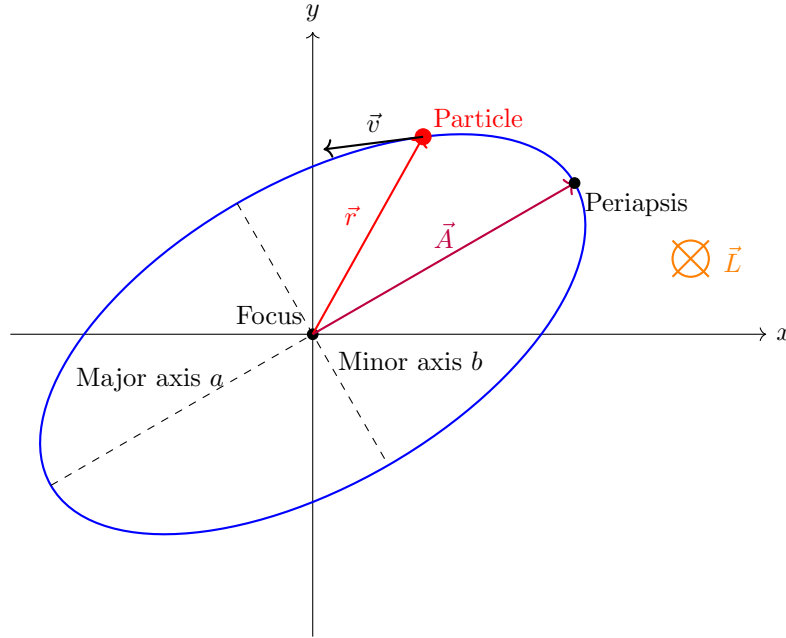


Figure 13: Tilted elliptic orbit with the focal point at the origin and the angular momentum vector perpendicular to the orbit plane. The Laplace-Runge-Lenz vector is here displayed pointing from the origin to the periapsis.

In order to better understand the significance of this new conserved quantity, let us study its properties, see Figure 13. First of all, recalling that $\mathbf{L} = \mathbf{r} \wedge \mathbf{p}$, it turns out that

$$\mathbf{A} \cdot \mathbf{L} = 0,$$

namely $\mathbf{A}$ lies on the orbital plane. By taking its scalar product with $\mathbf{r}$ the position vector of the particle on the elliptic orbit, we end up with

$$\mathbf{A} \cdot \mathbf{r} = (\mathbf{p} \wedge \mathbf{L}) \cdot \mathbf{r} - \mu\kappa r = (\mathbf{r} \wedge \mathbf{p}) \cdot \mathbf{L} - \mu\kappa r,$$

where we used Equation (65) in the last identity. In other words

$$\mathbf{A} \cdot \mathbf{r} = \mathbf{L} \cdot \mathbf{L} - \mu\kappa r = \ell^2 - \mu\kappa r \implies Ar \cos \theta = \ell^2 - \mu\kappa r, \quad (106)$$

where we used that the magnitude of the angular momentum vector is ℓ as per Equation (92) and where $A \doteq \sqrt{\|\mathbf{A}\|}$. We can rewrite this last identity as

$$Ar \cos \theta = \ell^2 - \mu\kappa r \implies \frac{1}{r} = \frac{\mu\kappa}{\ell^2} \left(1 + \frac{A}{\mu\kappa} \cos \theta \right), \quad (107)$$

which is exactly Equation (97) with $\varphi' = 0$ and

$$A = \mu\kappa e. \quad (108)$$

In addition, since θ is the angle between $\mathbf{A}$ and $\mathbf{r}$ with obtain that they are parallel when $\cos \theta = 1$, namely when the particle is at the periapsis of the ellipse defined by Equation (107).

Remark 18. *The Laplace-Runge-Lenz vector as per Equation (105) is a **hidden symmetry** since it cannot descend from Noether's Theorem 37. Most notably, even in its weakest form codified by Equation (73), the conserved quantity takes the form*

$$I_g(q, \dot{q}, t) = \sum_{k=1}^{\tilde{N}} \frac{\partial \mathcal{L}}{\partial \dot{q}^k} \frac{\partial q_\epsilon^{tk}}{\partial \epsilon} \Big|_{\epsilon=0} (t, q, \dot{q}) - g(t, q),$$

which tells us that I_g can at most depend linearly on the moment due to the term $\frac{\partial \mathcal{L}}{\partial \dot{q}^k}$ combined with Equations (34) and (35). Yet the Laplace-Runge-Lenz vector $\mathbf{A}$ contains a term of the form $\mathbf{p} \wedge \mathbf{L}$ which is quadratic in the momentum since $\mathbf{L} = \mathbf{r} \wedge \mathbf{p}$. Hence it cannot be a consequence of the celebrated Noether's theorem.

Dynamical Symmetries – Paradigm Shift (sponsored by Taffo)

The Laplace-Runge-Lenz vector as per Equation (105) has been highlighted as a hidden symmetry, which is good marketing, but it is highly unsatisfactory. In the literature there are several other models with vectors sharing similar features to those of $\mathbf{A}$, the most famous being still in the two body problem assuming in place of $V(r) = -\kappa r^{-1}$, the potential $\tilde{V}(r) = \frac{\mu\omega^2}{2}r^2$ with $\omega > 0$. In this case one can find an unexpected conserved quantity known as **Fradkin tensor**:

$$Q_{ij} = \dot{x}_i' \dot{x}_j' + \mu\omega^2 x_i' x_j', \quad i, j = 1, 2, 3$$

which cannot descend from Noether's theorem since it is quadratic in the conjugate momenta as the Laplace-Runge-Lenz vector. The reason for such abundance of hidden conserved quantities is not surprising if we accept the limitation of Noether's theorem. As a matter of fact this celebrated result characterizes and classifies **kinematic symmetries**. More precisely the line of reasoning is the following:

1. consider a physical system and the constraints it has to abide by,
2. establish whether it is Lagrangian and how many degrees of freedom are necessary. Hence classify the kinematic configurations,
3. in absence of constraints, on each point particle, kinematic symmetries are codified by the Galilei group. In other words, acting on a kinematically allowed configuration with a Galileian transformation yields another kinematically allowed configuration,
4. Constraints modify the configuration space and therefore only a subgroup of all the Galileian transformations are still symmetries.
5. Apply Noether's theorem to decide a priori whether a given potential is compatible or not with the surviving symmetry transformations. The remaining ones are the kinematically allowed symmetries of the system.

Yet, this reasoning fails in intercepting a different class of symmetries, namely transformations which map solutions of the equation of motion into other ones. These can a priori depend both on the Lagrangian coordinates and on the associated generalized velocities yielding quantities which are conserved only if these are evaluated on trajectories, solution of the underlying dynamics. This is what happened in deriving the Laplace-Runge-Lenz vector in Equation (103). In other words there exists a larger class of symmetries and of associated conserved quantities, called **dynamical**, which are extremely powerful tools in simplifying the complexity of the underlying equations of motion. Yet, as it stands, Lagrangian mechanics is not well suited to account for them as one needs to devise a generalization of Noether's theorem in which generalized velocities are truly independent variables. This is possible using the language of differential geometry, see [Mor20, Chap. 9.5] or working in a more general setting, known as **Hamiltonian mechanics** which will be the heart of the third part of this course.

5 Perturbation Theory

Let's work the problem, people. Let's not make things worse by guessing.

Apollo13

One of the greatest open problems in physics and, to many extents, one of biggest challenges in the years ahead is to develop a reliable and efficient set of tools and results to deal with nonlinearities in physics. Already in the several examples given in these lecture notes, we have been faced with several non linear ordinary differential equations. The difficulties in solving them have been bypassed using energy conservation, but this escamotage worked only because we looked at very specific classes of exercises where I knew it would work. In general, even after taking into account all underlying conserved quantities, we are forced to deal with a set of nonlinear ordinary differential equations. This is a general feature of several models in physics with the complication that the dynamics is ruled by partial differential equations, adding an extra layer of difficulty.

It would be unfair and actually utterly wrong stating that nothing is known concerning this class of equations and actually a lot of work has been done and terrific results have been obtained. Yet, even restricting the attention to ordinary differential equations, in many cases, what we really have available are existence and uniqueness results, *e.g.*, the Picard-Lindelöf theorem. Yet reliable methods to construct explicitly the solutions are largely unavailable or limited to rather specific cases.

For this reason, it is imperative to develop at least efficient schemes which allow to construct an approximation of the sought solution. On the one hand, one might resort to numerical methods, but, on the other hand, one can also rely on analytic procedures which are often collectively denoted as *perturbation theory*. Let us discuss the basic features in the context of Lagrangian mechanics.

Heuristically speaking, the underlying rationale starts from considering the behaviour of this toy model. Suppose that a point particle is moving in a one-dimensional configuration space, say $\mathbb{R}$ with coordinate q , and that all active forces descend from a potential function $V(q)$, see, *e.g.*, Figure 14. Let us suppose that this possesses one or more local minima. If the particle, say P , is at a given instant of time at one of the minima, say q_0 , then three scenarios can occur:

1. if the initial velocity vanishes, then the particle is at equilibrium and $q(t) = q_0$ for all $t \in \mathbb{R}$,
2. if the initial velocity is small, then the particle moves from q_0 , but the active force $\vec{F} = -\nabla V$ forces it back towards the minimum and it starts oscillating forth and back,
3. if the initial velocity is large, then the particle moves from q_0 , but the active force is too weak to counteract the velocity and the motion becomes largely unpredictable.

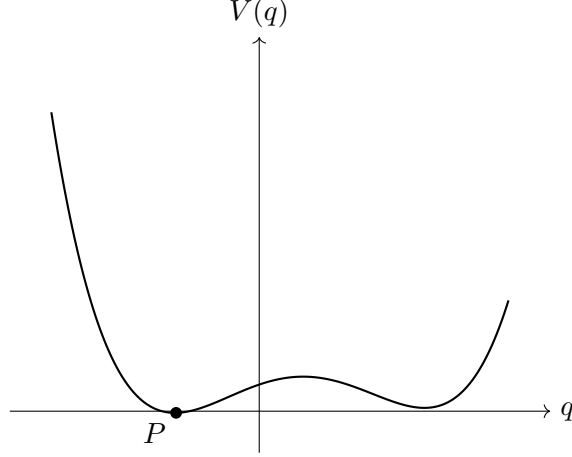


Figure 14: Example of a potential with two local minima and one local maximum.

In the following we generalize this reasoning to Lagrangian mechanics with an arbitrary but finite number of degrees of freedom.

Definition 42. Consider a Lagrangian system as per Nomenclature 2 which has only time-independent constraints. Assume in addition that the active forces all descend from a time-independent potential $V(q^1, \dots, q^{\tilde{N}})$ and denote by $\mathbf{q}$ the generic point $(q^1, \dots, q^{\tilde{N}})$ in the configuration space. We call $\mathbf{q}_0 = (q_0^1, \dots, q_0^{\tilde{N}})$ an **equilibrium configuration** if

$$\nabla V|_{\mathbf{q}=\mathbf{q}_0} = 0 \quad \text{and} \quad \mathbf{H}[V]|_{\mathbf{q}=\mathbf{q}_0} > 0,$$

where $\mathbf{H}[V]$ denotes the Hessian matrix whose generic entry reads

$$\mathbf{H}[V]_{ij} = \frac{\partial^2 V}{\partial q^i \partial q^j} \quad \forall i, j = 1, \dots, \tilde{N} \quad (109)$$

Our next step consists of considering the underlying Lagrangian $\mathcal{L} \equiv \mathcal{L}(\mathbf{q}, \dot{\mathbf{q}})$ and we look for an approximation which is reliable in a neighbourhood of an equilibrium point assuming also small (generalized) velocities. In other words we are interested in its Taylor series:

$$\mathcal{L} = \mathcal{L}(\mathbf{q}_0, 0) + \mathcal{L}_1 + \frac{1}{2}\mathcal{L}_2 + \mathcal{O}(\|\mathbf{q} - \mathbf{q}_0, \dot{\mathbf{q}} - \dot{\mathbf{q}}_0\|^3), \quad (110)$$

where the $\mathcal{O}$ -term entails that we are neglecting all terms which are of order three or higher in the Lagrangian variables $\mathbf{q}$ and $\dot{\mathbf{q}}$, while

$$\mathcal{L}_1 = \sum_{i=1}^{\tilde{N}} \left. \frac{\partial \mathcal{L}}{\partial q^i} \right|_{\substack{\mathbf{q}=\mathbf{q}_0 \\ \dot{\mathbf{q}}=0}} (q^i - q_0^i) + \sum_{j=1}^{\tilde{N}} \left. \frac{\partial \mathcal{L}}{\partial \dot{q}^j} \right|_{\substack{\mathbf{q}=\mathbf{q}_0 \\ \dot{\mathbf{q}}=0}} \dot{q}^j, \quad (111a)$$

$$\mathcal{L}_2 = \sum_{i,j=1}^{\tilde{N}} \frac{\partial \mathcal{L}}{\partial q^i \partial q^j} \Big|_{\substack{\mathbf{q}=\mathbf{q}_0 \\ \dot{\mathbf{q}}=0}} (q^i - q_0^i)(q^j - q_0^j) + \sum_{i,j=1}^{\tilde{N}} \frac{\partial \mathcal{L}}{\partial \dot{q}^i \partial q^j} \Big|_{\substack{\mathbf{q}=\mathbf{q}_0 \\ \dot{\mathbf{q}}=0}} \dot{q}^i (q^j - q_0^j) + \sum_{i,j=1}^{\tilde{N}} \frac{\partial \mathcal{L}}{\partial \dot{q}^i \partial \dot{q}^j} \Big|_{\substack{\mathbf{q}=\mathbf{q}_0 \\ \dot{\mathbf{q}}=0}} \dot{q}^i \dot{q}^j. \quad (111b)$$

As one knows from calculus, going to higher orders in the Taylor expansion allows a better approximation of the expanded function, in the sense that the remainder grows smaller paying at most the price of shrinking the domain of validity for such expansion. Yet, a close investigation of Equation (43) shows that only first derivatives in the generalized velocities and coordinates appear in the Euler-Lagrange equations. Hence, using Equation (110) guarantees that the equations of motion using $\mathcal{L}_0 + \mathcal{L}_1 + \mathcal{L}_2$ in place of the full Lagrangian $\mathcal{L}$ are linear in the underlying variables, while this cannot occur at higher orders. Yet we can further simplify the scenario thanks to the assumptions taken:

- $\mathcal{L}_0$ is a constant since it evaluates the Lagrangian at a fixed configuration and vanishing velocities. Hence this term can be discarded since it cannot contribute to the dynamics.
- $\mathcal{L}_1 = 0$ for the following two reasons combined with the fact that $\mathcal{L} = T - V(\mathbf{q})$:

1. on account of Definition 42, $\nabla V|_{\substack{\mathbf{q}=\mathbf{q}_0 \\ \dot{\mathbf{q}}=0}} = 0$ while,

2. on account of Equations (34) and Equation (35) $T = \sum_{k=1}^N \frac{m_k}{2} \sum_{h,l=1}^{\tilde{N}} a_{hl} \frac{dq^h}{dt} \frac{dq^l}{dt}$, which yields

$$\nabla T|_{\substack{\mathbf{q}=\mathbf{q}_0 \\ \dot{\mathbf{q}}=0}} = 0,$$

since T is quadratic in the velocities and

$$\frac{\partial T}{\partial \dot{q}^i \partial q^j} \Big|_{\substack{\mathbf{q}=\mathbf{q}_0 \\ \dot{\mathbf{q}}=0}} = \sum_{k=1}^N \frac{m_k}{2} \sum_{h,l=1}^{\tilde{N}} \frac{\partial a_{hl}}{\partial q^j} \dot{q}^k \Big|_{\substack{\mathbf{q}=\mathbf{q}_0 \\ \dot{\mathbf{q}}=0}} = 0,$$

- in $\mathcal{L}_2$ for the same reason as in the previous point

$$\sum_{i,j=1}^{\tilde{N}} \frac{\partial \mathcal{L}_2}{\partial \dot{q}^i \partial q^j} \Big|_{\substack{\mathbf{q}=\mathbf{q}_0 \\ \dot{\mathbf{q}}=0}} \dot{q}^i (q^j - q_0^j) = 0.$$

Gathering together all these data we end up rewriting Equation (110) as

$$\mathcal{L} = \sum_{k=1}^N \frac{m_k}{2} \sum_{i,j=1}^{\tilde{N}} a_{ij}(\mathbf{q}_0) \dot{q}^i \dot{q}^j - \sum_{i,j=1}^{\tilde{N}} \frac{H[V]_{ij}(\mathbf{q}_0)}{2} (q^i - q_0^i)(q^j - q_0^j) + \mathcal{O}(\|\mathbf{q} - \mathbf{q}_0, \dot{\mathbf{q}} - \dot{\mathbf{q}}_0\|^3), \quad (112)$$

where we recall that there is an implicit dependence on k in $a_{hl}(\mathbf{q}_0)$. It is quite convenient to write this last expression in a matrix form, namely

$$2\mathcal{L} = \dot{\mathbf{q}}^\top A \dot{\mathbf{q}} - (\mathbf{q} - \mathbf{q}_0)^\top H[V](\mathbf{q} - \mathbf{q}_0),$$

where A is a matrix whose generic entry reads

$$A_{ij} = \sum_{k=1}^N \frac{m_k}{2} a_{ij}(\mathbf{q}_0). \quad (113)$$

The Euler-Lagrange equations acquire therefore a simple form, also referred to as *linearization procedure*,

$$A\ddot{\mathbf{q}} = \mathbf{H}[V](\mathbf{q} - \mathbf{q}_0). \quad (114)$$

Setting $\mathbf{q}_0 = 0$ for simplicity or changing coordinates as $\tilde{\mathbf{q}} \doteq \mathbf{q} - \mathbf{q}_0$, we end up with the final form of the linearized version of the Euler-Lagrange equations

$$\ddot{\mathbf{q}} = A^{-1}\mathbf{H}[V]\mathbf{q}, \quad (115)$$

where we used that, per construction A is invertible, see Section 3.1. Observe that this is a system of $\tilde{N}$, coupled, linear ordinary differential equations. Yet, we can set up a diagonalization procedure. We introduce an invertible matrix $K \in \text{GL}(n, \mathbb{R}) = \{A \in \mathcal{M}(n, \mathbb{R}) \mid \det A \neq 0\}$ and, setting $\mathbf{Q} = K\mathbf{q}$, Equation (115) becomes

$$K^{-1}\ddot{\mathbf{Q}} = A^{-1}\mathbf{H}[V]K^{-1}\mathbf{Q} \implies \ddot{\mathbf{Q}} = KA^{-1}\mathbf{H}[V]K^{-1}\mathbf{Q}. \quad (116)$$

We prove a notable result of linear algebra.

Proposition 43. *Given $A, B \in \mathcal{M}(n, \mathbb{R})$, $n \geq 1$, which are both symmetric while $A > 0$, then $\exists K \in \text{GL}(n, \mathbb{R})$ such that*

$$KA^{-1}BK = \text{diag}(\lambda_1, \dots, \lambda_n).$$

In addition, for all $k = 1, \dots, n$, $\lambda_k \geq 0$ if and only if $B \geq 0$, while $\lambda_k > 0$ if and only if $B > 0$.

We omit the long and tedious proof, referring to [Mor20, Prop. 10.22]. We observe that Equation (116) or, more properly, A and $\mathbf{H}[V]$ abide by the hypotheses of Proposition 43. Hence Equation (116) can be transformed into its final form:

$$\ddot{\mathbf{Q}} = D\mathbf{Q}. \quad D = \begin{bmatrix} \omega_1^2 & 0 & \cdots & 0 \\ 0 & \omega_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \omega_n^2 \end{bmatrix}, \quad (117)$$

where the eigenvalues $\omega_1^2, \dots, \omega_n^2$ are called **normal modes** and they are the solution of the secular equation

$$\det(A^{-1}\mathbf{H}[V] - \omega^2\mathbb{I}) = \det(\mathbf{H}[V] - \omega^2A) = 0, \quad (118)$$

where $\mathbb{I}$ is the $n \times n$ -identity matrix and where we used implicitly the Binnet formula combined with the information that $\det A^{-1} \neq 0$. The associated eigenvectors are called **normal coordinates**.

Remark 19. *It is worth stressing that the coordinates $\mathbf{Q}$ which arise out of the diagonalization procedure leading to Equation (117) might not have an immediate or distinguished geometrical significance. As a matter of fact, The coordinates $\mathbf{q}$ are typically chosen or even inspired according to the underlying geometry of the problem and they usually refer to the position in space of the individual point particles, once the underlying rehomomic constraints are taken into account. On the contrary their companions, $\mathbf{Q}$, describe collective oscillation modes and, as the following examples shall elucidate, they are not necessarily pinpointing the position of a single point particle in space.*

Remark 20. *The normal modes and Equation (117) are the consequence of a Taylor expansion of the Lagrangian as per Equation (28). A first and often ignored question is whether this entails that a solution of Equation (117) is a good approximation of the exact solution of the underlying Euler-Lagrange equation. The answer is yes and the proof is deeply rooted in the Picard-Lindelöf theorem and in the underlying Banach fixed point argument. Yet this is not the real point also because the answer is yes.*

The true, never asked, question is whether, going further in the Taylor series, or as it is commonly stated, in the perturbative series leads to a better approximation of the solutions, despite the challenges which the nonlinear contributions bring from the third order and beyond. The answer is positive and, therefore, all is well what it ends well. Nothing ever ends well when you are coping with perturbation series.

*The never asked question is whether the series that you obtain converges to the real solution. Of course there is no generic answer and it depends on the case in hand, but, a good rule of the thumb is that it **does not**! It is like asking to the Taylor series to be convergent at all orders for all values of the underlying variable. You are asking yourself if the function is real analytic, which it could be. Alas, real analytic functions are very rare even in the class of smooth functions. The same applies with physical models: only a few and usually the least interesting ones have a perturbative series which converges. What happens is that the higher is the order of the expansion of the Lagrangian, the better your approximation gets, but smaller it becomes the range of validity of such an approximation. In other words, if you start from $\mathbf{q}_0$, the higher is the order in the Taylor series the closer you have to be to the expansion point, for the series to become meaningful. In the limit, your range of validity shrinks to $\mathbf{q}_0$. In classical mechanics this is often under control, but mark these words since you will be seeing this same procedure in quantum mechanics, in solid state physics, in statistical mechanics, in quantum field theory, in general relativity and so on and so forth. Does it ever get better? No, if possible and it is, it gets worse.*

So why do we rely on perturbation theory? Simply because, to a certain extent, namely to low orders, it works very well for the range of energies and for the sizes that can be probed by experiments. In addition this is the best we can do and therefore we have to live with it.

Example – masses with wall springs and a coupling spring

We study two prototypical examples of applications of the theory of normal modes. The first model is strongly inspired by models describing the vibration frequencies of molecules and similar

calculations will be appearing during more advanced courses at the third year of your studies.

Exercise: In $\mathbb{R}^2$ consider a reference system as per Figure 15 and two point particles of mass m_1 and m_2 moving along the horizontal axis. Among them there is a spring of elastic constant $\kappa_c > 0$ and vanishing rest length. The masses m_1 and m_2 are each attached to a spring of elastic constant $\kappa > 0$ and vanishing rest length, the first pulling m_1 towards the origin, the second pulling m_2 towards a wall at $L > 0$. Determine the quasi normal models of the system. In the limit case $m_1 = m_2$ compute also the eigenvectors associated to the modes, the solution of their equation of motion and establish a physical interpretation.

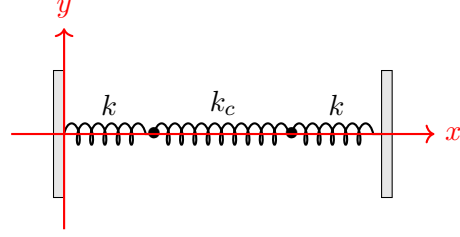


Figure 15: Two masses connected by three springs.

Solution: Observe that we are considering a system subject to the ideal constraints $y_1 = y_2 = 0$, the subscript labeling the underlying point particles. Hence we have 2 degrees of freedom and the geometry of the problem suggest to consider as Lagrangian coordinate x_1 and x_2 the position along the horizontal axis. Accordingly the kinetic and potential energy read:

$$T = \frac{m_1}{2} \dot{x}_1^2 + \frac{m_2}{2} \dot{x}_2^2 \quad \text{and} \quad V = \frac{\kappa}{2} x_1^2 + \frac{\kappa}{2} (L - x_2)^2 + \frac{\kappa_c}{2} (x_2 - x_1)^2.$$

In order to individuate the equilibrium position we compute

$$\nabla V = \begin{pmatrix} \frac{\partial V}{\partial x_1} \\ \frac{\partial V}{\partial x_2} \end{pmatrix} = \begin{pmatrix} \kappa x_1 - \kappa_c (x_2 - x_1) \\ -\kappa (L - x_2) + \kappa_c (x_2 - x_1) \end{pmatrix} = 0,$$

which entails

$$x_{1,0} = \frac{\kappa_c}{\kappa + 2\kappa_c} L, \quad x_{2,0} = \frac{\kappa + \kappa_c}{\kappa + 2\kappa_c} L.$$

To verify whether this a stable equilibrium point we compute the Hessian of the potential, namely

$$\mathbf{H}[V]_{ij} = \begin{bmatrix} \frac{\partial^2 V}{\partial x_1^2} & \frac{\partial^2 V}{\partial x_1 \partial x_2} \\ \frac{\partial^2 V}{\partial x_2 \partial x_1} & \frac{\partial^2 V}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} \kappa + \kappa_c & -\kappa_c \\ -\kappa_c & \kappa + \kappa_c \end{bmatrix}.$$

Since it is a 2×2 matrix, being positive is equivalent to the positivity of both the trace and the determinant which read in this case

$$\text{Tr}(\mathbf{H}[V]) = 2(\kappa + \kappa_c) > 0 \quad \text{and} \quad \det(\mathbf{H}[V]) = \kappa^2 + 2\kappa\kappa_c > 0.$$

Hence in order to compute the normal modes, we are left with computing the matrix A as per Equation (113):

$$A = \begin{bmatrix} \frac{m_1}{2} & 0 \\ 0 & \frac{m_2}{2} \end{bmatrix}.$$

Hence we are left with solving the secular equation (118), namely

$$\det \left(\begin{bmatrix} \kappa + \kappa_c & -\kappa_c \\ -\kappa_c & \kappa + \kappa_c \end{bmatrix} - \omega^2 \begin{bmatrix} \frac{m_1}{2} & 0 \\ 0 & \frac{m_2}{2} \end{bmatrix} \right) = 0,$$

or, equivalently,

$$\frac{m_1 m_2}{4} \omega^4 - \frac{(m_1 + m_2)}{2} (\kappa + \kappa_c) \omega^2 + \kappa^2 + 2\kappa\kappa_c = 0.$$

In other words

$$\omega_{\pm}^2 = \frac{(\kappa + \kappa_c)(m_1 + m_2) \pm \sqrt{(\kappa + \kappa_c)^2(m_1 + m_2)^2 - 4m_1 m_2 (\kappa^2 + 2\kappa\kappa_c)}}{m_1 m_2}.$$

We can now take the limiting case $m_1 = m_2 = m$ where this last identity reduces to

$$\omega_-^2 = \frac{\kappa}{m} \quad \text{and} \quad \omega_+^2 = \frac{\kappa + 2\kappa_c}{m}.$$

In order to determine the eigenvectors, we start from ω_-^2 and we look for $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ such that

$$(\mathbf{H}[V] - \omega_-^2 \mathbf{A}) \mathbf{v} = \begin{bmatrix} \kappa_c & -\kappa_c \\ -\kappa_c & \kappa_c \end{bmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \implies \mathbf{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Rewriting this expression in terms of Lagrangian coordinates is tantamount to saying that the first *normal coordinate* is

$$Q_1 \doteq x_1 - x_2,$$

which abides by Equation (117), hence

$$Q_1(t) = Q_1(0) \cos(\omega_- t) + \frac{\dot{Q}_1(0)}{\omega_-} \sin(\omega_- t),$$

where the values of $Q_1(0)$ and $\dot{Q}_1(0)$ depend on the assigned initial conditions. Repeating the same procedure with ω_+^2 we are looking now for a vector $\mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$ such that

$$(\mathbf{H}[V] - \omega_+^2 \mathbf{A}) \mathbf{w} = \begin{bmatrix} -\kappa_c & -\kappa_c \\ -\kappa_c & -\kappa_c \end{bmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = 0 \implies \mathbf{w} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Rewriting this expression in terms of Lagrangian coordinates is tantamount to saying that the first *normal coordinate* is

$$Q_2 \doteq x_1 + x_2,$$

which abides by Equation (117), hence

$$Q_2(t) = Q_2(0) \cos(\omega_+ t) + \frac{\dot{Q}_2(0)}{\omega_+} \sin(\omega_+ t),$$

where the values of $Q_2(0)$ and $\dot{Q}_2(0)$ depend on the assigned initial conditions. The physical interpretation is that Q_1 describes the *in-phase* mode, namely both masses are moving together in the same direction, while Q_2 is associated to the *out-phase* mode namely the two masses are moving in opposite directions.

Example – Double Pendulum

We discuss now a second very traditional example of system exhibiting normal modes, the double pendulum, see Figure 16.

Exercise: In $\mathbb{R}^2$ consider a reference system with origin labeled by O as per Figure 16. A pendulum of negligible mass and length $\ell > 0$ is constrained on one end at O . At the other hand it is constrained a point particle of mass $m_1 > 0$. A second pendulum still of length $\ell > 0$ is constrained on one end at the second end of the first pendulum. At the other end it is attached a point particle of mass $m_2 > 0$. On the system it is active only a constant gravitational force directed along the negative y -direction, whose associated acceleration is $g > 0$. Determine the normal modes and the normal coordinates of the system.

Solution: Observe that we are considering a system of two point particles with two ideal constraints, namely, denoting the position vectors of the two masses $\mathbf{x}_1 = (x_1, y_1)$ and $\mathbf{x}_2 = (x_2, y_2)$, it turns out that $x_1^2 + y_1^2 - \ell^2 = 0$ while $(x_2 - x_1)^2 + (y_2 - y_1)^2 - \ell^2 = 0$. In other words the system has two degrees of freedom and the geometry suggests to choose as Lagrangian coordinates the angles θ_1, θ_2 as depicted in Figure 16. Accordingly the position of the two masses is

$$\mathbf{x}_1 = (\ell \cos \theta_1, -\ell \sin \theta_1) \quad \text{and} \quad \mathbf{x}_2 = (\ell(\cos \theta_1 - \sin \theta_2), -\ell(\sin \theta_1 + \cos \theta_2)),$$

while the velocity vectors are

$$\dot{\mathbf{x}}_1 = \ell(-\sin \theta_1 \dot{\theta}_1, -\cos \theta_1 \dot{\theta}_1) \quad \text{and} \quad \dot{\mathbf{x}}_2 = \ell(-\sin \theta_1 \dot{\theta}_1 - \cos \theta_2 \dot{\theta}_2, -\cos \theta_1 \dot{\theta}_1 + \sin \theta_2 \dot{\theta}_2),$$

from which it descends that the kinetic energy is

$$T = \frac{1}{2} m_1 \ell^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 \ell^2 (\dot{\theta}_1^2 + \dot{\theta}_2^2 - 2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2)).$$

Using the same rationale as in the first example of Section 3.2, the potential energy is

$$V = -m_1 g \ell \sin \theta_1 - m_2 g \ell (\sin \theta_1 + \cos \theta_2).$$

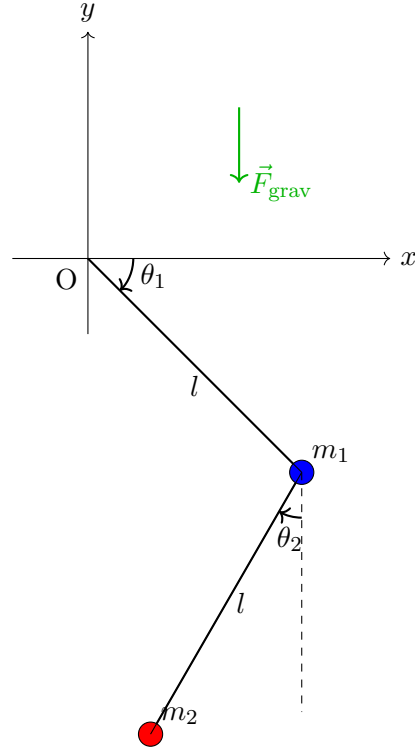


Figure 16: Each mass is attached at a pendulum of length $\ell > 0$ and a gravitational force $\vec{F} = -mg\mathbf{e}_y$ acts on the system.

The next step consists of establishing the equilibrium configuration starting from

$$\nabla V = \begin{pmatrix} \frac{\partial V}{\partial \theta_1} \\ \frac{\partial V}{\partial \theta_2} \end{pmatrix} = \begin{pmatrix} -(m_1 + m_2)g\ell \cos \theta_1 \\ m_2 g\ell \sin \theta_2 \end{pmatrix} = 0 \implies \begin{cases} \theta_1 = \pm \frac{\pi}{2} \\ \theta_2 = 0, \pi \end{cases}.$$

We have four possible equilibrium points $(-\frac{\pi}{2}, 0), (-\frac{\pi}{2}, \pi), (\frac{\pi}{2}, 0)$ and $(\frac{\pi}{2}, \pi)$ and, to select the correct ones we compute the Hessian

$$H[V]_{ij} = \begin{bmatrix} \frac{\partial^2 V}{\partial \theta_1^2} & \frac{\partial^2 V}{\partial \theta_1 \partial \theta_2} \\ \frac{\partial^2 V}{\partial \theta_2 \partial \theta_1} & \frac{\partial^2 V}{\partial \theta_2^2} \end{bmatrix} = \begin{bmatrix} (m_1 + m_2)g\ell \sin \theta_1 & 0 \\ 0 & m_2 g\ell \cos \theta_2 \end{bmatrix}.$$

Recalling that a 2×2 matrix is positive if and only if both the trace and the determinant are greater than 0, we obtain

$$\text{Tr}(H[V]) = g\ell((m_1 + m_2)\sin \theta_1 + m_2 \cos \theta_2) \quad \text{and} \quad \det(H[V]) = (m_1 + m_2)m_2 g^2 \ell^2 \sin \theta_1 \cos \theta_2,$$

which selects $(\frac{\pi}{2}, 0)$ as the only stable point. Hence in order to find the normal modes, we are left with computing the matrix A as per Equation (113):

$$A = \ell^2 \begin{bmatrix} m_1 + m_2 & m_2 \\ m_2 & m_2 \end{bmatrix},$$

while, at the equilibrium point the Hessian reads

$$H[V]\left(\frac{\pi}{2}, 0\right) = \begin{bmatrix} (m_1 + m_2)g\ell & 0 \\ 0 & m_2 g\ell \end{bmatrix}.$$

Hence we are left with solving the secular equation (118), namely

$$\det\left(\ell^2 \begin{bmatrix} m_1 + m_2 & m_2 \\ m_2 & m_2 \end{bmatrix} - \omega^2 \begin{bmatrix} (m_1 + m_2)g\ell & 0 \\ 0 & m_2 g\ell \end{bmatrix}\right) = 0,$$

which yields

$$(m_1 + m_2)(g - \omega^2)^2 - m_2 \omega^4 = 0.$$

Hence the *normal modes* are

$$\omega_{\pm}^2 = \frac{g}{m_1} \left((m_1 + m_2) \pm \sqrt{m_2(m_1 + m_2)} \right).$$

As in the previous example we can also construct the *normal coordinates*, taking $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ such that

$$(H[V] - \omega^2 A) \mathbf{v} = 0,$$

where ω^2 is one of the normal modes. A direct calculation which is left to the reader leads to $\mathbf{v}_\pm$ associated to $\omega_\pm^2$ whose expression is

$$\mathbf{v}_- = \left(\frac{1}{\frac{(m_1+m_2)(g-\omega_-^2)}{m_2\omega_-^2}} \right) \quad \text{and} \quad \mathbf{v}_+ = \left(\frac{1}{\frac{(m_1+m_2)(g-\omega_+^2)}{m_2\omega_+^2}} \right),$$

or, working at the level of Lagrangian coordinates

$$Q_1 = \theta_1 + \frac{(m_1+m_2)(g-\omega_-^2)}{m_2\omega_-^2}\theta_2 \quad \text{and} \quad Q_2 = \theta_1 + \frac{(m_1+m_2)(g-\omega_+^2)}{m_2\omega_+^2}\theta_2.$$

Not yet. We're only scratching the surface.

Interstellar

A Rotations in $\mathbb{R}^3$

In this short appendix we review how to parameterize rotations of vectors. Let us start from $\mathbb{R}^2$ and let us consider two unit vectors $\mathbf{v} = (v_1, v_2)$ and $\mathbf{w} = (w_1, w_2)$ as per Figure 17

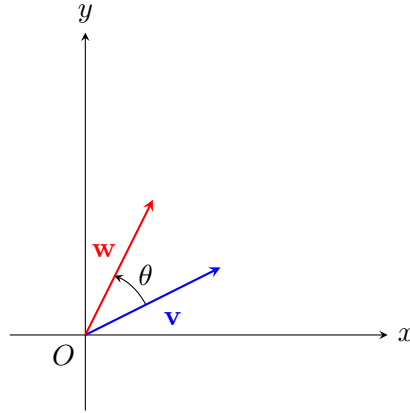


Figure 17: Two unit vectors in $\mathbb{R}^2$ with the relative angle displayed

Using standard trigonometry one can rewrite the coordinates of $\mathbf{w}$ in terms of those of $\mathbf{v}$, namely

$$\begin{cases} w_1 = v_1 \cos \theta + v_2 \sin \theta \\ w_2 = -v_1 \sin \theta + v_2 \cos \theta \end{cases},$$

which can be rewritten in a matrix form:

$$\mathbf{w} = R(\theta)\mathbf{v} \implies \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix},$$

where $R(\theta)$ is known as **rotation matrix**

Exercise 44. Denote by $SO(2, \mathbb{R}) \doteq \{A \in \mathcal{M}(2, \mathbb{R}) \mid A^\top = A^{-1} \text{ and } \det(A) = 1\}$ the *orthogonal group* on $\mathbb{R}^2$. Prove that any element therein can be written as a rotation matrix.

Let us move our attention to $\mathbb{R}^3$ and in this case, if you want to think of all possible rotations, before moving to a more mathematical description, think of an airplane:

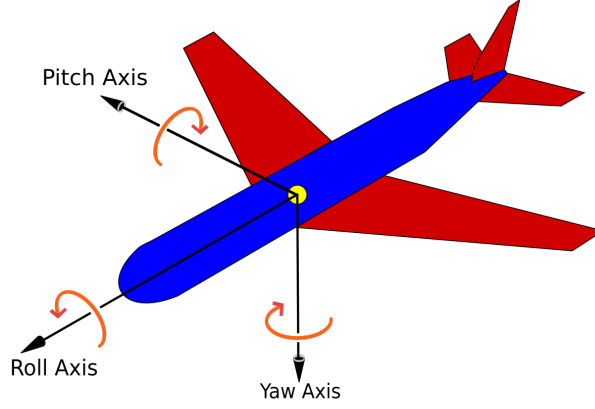


Figure 18: Yaw, pitching and rolling axis of a plane

As a matter of fact you expect that every possible rotation can be realized as a combination of one or more among the basic rotations, which are known as yawing (rotation in the x-y plane), pitching (rotation in the x-z plane) and rolling (rotation in the y-z plane), assuming that the plane points along the x-axis. Let us translate this statement in a mathematical language. Assume therefore that we have a Cartesian reference frame centered at O and whose axis are labeled x, y, z . Let us introduce the following matrices

$$D(\varphi) = \begin{bmatrix} \cos \varphi & \sin \varphi & 0 \\ -\sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad C(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix}, \quad B(\psi) = \begin{bmatrix} \cos \psi & 0 & \sin \psi \\ 0 & 1 & 0 \\ -\sin \psi & 0 & \cos \psi \end{bmatrix}.$$

If we multiply together this matrices in the following strict order we obtain

$$B(\psi)C(\theta)D(\varphi) = \begin{bmatrix} \cos \psi \cos \varphi & -\cos \psi \sin \varphi & \sin \psi \\ \sin \theta \sin \psi \cos \varphi + \cos \theta \sin \varphi & -\sin \theta \sin \psi \sin \varphi + \cos \theta \cos \varphi & -\sin \theta \cos \psi \\ -\cos \theta \sin \psi \cos \varphi + \sin \theta \sin \varphi & \cos \theta \sin \psi \sin \varphi + \sin \theta \cos \varphi & \cos \theta \cos \psi \end{bmatrix}.$$

Proposition 45. Given the unit 2-sphere $\mathbb{S}^2 \doteq \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$, for any $\mathbf{v}, \mathbf{w} \in \mathbb{S}^2$, there exists $A \in SO(3, \mathbb{R})$ such that $\mathbf{v} = A\mathbf{w}$, where

$$SO(3, \mathbb{R}) \doteq \{A \in \mathcal{M}(3, \mathbb{R}) \mid A^\top = A^{-1} \text{ and } \det(A) = 1\}.$$

Proof. Let us consider $\mathbf{v} \neq \pm \mathbf{w}$, since these are trivial cases. In the first case, it suffices to choose $A = \mathbb{I}$, the identity matrix, the second will be dealt with at the end of the proof.

In all other case, we know that there exists a plane containing both $\mathbf{v}$ and $\mathbf{w}$ as well as that a normalized vector orthogonal to such a plane is

$$\mathbf{u} = \frac{\mathbf{v} \wedge \mathbf{w}}{\|\mathbf{v} \wedge \mathbf{u}\|},$$

where $\wedge$ still denotes the vector product. Denoting the components as (u_x, u_y, u_z) , we can construct the auxiliary matrix

$$K = \begin{bmatrix} 0 & -u_z & u_y \\ u_z & 0 & -u_x \\ -u_y & u_x & 0 \end{bmatrix},$$

and we can introduce the angle $\theta = \arccos(\mathbf{v} \cdot \mathbf{w})$. Consider the matrix

$$A = \mathbb{I} + \sin \theta K + (1 - \cos \theta) K^2,$$

which lies in $SO(3, \mathbb{R})$ since

$$AA^\top = (\mathbb{I} + \sin \theta K + (1 - \cos \theta) K^2) (\mathbb{I} - \sin \theta K + (1 - \cos \theta) K^2) = \mathbb{I},$$

where we used that $K^\top = -K$ and that $K^3 = -K$. Observe that this entails that $\det(A) = \pm 1$. Yet $A \equiv A(\theta)$ is a continuous function with respect to θ and so is $\det A(\theta)$. Accordingly, since it cannot change sign varying θ , the price being the failure of continuity, it suffices to compute the determinant for a single value of the angle. Setting $\theta = 0$ yields $A(0) = \mathbb{I}$ and, therefore $\det A = 1$. To conclude the proof one needs to evaluate

$$A\mathbf{w} = \mathbf{v},$$

which is left as an exercise to the reader. □

Exercise 46. Prove that any $A \in SO(3, \mathbb{R})$ can be written as the product of three rotation matrices, namely $A = B(\psi)C(\theta)D(\varphi)$.

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